# Epic-FHIR — Integration Playbooks (sections 2 – 2.55)

Source: Epic on FHIR (fhir.epic.com) documentation, mirrored 2026-11. Text-only edition for AI ingestion (NotebookLM); pages follow the source order (its table of contents when one exists).

## 2 Integration Playbooks

File: site/docs/02-playbooks/index.html

* Vaccine Credentials
* Remote Patient Monitoring
* International Patient Summary Playbook
* Wound Care Apps
* JWT Authentication Troubleshooting
* Payer Connections to Epic Provider Systems
* Implantable Cardiac Devices Playbook
* Computer Telephony Integrations (CTI)
* Payment Device Integrations Playbook
* Bayesian Medication Dosing Decision Support Playbook
* Medication Ordering Decision Support Playbook
* Fully Autonomous Coding Playbook
* Automated Dispensing System Playbook
* Hemodynamics System Playbook
* ECG Management System Playbook
* Point-of-care Lab Middleware Integrations Playbook
* Standard Lab Middleware Integrations Playbook
* Waveform Visualization Playbook
* RTLS Playbook
* Surgical Materials Playbook
* Digital Pathology Playbook
* Wearable Cardiac Devices Playbook
* Clinical Knowledge Set Apps Playbook
* Total Parenteral Nutrition Compounding Playbook
* Encoder Playbook
* Medication Carousel Playbook
* End-User Authentication Playbook
* Bedside Dining Playbook
* Printing Output Management Playbook
* Wayfinding Playbook
* Infusion Pump Playbook
* Patient-Facing / Consumer Health Apps
* Endoscope Data Capture Playbook
* Alert Manager Playbook
* Telehealth Providers Playbook
* Vitals Gateway and Middleware Device Integration Playbook
* Remote Patient Monitoring with Devices Playbook
* External Health Records for Patient Treatment
* Building a Patient Roster
* Reference Labs
* Perioperative Workflows Playbook
* Deploying Predictive Models Playbook
* Clinical Decision Support Playbook
* ED Pre-Arrival EMS Playbook
* Specialty Registries Playbook
* Validating AI Features Playbook

* CMS-0057 Prior Authorization Playbook
* CMS-0057 Access Playbook
* Radiation Therapy Treatment Management System Playbook
* XRTS FHIR API Tutorial
* Histocompatibility Laboratory Information System Playbook
* Staff Duress Playbook
* Diagnostic Image Exchange Playbook
* Payer Medical Necessity Guidelines Playbook
* Hospital Utilization Review Creation Playbook

# 2.1 Vaccine Credentials

File: site/docs/02-playbooks/01-vaccinecredential.html

Source: https://fhir.epic.com/Documentation?docId=vaccinecredential

Epic's support for vaccine credentials using the SMART Health Cards specification is evolving. The content on this page might change without notice.

What's a "Vaccine Credential?

* Related Terms & Definitions
* Audience and Intent

What's a SMART Health Card?

* Accessing Epic-Generated SMART Health Cards

Reading SMART Health Cards

QR codes

* Numerical Decoding
* Chunked QR Codes
* .smart-health-card Files

Consuming SMART Health Cards

* JOSE Header

Payload

* Inflation
* SMART Health Card Header
* FHIR Bundle

FHIR Patient

* Considerations For Patient Identity

FHIR Immunization

* Example Immunization

FHIR Observation

* Example Observation

SMART Health Cards & Trust

* Verify the Signed Web Token
* Open Source Libraries & References

## What's a "Vaccine Credential"?

A vaccine credential binds an individual's demographic information with information about their lab tests or vaccine history into a package that is machine-readable and digitally signed by the issuer. Unlike a paper vaccine card, a vaccine credential is verifiable and therefore much more difficult to tamper with or counterfeit.

A CDC vaccine card, widely issued in the United States to patients when they receive a COVID-19 vaccination.

## Related Terms & Definitions

* Vaccine or immunity passport. Passports are issued by governments. An immunity passport asserts immunity. In contrast, a vaccine credential is just an alternative way of representing information documented about a patient--specifically, their vaccination or lab history.

* Health Wallet. An app available for install and use on a consumer's device that aggregates and stores vaccine credentials from different sources. A health wallet is a convenient way to store multiple credentials, but is not strictly required.

* Pass App. An app available for install and use on a consumer's device that reads one or more vaccine credentials to determine if the user meets entry criteria for a specific venue or building, for example. might also perform some form of

identity verification and might also commonly have health wallet-type functionality. A pass app might be useful in creating "fast lanes" in which a consumer's identity would not need to be re-verified, but a verifier should not require the patient to have a pass app.

* Verifier. A consumer of a SMART Health Card.

## Audience and Intent

This document was designed to support health wallets, pass apps, and verifiers, in designing apps and technologies that can consume a patient-provided SMART Health Card. We note particularly that any developer creating a pass app should also consider how verifiers using that pass app will interact with patients who do not have access to technology and present only with their ID and a printed SMART Health Card. Any verifier that requires patients to download a specific type of pass app (and thus, cannot read and interact with printed SMART Health Cards) is disadvantaging individuals without technology. SMART Health Cards were specifically designed to provide patient access regardless of technological ability or sophistication.

In other words, if you require an individual to download and present a certain type of app to enter your venue, you have not implemented the specification correctly.

## What's a SMART Health Card?

A SMART Health Card is a standard for representing a vaccine credential. The SMART Health Cards standard is open, interoperable, and built on top of HL7 FHIR and compatible with W3C's Verifiable Credentials. It's designed to be:

* Open – The standard is open source.

* Interoperable – It's built on top of FHIR.

* Privacy-preserving – Clinical content is embedded within SMART Health Cards, ensuring that a patient can't be tracked based upon where they use their vaccine credential.

* Equitable – It can be represented entirely on paper.

A SMART Health Card contains very limited patient demographics, and either information about the patient's COVID-19 immunization or COVID-19 lab result.

Technically, a SMART Health Card is a JSON object containing a JWS of compressed JSON, consisting of a FHIR Patient and either one or two FHIR Immunizations; or a FHIR Observation resource. The card can be represented as a file ending in the .smart-health-card extension, or as a QR code printed on paper or displayed digitally.

You can learn more about SMART Health Cards by visiting the following websites:

* https://vci.org/

* https://smarthealth.cards/

* http://build.fhir.org/ig/dvci/vaccine-credential-ig/branches/main/

## Accessing Epic-Generated SMART Health Cards

Epic generates SMART Health Cards in two different ways and via a few different workflows:

* Patients can access SMART Health Cards as either a QR code displayed in or a file downloaded from the MyChart website or the MyChart mobile apps.

* Health systems using Epic can generate and print a QR code containing a patient's SMART Health Card in the clinician-facing interface, Hyperspace. Typically, this workflow would be used to provide SMART Health Cards to patients without access to MyChart.

Patients can share QR codes either directly from a mobile device or from a piece of paper. You should test and validate that your app can consume the QR code from both a mobile device and from a piece of paper.

A patient can download their SMART Health Card from MyChart onto their device. If you're developing an app that can run on a patient's device, alongside MyChart, you should associate your app with the .smart-health-card filename extension.

## Reading SMART Health Cards

### QR codes

Every SMART Health Card can be represented as a QR code. Typical QR scanner apps expect the QR code to contain a URL that can be navigated to in a browser, but a SMART Health Card's QR code does not contain an http:// URL. Rather, the content begins with an shc:/ prefix, followed by a series of numbers and optionally forward slashes (/).

For example, a SMART Health Card QR code could contain this example content after it is scanned:


```
shc:/567629595326546034602925407728043360287028647167452228092864456652064567322455425662713655612967382737094440447142272560543945364126252454415527384355065405373633654575390303603122290952432060346029243740446057360106413733531270742335036958065345203372766522453157257438663242742309774531624034227003622673375938313137693259503100426236772202335560503640358003561030076650858305837427006053253525808100334720725083211056752443072582026570761440666744164312435356975435537617044773254106560764243695628297630727624737442665732682773614325600457746054112123096876672930040634423253734570636732752736052662062603077704743466272159075371252529575571715771240430536555305000757007062625276232697334206803333132397043325945353239607168401108387764601158554206752527675040356974657404250043212758306560270029683460506837412661763738303203115327353229555240056465566937404476720828374369077767612600436157602561286826297610761176744250457077312409772972613464700060063524074066447556263440056932422923764107223028717704747432072167635507356054205837774364737133521250632007730035294059084074092650297428443462657163210643750776040812622734417303523427222465085925
```

```
    8442455033007273821670967547629685929084256416326620010003040772650374024334366606328710068067729684128676639665731687
    8714052274132636211277761616474430568250469036563675600425527541073093326122904354331506352505556004167345057310352415
    1562857000866406707250066585029295368623020723771290300716970507307715829230409632369722208220022421235364507226273407
    0762363767326612510255232725503432227582401080067696650706662742033521070103752057642223061237175612365276528776200744
    4440021316664703977272245005641643254373305420952023770334642853457603077175713340643036122438542 0
```

The numbers following the shc:/ prefix are the content of the SMART Health Card and must be numerically decoded.

### Numerical Decoding

SMART Health Cards' QR codes represent characters as their ASCII code. The SMART Health Cards specification defines a simple algorithm to represent each character from the JWS as a two-digit number. Following the prefix, each pair of digits is converted into a single character by adding 45 to determine the encoded letter that the integer represents.

For example, the first pair of digits in the numerically encoded SMART Health Card example above is 56. By adding our offset constant 45 to 56, the value translates to ASCII's lower-case e (Chr(101) = "e"). The constant 45 was chosen because 45 is the ASCII value for "-", which is the lowest character that can exist in our Base64-URL encoded content.

Decoding the string proceeds as follows:

56 + 45 = e

76 + 45 = y

29 + 45 = J

Etc.

When parsing is complete, the above shc:/ string resolves to the following JWS:

```
eyJhbGciOiJFUzI1NiIsImtpZCI6ImZoa3ZpMEdWektQdjJpSHR6YUYtWHFicTZQVGFEcVdHSXd3c2RQNnZxT00iLCJ6aXAiOiJERUYifQ.3VRNb9swDP0rg
    3bZANuynCZLfFwSoMWwD6zZLkUOCs0kGvRhSLLRrMh_L-WkQzC0Pe20m0g-Pj0-
    yn5gKgRWs32Mbag570Ou4F5M82paYKugAGf4jY3owVmLEPPrxXdRjsYzMc7niyWXreIJyKuyEvwWofMqHvjXFi1fwic8BD6qypJK13OWMbvZslpMxHQ2
    Gk3G04z1wOoHBh4btFFJfdttftE1KbndK_-xs43GFHkMrvOAq0NLMTsXMhZPMTitqU85Szmi8gdW3xFHp_UPrwnw1F-XBHgKniH-
    JqOi_qRVGjyRSKM08bHPMJdaU2mnerRUYyu5IRXr4zpjG-XjfiFjIqGJy7y8ysWYHY_ZszLE6zJujOms-
    i3PE4UoYxeGOU2rMWJDyV4CKItz1wwM4Bpld4PicAgRzXmvtNa9_lA4v-
    PJUh5Uw6G_JwIYOkntlB3Xx4y159kHOVv0aOHCEn5hIYEd0K4HSBp6pcyJqhJ5WeVlGk-7-KUzG_RUENXoilIt-
    q3zJqVIpoTofLqtUaHVMlk8HzjjmwX2qF1r0nlpe-WdHc7v6NG9J1PXL_la_de-VpO_fL0aVeIf-5oUp4F-og_JJbqkKAuRtJ0-
    trs_v4tgJD16lDruC5C-CW9PQZ4CkvUyDlyvGjF7FaMud0XCHgE.5-pro_sokwANa7s7Ra2yWCKjDtxjDnHnIzk-wY-BLomsTzHCZ-
    eVmMcRN2W6a_Dz0OmIbZy04txtNUmKQ9EScA
```

See the SMART Health Cards specification for an alternate explanation of numerical encoding.

### Chunked QR Codes

In some cases, the underlying demographic and clinical data stored in a SMART Health Card QR code might simply be too large for a single QR code. If so, the SMART Health Card is "chunked" -- split across multiple QR codes that must be scanned and reassembled before any data can be accessed. In this case, the shc:/ prefix is appended with two additional values, delimited by forward slashes ("/"). The first value defines the order of the current QR code for reassembly and the second is the total number of QR codes required to reassemble the data.

For example, a SMART Health Card of 3500 characters is split across three distinct QR codes. The prefix of the first QR code will be: shc:/1/3/, the second: shc:/2/3/ and the third: shc:/3/3/. The contents of all three QR codes (following the final forward slash) must be concatenated in order before inflation.

Requiring multiple QR codes to receive a single SMART Health Card is undesirable and likely confusing to consumers. Epic will rarely place so much information in a single SMART Health Card such that chunking the data across multiple QR codes is necessary. However, apps consuming these QR codes must be prepared to scan and assemble multiple QR codes.

### .smart-health-card Files

In addition to presenting a QR code to a scanner, a consumer can also download a file to share a SMART Health Card. In this case, there's no QR code, no numeric encoding, and no chunking. The SMART Health Card is represented as a JWS wrapped in a simple JSON array. The filename extension of this file will always be ".smart-health-card".

Here's an example of a .smart-health-card file containing a single SMART Health Card:

```
{
  "verifiableCredential": [
"eyJhbGciOiJFUzI1NiIsImtpZCI6ImZoa3ZpMEdWektQdjJpSHR6YUYtWHFicTZQVGFEcVdHSXd3c2RQNnZxT00iLCJ6aXAiOiJERUYifQ.3VRNb9swDP0r
    g3bZANuynCZLfFwSoMWwD6zZLkUOCs0kGvRhSLLRrMh_L-WkQzC0Pe20m0g-Pj0-
    yn5gKgRWs32Mbag570Ou4F5M82paYKugAGf4jY3owVmLEPPrxXdRjsYzMc7niyWXreIJyKuyEvwWofMqHvjXFi1fwic8BD6qypJK13OWMbvZslpMxHQ2
    Gk3G04z1wOoHBh4btFFJfdttftE1KbndK_-xs43GFHkMrvOAq0NLMTsXMhZPMTitqU85Szmi8gdW3xFHp_UPrwnw1F-XBHgKniH-
    JqOi_qRVGjyRSKM08bHPMJdaU2mnerRUYyu5IRXr4zpjG-XjfiFjIqGJy7y8ysWYHY_ZszLE6zJujOms-
    i3PE4UoYxeGOU2rMWJDyV4CKItz1wwM4Bpld4PicAgRzXmvtNa9_lA4v-
    PJUh5Uw6G_JwIYOkntlB3Xx4y159kHOVv0aOHCEn5hIYEd0K4HSBp6pcyJqhJ5WeVlGk-7-KUzG_RUENXoilIt-
    q3zJqVIpoTofLqtUaHVMlk8HzjjmwX2qF1r0nlpe-WdHc7v6NG9J1PXL_la_de-VpO_fL0aVeIf-5oUp4F-og_JJbqkKAuRtJ0-
    trs_v4tgJD16lDruC5C-CW9PQZ4CkvUyDlyvGjF7FaMud0XCHgE.5-pro_sokwANa7s7Ra2yWCKjDtxjDnHnIzk-wY-BLomsTzHCZ-
    eVmMcRN2W6a_Dz0OmIbZy04txtNUmKQ9EScA"
  ]
}
```

The .smart-health-card files downloaded from MyChart web or mobile will contain one or two entries in the verifiableCredential JSON array. Each array entry is a self-contained SMART Health Card.

## Consuming SMART Health Cards

Regardless of how a SMART Health Card is received -- a QR code scan or a file download, the same steps are used to verify and access the information contained within it.

A JWS, including our SMART Health Card JWS, is made up of three, Base64-encoded parts: JOSE header, a payload, and a signature. Per RFC 7515, these three parts are concatenated together with periods.

### JOSE Header

The JOSE (Javascript Object Signing and Encryption) Header identifies the algorithms used to sign and compress the payload and a unique identifier of the public key to be used to verify the signature.

For example, the JOSE header from the above JWS is:

eyJhbGciOiJFUzI1NiIsImtpZCI6ImZoa3ZpMEdWektQdjJpSHR6YUYtWHFicTZQVGFEcVdHSXd3c2RQNnZxT00iLCJ6aXAiOiJERUYifQ

Upon Base64-decoding it (and formatting for readability):

```
{
"alg": "ES256",
"kid": "fhkvi0GVzKPv2iHtzaF-Xqbq6PTaDqWGIwwsdP6vqOM",
"zip": "DEF"
}
```

* alg – the algorithm used to sign the JWS.

* kid – a "key identifier" which is used to find the public key for validating the signature.

* zip – the compression algorithm used to deflate the payload. Note that the JWS standard does not define the zip header; the SMART Health Card uses borrows it from the closely related JWT standard.

In the SMART Health Card standard, the alg header will always be "ES256" and the zip header will always be "DEF". In Epic's implementation of the standard, only the kid header will change, as it uniquely identifies the public key used by the healthcare delivery organization that issued the SMART Health Card. The public key may be rotated periodically.

### Payload

The payload of a SMART Health Card JWS (that is, the second of the three period-delimited pieces in the JWS) is base64-encoded, compressed JSON. The JSON is compressed with the DEFLATE algorithm.

### Inflation

The DEFLATE algorithm is open source and has significant library support. DEFLATE is the same compression algorithm used by gzip but doesn't make use of the gzip header and footer. See RFC 1951 for more information about this compression algorithm.

As an example, starting with this JWS payload from our example above,


```
3VRNb9swDP0rg3bZANuynCZLfFwSoMWwD6zZLkUOCs0kGvRhSLLRrMh_L-WkQzC0Pe20m0g-Pj0-
    yn5gKgRWs32Mbag570Ou4F5M82paYKugAGf4jY3owVmLEPPrxXdRjsYzMc7niyWXreIJyKuyEvwWofMqHvjXFi1fwic8BD6qypJK13OWMbvZslpMxHQ2
    Gk3G04z1wOoHBh4btFFJfdttftE1KbndK_-xs43GFHkMrvOAq0NLMTsXMhZPMTitqU85Szmi8gdW3xFHp_UPrwnw1F-XBHgKniH-
    JqOi_qRVGjyRSKM08bHPMJdaU2mnerRUYyu5IRXr4zpjG-XjfiFjIqGJy7y8ysWYHY_ZszLE6zJujOms-
    i3PE4UoYxeGOU2rMWJDyV4CKItz1wwM4Bpld4PicAgRzXmvtNa9_lA4v-
    PJUh5Uw6G_JwIYOkntlB3Xx4y159kHOVv0aOHCEn5hIYEd0K4HSBp6pcyJqhJ5WeVlGk-7-KUzG_RUENXoilIt-
    q3zJqVIpoTofLqtUaHVMlk8HzjjmwX2qF1r0nlpe-WdHc7v6NG9J1PXL_la_de-
    VpO_fL0aVeIf-5oUp4F-og_JJbqkKAuRtJ0-trs_v4tgJD16lDruC5C-CW9PQZ4CkvUyDlyvGjF7FaMud0XCHgE
```


we Base64 URL-decode and inflate to arrive at the actual JSON content. The JSON of a SMART Health Card is made up of a header and a few FHIR resources.

### SMART Health Card Header

A SMART Health Card's header elements identify if the card contains immunization or lab result information, a unique url for the organization that issued the card and the time at which the card was issued.

* iss – Append /.well-known/jwks.json to this url to access the JSON Wek Key Set containing a public key for verification of the JWS signature. See SMART Health Cards & Trust for additional information.

* nbf – The time at which the SMART Health Card was issued (per RFC 7519). Epic generates SMART Health Cards on-demand, so this timestamp will correspond to the time at which the card was accessed by the patient.

* – Identifies the SMART Health Card as containing either COVID-19 vaccination or lab result. Epic-generated SMART Health Cards will always be one of the following:

type

* ["https://smarthealth.cards#health-card", "https://smarthealth.cards#covid19", "https://smarthealth.cards#immunization"]

* ["https://smarthealth.cards#health-card", "https://smarthealth.cards#covid19", "https://smarthealth.cards#laboratory"]

* credentialSubject – Contains a FHIR Bundle of a Patient resource, either one or two FHIR Immunization resources, or an Observation resource representing a lab result.

This is an example of the SMART Health Card header describing an immunization card, issued on April 20, 2021.

```
{
  "iss": "https://vs-icx18-28.epic.com/Interconnect-HDR1035915-CDE/api/epic/2021/Security/Open/EcKeys/32001/SHC",
  "nbf": 1618933658,
  "type": ["https://smarthealth.cards#health-card", "https://smarthealth.cards#covid19",
    "https://smarthealth.cards#immunization"],
  "vc": {
```

```
  "credentialSubject": {
      //payload removed for readability
    }
  }
}
```

### FHIR Bundle

The credentialSubject element from the SMART Health Card header defines the version of FHIR and contains a Bundle of FHIR resources.

* fhirVersion – Version of FHIR resources contained in this SMART Health Card. Epic generated SMART Cards will always use FHIR R4 resources, represented in this field as “4.0.1”.

* fhirBundle – A Bundle of type collection containing two to three FHIR resources. Each entry in the Bundle is a distinct FHIR resource.

Here's an example of a credentialSubject containing a fhirBundle:

```
{
  "credentialSubject": {
    "fhirVersion": "4.0.1",
    "fhirBundle": {
      "resourceType": "Bundle",
      "type": "collection",
      "entry": [
        { //FHIR resource },
        { //FHIR resource #2 },
        { //another FHIR resource! },
      }
    }
  }
```

### FHIR Patient

SMART Health Cards always contain a FHIR Patient resource. Unlike a typical Patient resource, the only information in the resource is the patient's given and family name, and date of birth.

When consuming a SMART Health Card, the demographics in this Patient resource are verified against the user's identity.

```
{
  "resource": {
    "resourceType": "Patient",
    "name": [{
      "family": "McCall",
      "given": ["Table"]
    }],
    "birthDate": "2000-04-15"
  }
}
```

### Considerations For Patient Identity

Importantly, possession of a SMART Health Card does not prove a patient's identity. In fact, patient caregivers, for example: parents, may access and download the SMART Health Cards of their wards. Pass apps and verifiers must verify identity.

When a patient shares a SMART Health Card with a verifier, the verifier is expected to validate the patient's identity independently. For example, if a patient presents their driver's license along with their SMART Health Card, the verifier should confirm that the demographics within the SMART Health Card (specifically, the name and date of birth), match those on the driver's license.

### FHIR Immunization

A SMART Health Card documenting a vaccination contains one or two FHIR Immunization resources. The below example describes a Pfizer immunizations completed on February 1.

* status – contains the value “completed”

* vaccineCode – contains a CVX code identifying the given vaccine.

* occurrenceDateTime – the date on which the vaccine was administered

* lotNumber – the product lot number, further identifying the vaccine

* performer – contains the name of the organization at which the vaccine was administered. Will not be present if administering organization is unknown.

### Example Immunization

```
{
  "resource": {
    "resourceType": "Immunization",
    "status": "completed",
    "vaccineCode": {
      "coding": [{
        "system": "http://hl7.org/fhir/sid/cvx",
        "code": "208"
      }]
    },
    "patient": {
      "reference": "Patient/resource:0"
```

```
    },
    "occurrenceDateTime": "2021-02-01",
    "lotNumber": "1234",
    "performer": [{
      "actor": {
        "display": "Current Development Environment (CDE)"
      }
    }]
  }
}
```

### FHIR Observation

A SMART Health Card documenting a lab result contains one FHIR Observation resource. The below example describes a negative lab test resulted on November 23, 202.

* status – contains the value "final"

* code – contains a LOINC code identifying the lab test, if known

* effectiveDateTime – the time at which the lab test was resulted

* performer – contains the name of the organization at which the test was performed.

* value – the result of a COVID-19 lab test may be represented as a SNOMED code, a quantity, or a string.

### Example Observation

```
{
  "resource": {
    "resourceType": "Observation",
    "status": "final",
    "code": {
      "coding": [{
        "system": "http://loinc.org",
        "code": "882-1"
      }]
    },
    "subject": {
      "reference": "Patient/resource:0"
    },
    "effectiveDateTime": "2020-11-23T06:00:00Z",
    "performer": [{
      "display": "Health system"
    }],
    "valueString": "Negative"
  }
 }
```

### SMART Health Cards & Trust

A SMART Health Card is verifiable because its signed payload can be verified against a trusted public key. Each issuer publishes a JSON Web Key Set (JWKS) containing at least one public key. Typically, each healthcare organization that uses Epic acts as a distinct issuer.

A verifier must determine which issuers to trust. Issuers participating in the Vaccine Credential Initiative have shared their publicly accessible JSON Web Key Set URLs in this directory:

* https://raw.githubusercontent.com/the-commons-project/vci-directory/main/vci-issuers.json

Verifiers should consider trusting and caching all of the public keys provided in the key sets from this directory.

### Verify the Signed Web Token

An issuer's key set is discoverable during verification by appending "/.well-known/jwks.json" to a SMART Health Card's iss value. For example, iss value in the above SMART Health Cards is an inaccessible, url internal to Epic: https://vs-icx18-28.epic.com/Interconnect-HDR1035915-CDE/api/epic/2021/Security/Open/EcKeys/32001/SHC

```
HTTP Get https://vs-icx18-28.epic.com/Interconnect-HDR1035915-CDE/api/epic/2021/Security/Open/EcKeys/32001/SHC/.well-
    known/jwks.json

{
  "keys": [{
    "x": "gfoeoAb2l-hqEHXyZIBqa1OObZ7Tunx5hlYRXtnPJwg",
    "y": "HcBgiLJ53BFqOp5lxiMUz5Jsik9DtM286pyWv2hVKNA",
    "kid": "fhkvi0GVzKPv2iHtzaF-Xqbq6PTaDqWGIwwsdP6vqOM",
    "use": "sig",
    "kty": "EC",
    "alg": "ES256",
    "crv": "P-256"
  }]
}
```

The key identifier from the JWKS's kid element corresponds to kid element in the header of the SMART Health Card's JWS. Epic issued SMART Health Cards are signed using ECDSA P-256 SHA-256.

Verifiers should use standard security libraries to verify SMART Health Card signatures. Refer to Appendix A.3 of RFC 7515 to understand how Epic's signature is calculated and formatted.

## Open Source Libraries & References

While it is not an exhaustive list, nor maintained or affiliated with Epic, the following resources might be helpful:

* Reference implementation: https://c19.cards/venue and open source code: https://github.com/smart-on-fhir/health-cards-tests

* Microsoft-developed SMART Health Card validation SDK: https://github.com/microsoft/health-cards-validation-SDK

* MITRE-developed SMART Health Cards tutorial Jupyter Notebook: https://github.com/dvci/health-cards-walkthrough/blob/main/SMART%20Health%20Cards.ipynb

### Get Started

These connection practices have enabled many successful connections across our community. As every product has unique goals, we encourage you to review this approach with your customers to confirm the best fit. If you're just getting started, check out our step-by-step Developer Guide. Want a little extra help? Vendor Services offers an optional suite of benefits including the ability to request individualized assistance with your app's data exchange. In addition, some product categories qualify for Toolbox support and designation on Showroom, which can accelerate implementation and enable you to highlight your product's strengths. Explore the list of eligible categories and share your interest with us through the Technology Guidance form.

## 2.2 Remote Patient Monitoring

File: site/docs/02-playbooks/02-rpm-devices-software.html
Source: https://fhir.epic.com/Documentation?docId=rpm_devices_software

### Remote Patient Monitoring

### Overview

In Epic, the designated data structure for discrete data is called a flowsheet. For RPM use cases, we use patient-entered flowsheets (PEFs) to indicate data that was sourced from a patient, either directly from a patient-owned device, or manually entered by the patient. The following document steps through how an RPM program or Home Dialysis program can connect to Epic securely using HL7v2 and FHIR standards.

By filing RPM data to the patient’s chart, you ensure the full context of their at-home care is available for providers, and you enable downstream processes in Epic such as triggering escalations enabling the automatic calculation of RPM billing.

### Workflow Diagram

This diagram illustrates an example workflow using HL7v2 as the primary method to file flowsheet readings.

#### Create an RPM App

Copy Template App ->

One click creates a copy of this app with the context, use case, and all required scopes already selected for an RPM over HL7v2 integration.

### Enrollment

On Epic, PEFs are not ready-to-file by default. There is an enrollment workflow that opens PEF data structures for a specific patient and time period, known as an episode. This enrollment workflow usually involves a provider placing an order for an RPM program in Epic and optionally entering additional data specific to that order (order-specific questions) such as:

* High/Low normal ranges for individual vitals - Note if the RPM device files an abnormal vital an In Basket message can be sent to the provider through MyChart functionality.

* How often (in days) the provider would like to be notified of a patient's flowsheet entries

* The supporting diagnoses

After an RPM program order is placed, a notification can be sent to the RPM system in the form of an HL7v2 Outgoing Ancillary Orders message. for more details, and refer to the example message in the appendix. This notification contains information such as:

* The Patient's identifiers and demographics

* The signing provider's identifiers and demographics

* An identifier for the corresponding order record in Epic

* Optionally, the additional order-specific data above captured by the provider during enrollment

Depending on the customer’s workflow, there may be two Episodes created. This will become relevant in the Ending Enrollment section below.

* One episode is explicitly tied to the flowsheet, which we’ll call the PEF Episode. This should be created for all RPM workflows.

* The optional 2nd episode is the Compass Rose episode, which represents that the patient is being actively monitored by clinicians.

## Filing Vitals (HL7v2)

* This interface is designed to ingest data straight from a device. The data it files is read-only for patients in MyChart. (They cannot manually enter/modify the readings after you file them, but they can delete them.)

If your app does not have a patient-facing UI where patients manually enter/modify readings, or it needs to file multiple vitals readings at once, then Epic recommends using the HL7 Incoming Clinical Documentation Flowsheet Data – Patient Entered interface to file readings.

* PEFs filed via this interface do not require patients to have an active MyChart account, unlike PEFs with Observation.Create discussed below.

* If you are using HL7v2 over HTTP(s), “HL7v2” must be added as an Incoming API on an app with a primary user type of “Backend Systems.” Refer to the Application User Context tutorial for additional background.

You need the patient’s name and an identifier for the interface to file readings to the correct patient. You can use the MRN and other identifiers from the Outgoing Ancillary Orders message, or you can use the Patient.Search API to obtain other identifiers.

* In the sandbox, you can use an ID system of urn:oid:1.2.840.114350.1.13.0.1.7.5.737384.14 to search by MRN with Patient.Search. Refer to the for more information.

Note: this system value will differ in each customer environment.

ID Types for APIs tutorial

* Ex: /api/FHIR/R4/Patient?identifier=urn:oid:1.2.840.114350.1.13.0.1.7.5.737384.14|<Insert MRN>

* If you already have the patient’s FHIR ID, you can instead make a call. The response includes an “identifiers” array that includes the patient’s MRN with that same system: urn:oid:1.2.840.114350.1.13.0.1.7.5.737384.14

Patient.Read

```
{
   "use": "usual",
   "type": {
     "text": "MRN" //Do NOT rely on this saying ‘MRN’ at every customer!
   },
   "system": "urn:oid:1.2.840.114350.1.13.0.1.7.5.737384.14",
   "value": "206919"
 }
```

* Starting in the November 2025 version of Epic, the Incoming Flowsheet Data interfaces supports a flag in OBX-17 to indicate whether the data was manually entered. The actual values you'll send in this field are mapped during customer implementation.

* To allow an interface to write to a given flowsheet row, customers need to perform some ID mapping in Epic. Some flowsheet rows have LOINC codes mapped to them by default, which you can use in your interface message to indicate which flowsheet rows to file to. Work with your customer to determine the IDs mapped to the flowsheet rows you want to use

* Refer to the appendix for a sample interface message.

## Filing Vitals (FHIR Observation.Create)

* If your app has a patient-facing UI such as a website or mobile app that patients need to log into to file their readings, Epic recommends the use of the Observation.Create API on a patient-facing app for filing readings authorized with the patient’s credentials.

* This workflow has a few key requirements:

* Patients must have active MyChart accounts. Epic customers will be responsible for provisioning MyChart accounts for their patients before enrolling them in an RPM program that uses Observation.Create.

* You need to add the Observation.Create API to an app with “Patients” as the primary user type. This enables the API to write to PEFs as opposed to other types of flowsheets. Refer to the Application User Context tutorial for additional background.

* You must authenticate your Observation.Create calls using OAuth 2.0, as described in an upcoming section.

### Account Linking

After the provider has placed the enrollment order and the notification has been sent to your system, the patient opens MyChart and launches your account linking portal through an embedded EHR launch. At the end of this launch sequence, your portal receives the following data:

* An OAuth 2.0 access token (for use with all API calls, such as Observation.Create for filing vitals)

* An OAuth 2.0 refresh token (for obtaining a new access token periodically)

* The patient's FHIR ID (for use with Observation.Create and Patient.Read)

* The response to Patient.Read contains the Patient's MRN, which can be used to match to the MRN received in the HL7 enrollment notification.

* It's important that your portal is properly setup with JWT authentication to receive a refresh token, so that patients do not have to login each time you file a reading.

### Observation.Create (Vitals)

The main inputs to this API are:

* The patient's FHIR ID, as a unique identifier for that Patient in Epic

* Your Epic customer's specific "code.coding.system" value, identifying you are filing to the flowsheet data structure

* The ID of the specific flowsheet row that you are filing to, where each row corresponds to a different type of vital for example, systolic blood pressure or heart rate). This will be used as the "code.coding.code" value in the payload.

* The value of the reading, with an optional unit assigned. The unit must either be omitted or match the understood unit for that vital (listed in the appendix).

* Any relevant metadata that should be stored with the reading such as whether the reading was manually entered. (The Component field)

## Ending Enrollment

* Providers control the RPM Episode(s) from Hyperspace and can end them at any time. After the episode has ended, your application cannot and should not attempt to file more readings to the patient's chart.

The EpisodeOfCare resource is used to signal when the patient's enrollment has been ended by the provider. For each patient enrolled in your program, you can periodically (example: once per day) query to check whether the status has changed from "active" to "finished".

On enrollment, you can use the EpisodeOfCare.Search resource, filtered to the Patient's MyChart Episodes using the following search: EpisodeOfCare?patient={patient}&type=urn:oid:1.2.840.114350.1.13.0.1.7.2.726668|3200

Note the "type" field varies by Epic environment. Refer to the ID Types for APIs tutorial for background.

* If the customer is also using a Compass Rose episode to track active monitoring in addition to the PEF episode, you will also need to check the status of that episode. If either episode is not active, you should stop filing RPM data.

Because this patient might be on multiple RPM programs, you need to identify your specific episodes. You can do this using the "episode-name" extension, which has a unique valueString per RPM program. Work with your customers to determine which Episode names to check for (both PEF and Compass Rose episodes):

```
"extension": [
    {
        "valueString": "Sample RPM Template",
        "url": "http://open.epic.com/FHIR/StructureDefinition/extension/episode-name"
    }
]
```

The episode's FHIR ID is persistent, so you can store them after the initial EpisodeOfCare.Search for use in later queries to EpisodeofCare.Read instead

## Appendix

### FAQs

Q: Will recorded vitals show up in Hyperspace?

A: Yes. Clinicians can manage an Episode and review its associated flowsheets.

Q: Will recorded vitals show up in MyChart?

A: Yes. They show up in MyChart and can be reviewed by patients. If your readings were filed with Observation.Create, patients can modify these values from MyChart. If they were filed with HL7v2, these values cannot be modified but can be deleted by the patient.

Q: Can I file more than one vital reading at once?

A: Not with Observation.Create. Vital readings must be individually posted as their own Observation.Create calls. To make several readings show up with the same timestamp in Epic, send multiple requests where each payload has the same "effectiveDateTime" value. The incoming flowsheet interface can file multiple readings at once.

Q: Can I file both systolic and diastolic blood pressure at the same time?

A: Yes. it’s possible to file systolic and diastolic blood pressure at the same time with HL7v2 if using the right linked flowsheet rows. For Observation.Create, include the Observation.Component field and the appropriate metadata.

Q: Should I file readings in real time?

A: Yes, you should. Your filed readings can trigger messages to providers, so it’s important those are sent as soon as possible.

Q: Should I use the abnormal ranges sent on the HL7 messages?

A: No, you shouldn't, since providers can change those values after the fact, and there is no way to notify you of this.

Q: Should I send in charges for my RPM services?

A: In the November 2023 version of Epic, we released enhanced tools to automatically calculate RPM billing based on PEF/DEF device data and clinical time spent with the patient. You do not need to send in charges yourselves. (IE with a DFT interface).

### Example Outgoing HL7 Ancillary Orders Message

```
MSH|^~\&|EPIC||||20210806112409|ADSTRICK|ORM^O01|7|T|2.3
PID|1||E4035^^^EPIC^EPI~203739^^^MRN^MRN||MYCHART^PEG^^^^^D||19500101|M|||^^^^^US^L|||||||38454|000-00-0000||||||||||N
PV1||OUTPATIENT|10501101^^^10501^^^^^^^DEPID||||1000^FAMILY MEDICINE^PHYSICIAN^^^^^^PROVID^^^^PROVID~1627363736^FAMILY
    MEDICINE^PHYSICIAN^^^^^^NPI^^^^NPI||||||||||||38454|||||||||||||||||||||||||||||38454
ORC|NW|1078751^EPC||38454|||^^^20210806^^R^^||20210806112404|ADSTRICK^STRICKLAND^ADAM^^||1000^FAMILY
    MEDICINE^PHYSICIAN^^^^^^PROVID^^^^PROVID~1627363736^FAMILY
    MEDICINE^PHYSICIAN^^^^^^NPI^^^^NPI|10501101^^^10501^^^^^EMC FAMILY
    MEDICINE|(555)555-5555^^^^^555^5555555||||CLISUP^EPIC SUPPORT^^10101102^EMH MED SURG||||||123 Anywhere
    Street^^MADISON^WI^53719^^C|||||O
OBR|1|1078751^EPC||^SAMPLE RPM PROGRAM ENROLLMENT^^^SAMPLE RPM ENROLLMENT||20210806|||||Clinic|||||1000^FAMILY
    MEDICINE^PHYSICIAN^^^^^^PROVID^^^^PROVID~1627363736^FAMILY MEDICINE^PHYSICIAN^^^^^^NPI^^^^NPI|(555)555-
    5555^^^^^555^5555555|||||||Procedures|||^^^20210806^^R^^|||||||||20210806
NTE|1||This flowsheet is filled with data from your Sample RPM device. Please do
NTE|2||not manually enter vitals here.
NTE|3||After how many days would you like to receive a notification of this patient's flowsheet entries?->7
NTE|4||What is the highest normal systolic value for this patient?->180
NTE|5||What is the lowest normal systolic value for this patient?->100
NTE|6||What is the highest normal diastolic value for this patient?->100
NTE|7||What is the highest normal weight in lbs for this patient?->180
DG1|1|I10|I50.22^Chronic systolic (congestive) heart failure^I10|Chronic systolic (congestive) heart failure||^180;ORD
DG1|2|I10|E08.22^Diabetes mellitus due to underlying condition with diabetic chronic kidney disease^I10|Diabetes
    mellitus due to underlying condition with diabetic chronic kidney disease||^180;ORD
DG1|3|I10|N18.6^End stage renal disease^I10|End stage renal disease||^180;ORD
DG1|4|I10|Z79.4^Long term (current) use of insulin^I10|Long term (current) use of insulin||^180;ORD
DG1|5|I10|Z99.2^Dependence on renal dialysis^I10|Dependence on renal dialysis||^180;ORD
```

### Example Observation.Create Payload

```
{
  "resourceType": "Observation",
  "category": [
    {
      "coding": [
        {
          "system": "http://hl7.org/fhir/observation-category",
          "code": "vital-signs",
          "display": "Vital Signs"
        }
      ],
      "text": "Vital Signs"
    }
  ],
  "code": {
    "coding": [
      {
        "system": "urn:oid:1.2.840.114350.1.13.0.1.7.2.707679",
        "code": "8",
        "display": "Heart Rate"
      }
    ]
  },
  "subject": {
    "reference": "Patient/e63wRTbPfr1p8UW81d8Seiw3"
  },
 "component": [
        { "code": {
                "coding": [
                    {
                        "system": "urn:oid:1.2.840.114350.1.13.861.1.7.4.708368.1310",
                        "code": "6",
                        "display": "Manually Entered"
                    }
                ]
            },
            "valueBoolean": true
        }
    ],

  "valueQuantity": {
    "value": 75
  },
  "effectiveDateTime": "2025-03-18T21:59:55",
  "note":[{"text":"Sample RPM device"}]
}
```

### Example HL7 Incoming Clinical Documentation Flowsheet Data message

```
MSH|^~\&|||AM_RPM_POSTMAN|EPIC|20230208134707||ORU^R01|e5649b0a-59ab-4b5a-aefa-f63c2c744db7|T|2.3
PID|||<PATIENT MRN>^^^^MRN||<PATIENT LAST NAME>^<PATIENT FIRST NAME>||19970501|F|||||||||||||||||||||
OBR|||||||20230208134707|||<VENDOR-SPECIFIC IDENTIFIER>|
OBX|1||8480-6||140||
OBX|2||8462-4||90||
OBX|3||8867-4||100||
```

# 2.3 International Patient Summary Playbook

File: site/docs/02-playbooks/03-internationalpatientsummary.html

Source: https://fhir.epic.com/Documentation?docId=internationalpatientsummary

## About the International Patient Summary

The International Patient Summary (IPS) is a FHIR document that contains an essential set of healthcare information for a single patient, making it easy for healthcare organizations to exchange clinical data for patient care across borders and jurisdictions.

Organizations that use Epic can generate FHIR-based IPS documents for treatment-based exchange between providers as well as make those documents available for patients. Consider the following example use case:

Ontario native, 73-year-old Gwendolyn visited a Michigan emergency department during a beach vacation to Lake Erie with her grandchildren. The emergency physician was able to view her complete medical history, previous diagnoses, problem list, medications, allergies, and other crucial information from her regular physician – leading to faster diagnosis, reduced duplicate testing, and better-coordinated care for patients.

Epic is among the first organizations to support the International Patient Summary (IPS) FHIR specification, beginning in May 2025. IPS improves care for patients by making it simpler for healthcare organizations to exchange data securely with partner organizations down the street, provincially, or even across international borders. Support for IPS extends Epic's decades-long track record as a leader in universal interoperability.

IPS builds on Epic's existing Care Everywhere and CDA workflows to help customers coordinate care with groups that exclusively use FHIR. The new specification governs the creation of electronic health record FHIR documents containing essential healthcare information about a patient. It is endorsed by multinational organizations such as the G7 and the GDHP.

## IPS Data Types and Roadmap

Epic’s initial release includes all required IPS data types – problems, allergies, and medications – as well as immunizations. We plan to add results, procedures, and other optional data types over time. Organizations that use Epic can generate IPS documents in response to treatment-based requests from providers and make those documents available for patients.

Epic’s IPS contains both discrete, machine-readable data about the patient, and human-readable narrative text. Systems or scenarios that don’t support the standard discrete FHIR data should present the narrative contained within the IPS.

## Exchange Pattern

For system-to-system integrations, such as provider-to-provider exchange of the IPS for treatment, the IPS $summary operation is well suited for query-based document exchange at the point of care. Epic recommends the use of backend OAuth 2.0 for authorization and authentication of clients querying for the IPS as well as the FHIR $match operation for patient lookup.

## Technical Specification

In Epic, the IPS is generated at the time of each request to ensure that integrations receive a real-time snapshot of the patient's clinical record. Developers who integrate with Epic organizations can request IPS documents.

### Structure of an IPS

The International Patient Summary is available starting in the May 2025 version of Epic. The IPS is returned as a FHIR Bundle that always contains at minimum the following resources:

* A FHIR Composition resource that summarizes all other data in the IPS Bundle

* A FHIR Patient resource, representing the patient who is the subject of this IPS document

* A FHIR Organization resource, representing the healthcare organization that provided the data in the IPS document

Additionally, the IPS FHIR Bundle contains FHIR resources for each discrete piece of healthcare data referenced in the overall Composition – including the patient's problem list, allergies and intolerances, immunizations, and medications. These are returned as FHIR Condition, AllergyIntolerance, Immunization, MedicationRequest, and Medication resources.

### Narrative Content

The FHIR IPS document contains narrative content in XHTML format, which is intended to make it easy for integrations to show patient data to end users without needing to do extra work parsing the FHIR resources in the response. The narrative XHTML is unstyled, so depending on the type of integration, developers should consider supplying a stylesheet to supplement the IPS narrative. Narrative content is unique to the IPS and is found in several locations in the $summary response Bundle, including each of the following:

Element | Section | Description
Patient.text | | Patient narrative, including name and demographics, such as address, phone, email, preferred languages, primary care providers, and contacts
Composition.text | | IPS document overview, including details about the organization that generated this document and the time it was generated
Composition.section.text | Problems | A tabular list of the patient's active and recent problems, along with relevant dates, notes, and priorities
Composition.section.text | Allergies | A tabular list of the patient's active and recent allergies, along with relevant dates, criticality, and reaction information
Composition.section.text | Medications | A tabular list of the patient's medications, including the sig, start and end dates, status information, and data about refill and dispense quantities
Composition.section.text | Immunizations | A tabular list of the patient's immunizations, administration dates, and next due dates

## Sample Request and Response

To retrieve a patient's IPS document, you first need the patient's FHIR ID. Then, make a GET request to the Patient.$summary endpoint using the patient FHIR ID in the following format:

```
https://hostname/instance/api/FHIR/R4/Patient/<PATIENT_FHIR_ID>/$summary
```

A sample request and response for an example patient with all of the core IPS data types (problems, allergies, medications, and immunizations) is shown below:

### Request:

```
https://hostname/instance/api/FHIR/R4/Patient/eNO3wqOfAltfnWMfWBQ1WmQ3/$summary
```

### Response:


```
{
    "resourceType": "Bundle",
    "identifier": {
        "system": "urn:ietf:rfc:3986",
        "value": "urn:uuid:9b0990f1-5de7-423f-a215-8b65ac69e37d"
    },
    "type": "document",
    "timestamp": "2025-02-19T21:04:03Z",
    "entry": [
        {
            "fullUrl": "https://vs-icx18-25.epic.com/Interconnect-FHIR-Current-Stage1-Primary-
    Unsecured/api/FHIR/R4/Composition/feE1F.6OLpos2ZZAXV5UHxg4",
            "resource": {
                "resourceType": "Composition",
                "id": "feE1F.6OLpos2ZZAXV5UHxg4",
                "text": {
                    "status": "generated",
                    "div": "<div xmlns=\"http://www.w3.org/1999/xhtml\"><div class=\"Section\"
    idref=\"eBBFDl1F2A4AHQpheZzVucw3\"><div class=\"Header\"><h3>Author</h3></div><div class=\"Body\"><span>Epic Central
     Clinic<br />608-163-0003<br />410 HIGH POINT ROAD<br />MADISON WI 53711</span></div></div><br /><div
    class=\"Section\"><div class=\"Header\"><h3>Document Information</h3></div><div class=\"Body\"><div
    class=\"SingleLineField\"><span class=\"Prefix\"><strong>UTC offset: </strong></span><span
    class=\"Value\">UTC-06:00</span></div><div class=\"SingleLineField\"><span class=\"Prefix\"><strong>Date format:
    </strong></span><span class=\"Value\">M/d/yyyy</span></div><div class=\"SingleLineField\"><span
    class=\"Prefix\"><strong>Decimal separator: </strong></span><span class=\"Value\">.</span></div><div
    class=\"SingleLineField\"><span class=\"Prefix\"><strong>Date generated: </strong></span><span
    class=\"Value\">2/19/2025</span></div></div></div></div>"
                },
                "status": "final",
                "type": {
                    "coding": [
                        {
                            "system": "http://loinc.org",
                            "code": "60591-5",
                            "display": "Patient summary Document"
                        }
                    ],
                    "text": "International Patient Summary"
                },
                "subject": {
                    "reference": "Patient/eNO3wqOfAltfnWMfWBQ1WmQ3",
                    "display": "Jackson, Jayden"
                },
                "date": "2025-02-19T21:04:03Z",
                "author": [
                    {
                        "reference": "Organization/eBBFDl1F2A4AHQpheZzVucw3",
                        "display": "Epic Central Clinic"
                    }
                ],
                "title": "International Patient Summary for Jackson, Jayden",
                "custodian": {
                    "reference": "Organization/eBBFDl1F2A4AHQpheZzVucw3",
                    "display": "Epic Central Clinic"
                },
                "section": [
                    {
                        "title": "Problems",
                        "code": {
                            "coding": [
                                {
                                    "system": "http://loinc.org",
                                    "code": "11450-4",
                                    "display": "Problem list - Reported"
                                }
                            ],
                            "text": "Problem list - Reported"
                        },
                        "text": {
                            "status": "generated",
                            "div": "<div xmlns=\"http://www.w3.org/1999/xhtml\"><div class=\"ActiveProblems\"><h3>Active
     Problems</h3><table class=\"ActiveProblemTable\"><thead><tr><th>Problem</th><th>Priority</th><th>Noted
    Date</th><th>Diagnosed Date</th></tr></thead><tbody><tr><td class=\"Name\"><span class=\"ProblemName\">Moderate
    persistent asthma (HHS/HCC) (HHS)</span></td><td class=\"Priority\">-</td><td class=\"NotedDate\">2/17/2025</td><td
    class=\"DiagnosedDate\">-</td></tr></tbody></table></div><br /><div class=\"ResolvedProblems\"><h3>Resolved
    Problems</h3><table class=\"ResolvedProblemTable\"><thead><tr><th>Problem</th><th>Priority</th><th>Noted
```

```
Date</th><th>Diagnosed Date</th><th>Resolved Date</th></tr></thead><tbody><tr><td class=\"Name\"><span
class=\"ProblemName\">Struck by alligator</span><ul><li class=\"Note\"><span class=\"NoteTitle\">Overview (2/18/2018
 1:22 PM):</span><br /><span class=\"WarningMsg\"><em>Formatting of this note might be different from the
original.</em></span><br /><span class=\"NoteText\">It was a real big gator.<br /></span></li></ul></td><td
class=\"Priority\">High</td><td class=\"NotedDate\">2/18/2018</td><td class=\"DiagnosedDate\">2/18/2018</td><td
class=\"ResolvedDate\">2/18/2018</td></tr></tbody></table></div></div>"
                    },
                    "entry": [
                        {
                            "reference": "Condition/e-8nWH.bGEpZxDWq2oCoCLREUwM32POPS7KMw9NU04mg3"
                        },
                        {
                            "reference": "Condition/evRt9rC4iF4hQ8HrVrp6408lZdE-j24dngrOAo3aMboY3"
                        }
                    ]
                },
                {
                    "title": "Allergies",
                    "code": {
                        "coding": [
                            {
                                "system": "http://loinc.org",
                                "code": "48765-2",
                                "display": "Allergies and adverse reactions Document"
                            }
                        ],
                        "text": "Allergies and adverse reactions Document"
                    },
                    "text": {
                        "status": "generated",
                        "div": "<div xmlns=\"http://www.w3.org/1999/xhtml\"><table
class=\"ActiveAllergyTable\"><thead class=\"ActiveAllergyHeader\"><tr><th>Active Allergy</th><th>Allergen
Type</th><th>Reactions</th><th>Criticality</th><th>Noted Date</th><th>Comments</th></tr></thead><tbody
class=\"ActiveAllergy\"><tr class=\"ActiveAllergy\"><td class=\"Name\">NICKEL</td><td class=\"Type\">Allergy</td><td
 class=\"Reactions\">Hives</td><td class=\"Criticality\">High</td><td class=\"NotedDate\">9/13/2024</td><td
class=\"Comments\">-</td></tr><tr class=\"ActiveAllergy\"><td class=\"Name\">LACTOSE</td><td
class=\"Type\">Allergy</td><td class=\"Reactions\">Nausea</td><td class=\"Criticality\">Low</td><td
class=\"NotedDate\">1/24/2025</td><td class=\"Comments\">-</td></tr></tbody></table></div>"
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                        },
                        {
                            "reference": "AllergyIntolerance/ew9Q0B3JJOwNipaZ.Pb2J5sXukodDH.N271w777jlVLU3"
                        }
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                            }
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Quantity</th><th>Refills</th><th>Start Date</th><th>End Date</th><th>Status</th><tr><td>aspirin-acetaminophen-
caffeine (EXCEDRIN MIGRAINE) 250-250-65 MG per tablet<ul></ul></td><td>For headaches, take 1 pill by mouth every 6
hours as needed.</td><td>30 tablet</td><td>0</td><td>9/17/2024</td><td>-</td><td>On
Hold</td></tr></tbody></table></div>"
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Due</th></tr></thead><tbody><tr><td>Influenza Virus
Vaccine</td><td>1/7/2025</td><td>-</td></tr></tbody></table></div>"
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                        }
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            ]
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        "resource": {
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            "clinicalStatus": {
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                        "version": "4.0.0",
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            },
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                        "code": "confirmed",
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                        "code": "696BNE976J"
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                        "code": "J7B7WZ5ZMT"
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                        "code": "NUZ5T2QN2V"
                    }
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                "display": "Jackson, Jayden"
            },
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                                }
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                }
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                        "display": "Confirmed"
                    }
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                        "code": "6211",
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                        "code": "1305527"
                    },
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                        "system": "http://www.nlm.nih.gov/research/umls/rxnorm",
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                    },
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                        "code": "EWQ57Q8I5X"
                    },
                    {
                        "system": "urn:oid:2.16.840.1.113883.4.9",
                        "code": "J2B2A4N98G"
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                    }
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        "resource": {
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                }
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                        "code": "2000-1234-01"
                    }
                ],
                "text": "Influenza Virus Vaccine"
            },
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                "display": "Jackson, Jayden"
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                        "display": "Left arm"
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                        "code": "2",
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                "text": "Intramuscular"
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                    "display": "Emile Underwood, MD"
                }
            }
        ]
    }
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    "resource": {
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        "id": "eZ.jcFkxpptG67qxgRtMnjY5sQsmiFdGFbRTrWxSCDtU3",
        "identifier": [
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            }
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        "intent": "order",
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                        "display": "Community"
                    }
                ],
                "text": "Community"
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            "display": "aspirin-acetaminophen-caffeine (EXCEDRIN MIGRAINE) 250-250-65 MG per tablet"
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            "display": "Jackson, Jayden"
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            "identifier": {
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            "display": "Emile Underwood, MD"
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                    "display": "Short course (acute) therapy"
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            "text": "Short course (acute) therapy"
        },
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                "patientInstruction": "For headaches, take 1 pill by mouth every 6 hours as needed.",
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                        "frequency": 1,
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                "code": "15",
                "display": "Oral"
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        "text": "Oral"
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                "code": "419652001",
                "display": "Take"
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        "text": "Take"
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                "text": "calculated"
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                "text": "ordered"
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                    "unit": "tablet"
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                "expectedSupplyDuration": {
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                    "system": "http://unitsofmeasure.org",
                    "code": "d"
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                            "display": "Problem List Item"
                        }
                    ],
                    "text": "Problem List Item"
                }
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                        "display": "Severe"
                    }
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                "text": "High"
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                        "code": "W58.02XA",
                        "display": "Struck by alligator, initial encounter"
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                    {
                        "system": "http://snomed.info/sct",
                        "code": "418589001",
                        "display": "Accidental Physical Contact With Animal"
                    }
                ],
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                "text": "Struck by alligator"
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                "reference": "Patient/eng1-x9wfpFRTWZ1fXHxlhg3",
                "display": "Realistic, Jayden FHIR Jr."
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            "abatementDateTime": "2025-02-17",
            "recordedDate": "2018-02-18",
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                                        "code": "68608-9",
                                        "display": "Summary note"
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                                "text": "Problem Overview"
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                            "url": "http://hl7.org/fhir/StructureDefinition/annotationType"
                        }
                    ],
                    "authorReference": {
                        "reference": "Practitioner/eTgWia0U3VHmeYIDAJ2.kFw3",
                        "display": "Justin Ramm"
                    },
                    "time": "2018-02-18T19:22:47Z",
                    "text": "It was a real big gator.\r\n"
                }
            ]
        }
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                            "display": "Problem List Item"
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                    "text": "Problem List Item"
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                        "system": "urn:oid:2.16.840.1.113883.6.96",
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                        "display": "Moderate"
                    }
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                "text": "Med"
            },
            "code": {
                "coding": [
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                        "system": "http://hl7.org/fhir/sid/icd-10-cm",
                        "code": "J45.40",
                        "display": "Moderate persistent asthma, uncomplicated"
                    },
                    {
                        "system": "http://snomed.info/sct",
                        "code": "427295004",
                        "display": "Moderate Persistent Asthma"
                    }
                ],
                "text": "Moderate persistent asthma (HHS/HCC) (HHS)"
            },
            "subject": {
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                "display": "Realistic, Jayden FHIR Jr."
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            "recordedDate": "2025-02-17"
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    },
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cation/eFPOEegDKIjYsM3RkHequHJ5I2sLnb6kCobyvt.2bHUQMn5mMSA84ght6QXRPlsUrYrpRQOlGll1yESbZhar2gGHGBpCW6ZAZfG3I-
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                                "code": "308297"
                            },
                            {
                                "system": "http://www.nlm.nih.gov/research/umls/rxnorm",
                                "code": "466584"
                            }
                        ],
                        "text": "aspirin-acetaminophen-caffeine (EXCEDRIN MIGRAINE) 250-250-65 MG per tablet"
                    }
                }
            ]
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                    ],
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                }
            ]
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Jackson</span></div><div class=\"SingleLineField\"><span class=\"Prefix\"><strong>Preferred name:</strong></span>
<span class=\"Value\">Not on file</span></div><div class=\"SingleLineField\"><span
class=\"Prefix\"><strong>Age:</strong></span> <span class=\"Value\">33 y.o.</span></div><div
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class=\"Value\">Female</span></div><div class=\"SingleLineField\"><span class=\"Prefix\"><strong>Gender
identity:</strong></span> <span class=\"Value\">Non-binary</span></div><div class=\"SingleLineField\"><span
class=\"Prefix\"><strong>Birth date:</strong></span> <span class=\"Value\">8/23/1991</span></div></div><br /><div
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file</p></div></div><br /><div class=\"Section\"><div class=\"Header\"><h3>Addresses</h3></div><div
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class=\"Section\"><div class=\"Header\"><h3>Communication</h3></div><div class=\"Body\"><ol
class=\"unlist\"><li>fhir@email.com (Email)</li><li>+1 608-777-7777 (Work Phone)</li></ol></div></div><br /><div
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class=\"Header\"><h2>Contacts</h2></div><div class=\"Body\"><table><thead><tr><th>Contact
Name</th><th>Communication</th><th>Relationship to Patient</th></tr></thead><tbody><tr><td
class=\"ContactName\">Smith,Sally</td><td class=\"ContactTelecom\"><div>+1 608-222-2392 (Mobile Phone)</div></td><td
 class=\"ContactRelationship\">Spouse, Emergency Contact</td></tr><tr><td class=\"ContactName\">Brown,Ryan</td><td
class=\"ContactTelecom\"><div>+1 608-777-7777 (Mobile Phone)</div></td><td class=\"ContactRelationship\">Care giver,
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                            }
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                            "url": "text"
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                            "url": "ombCategory"
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                        {
                            "valueString": "Not Hispanic or Latino",
                            "url": "text"
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                            }
                        ]
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                                    }
                                ]
                            },
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                    "value": "YC82BDQ3VDJK6XX"
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        },
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            ]
        }
    ],
```

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        "rank": 1
    }
],
"gender": "female",
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"deceasedBoolean": false,
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        "district": "DANE",
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        }
    ],
    "text": "Married"
},
"contact": [
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                        "display": "Spouse"
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                ],
                "text": "Spouse"
            },
            {
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                    }
                ],
                "text": "Emergency Contact"
            }
        ],
        "name": {
            "use": "usual",
            "text": "Smith,Sally"
        },
        "telecom": [
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                "value": "+1 608-222-2392",
                "use": "mobile",
                "rank": 1
            }
        ]
    },
    {
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                "coding": [
                    {
```

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                        }
                    ],
                    "text": "Care giver"
                },
                {
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                            "display": "guardian"
                        }
                    ],
                    "text": "Legal Guardian"
                }
            ],
            "name": {
                "use": "usual",
                "text": "Brown,Ryan"
            },
            "telecom": [
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                    "value": "+1 608-777-7777",
                    "use": "mobile",
                    "rank": 1
                }
            ],
            "period": {
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            }
        },
        {
            "relationship": [
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                            "code": "E",
                            "display": "Employer"
                        }
                    ]
                }
            ],
            "telecom": [
                {
                    "system": "phone",
                    "value": "608-271-9000",
                    "use": "work"
                }
            ],
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                "text": "TOKAY BLVD\r\nMADISON WI 53719\r\nUnited States of America",
                "line": [
                    "TOKAY BLVD"
                ],
                "city": "MADISON",
                "district": "DANE",
                "state": "WI",
                "postalCode": "53719",
                "country": "USA"
            },
            "organization": {
                "display": "Epic Systems Corp"
            },
            "period": {
                "start": "2014-07-07"
            }
        }
    ],
    "communication": [
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            "extension": [
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                                "code": "ESP",
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                            },
```

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                    }
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                            "code": "en",
                            "display": "English"
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                    "text": "English"
                },
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                                "code": "verbal",
                                "display": "verbal"
                            },
                            "url": "http://hl7.org/fhir/StructureDefinition/patient-preferenceType"
                        }
                    ]
                }
            }
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                "type": "Practitioner",
                "display": "Emile Underwood, MD"
            }
        ],
        "managingOrganization": {
            "reference": "Organization/ePioZEeqCYv.F5W96swOaXA3",
            "display": "EDI Healthcare Management #%~&"
        },
        "_birthDate": {
            "extension": [
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                    "valueDateTime": "1991-08-23T23:23:00Z",
                    "url": "http://hl7.org/fhir/StructureDefinition/patient-birthTime"
                }
            ]
        }
    }
  ]
}
```


# 2.4 Wound Care Apps

File: site/docs/02-playbooks/04-woundcare.html

Source: https://fhir.epic.com/Documentation?docId=woundcare

Connecting wound care apps typically involve the following steps:

* Search for or create a wound in Epic.

* File a wound image in Epic.

* File wound data in Epic flowsheet rows.

## Search for or create a wound in Epic

An app can file data for an existing wound or a new wound. Either way, the app should search the patient's existing wounds to ensure it is not duplicating an existing wound. You can use the following web service to complete this search.

## Observation.Search (LDA-W)

To search for a patient's existing wounds, you need their Patient FHIR ID. You can get this ID either through a launch token during a SMART on FHIR launch (if your app is launched from Epic with patient context) or a search for the Patient FHIR ID using patient demographics.

When you have the Patient FHIR ID, you can call Observation.Search (LDA-W) to retrieve the patient's wounds. Each wound has an Observation FHIR ID, which can be used to file data to that wound.

## Observation.Create (LDA-W)

If the patient chart does not yet include the wound for which the app is storing data, you need to create it. To do this, you need to understand the difference between wound properties and wound assessments.

### Properties vs. Assessments

In Epic, wounds are documented in property flowsheet rows and assessment flowsheet rows. Properties are static data points, such as the wound type and location. These are used to identify the wound and are documented once per wound. Assessments are the data points that change over time and are documented many times while the patient has the wound, such as wound images and wound length, width, and depth.

You need to call Observation.Create (LDA-W) once to create the wound and document the wound properties before you can file the assessments to Epic. When you make an Observation.Create (LDA-W) request to create a new wound, the wound's Observation FHIR ID is returned. The Observation FHIR ID can then be used to file assessment data to that wound.

### Flowsheet Row IDs

All flowsheet rows, whether they are properties or assessments, have a flowsheet row ID. This ID is what you use in your Observation.Create (LDA-W) request to indicate which flowsheet row you are filing data to. When working in a customer environment, you should work with the customer to get the necessary flowsheet row names and IDs. If a flowsheet row doesn't already exist for the data point you need, a custom row can be created.

Below you can find the wound care flowsheet rows currently available in the sandbox. Note that these will likely not be the flowsheet rows or IDs used by your customers. If you need to test writing to a Category Type or Custom List flowsheet row, open a Sherlock ticket to request the flowsheet response options.

### Property Flowsheet Rows

Flowsheet Row | Flowsheet ID | Value Type
PLACEMENT DATE | 700 | Date
PLACEMENT TIME | 701 | Time
R WOUND PRESENT ON ORIGINAL ADMISSION | 1570010003 | Category Type
R WOUND APPROX AGE AT FIRST ASSESSMENT (WEEKS) | 1570010004 | Numeric Type
R HAND HYGIENE COMPLETED | 3040109763 | Custom List
WOUND PRIMARY WOUND TYPE | 1570010000 | Category Type
R WOUND LOCATION | 396060 | Custom List
R WOUND LOCATION ORIENTATION | 303740 | Custom List
R WOUND DESCRIPTION | 400633 | String Type
R NUMBER OF SUTURES PLACED | 3040109153 | Numeric Type
R NUMBER OF STAPLES PLACED | 3040109154 | Numeric Type
REMOVAL DATE | 702 | Date
REMOVAL TIME | 703 | Time
R WOUND OUTCOME | 1570010001 | Category Type

### Assessment Flowsheet Rows

Flowsheet Row | Flowsheet ID | Value Type
WOUND IMAGE | 705 | Image
R WOUND SITE ASSESSMENT | 303750 | Custom List
R PERI-WOUND ASSESSMENT | 7061190 | Custom List
R WOUND SHAPE | 7061200 | String Type
R WOUND LENGTH (CM) | 1570010005 | Numeric Type
R WOUND WIDTH (CM) | 1570010006 | Numeric Type
R WOUND DEPTH (CM) | 1570010007 | Numeric Type
R WOUND VOLUME | 1570010009 | Numeric Type
R WOUND HEALING AS % | 3040105062 | Numeric Type
R WOUND SITE CLOSURE | 303760 | Custom List
R WOUND DRAINAGE AMOUNT | 303770 | Custom List
R WOUND DRAINAGE COLOR | 303780 | Custom List
R NON-STAGED WOUND DESCRIPTION | 7061220 | Custom List
R WOUND TREATMENTS | 3040102846 | Custom List
R WOUND DRESSING | 303800 | Custom List
R WOUND DRESSING CHANGED | 303820 | Custom List
R DRESSING STATUS | 304850 | Custom List
R ADHESIVE CLOSURE STRIPS | 3040109151 | Custom List
R NUMBER OF ADHESIVE CLOSURE STRIPS | 3040109152 | Numeric Type
R SUTURES REMOVED INTACT | 3040109148 | Custom List
R NUMBER OF SUTURES REMOVED | 3040109149 | Numeric Type
R STAPLES REMOVED INTACT | 3040109178 | Custom List
R NUMBER OF STAPLES REMOVED | 3040109150 | Numeric Type
R WOUND TUNNELING | 7061250 | Numeric Type
R WOUND UNDERMINING | 7061260 | Numeric Type
R WOUND MARGINS | 7061270 | Custom List
R $ WOUND CARE CHARGES | 3040105063 | Custom List

## File a wound image in Epic

If your app takes photos of wounds, users likely want those photos available within their normal workflow in Epic. To send a wound image to Epic, you need to use the Incoming Scanned Document Link HL7v2 interface. This interface files the image

to a patient encounter so that it appears in the patient chart but does not file the image to the wound image flowsheet row. The image can be linked to the flowsheet row using Observation.Create (LDA-W). This step is described in the next section.

### File wound data in Epic flowsheet rows

Wound images and other wound assessment data can be filed to the Flowsheet activity using Observation.Create (LDA-W). In order to file assessment data, you need:

* The Observation FHIR ID of the wound being documented on, which is obtained either through a search or as a result of creating a new wound.

* The flowsheet row ID of the flowsheet row that you are writing data to. You'll get this ID from the customer during the app install.

* The value type (string type, numeric type, custom list, etc.) of the flowsheet rows to which you are writing data. If you are writing to a custom list flowsheet row, you also need to know the custom list values. You can get these from the customer during the app install.

One Observation.Create request can file one assessment to Epic. For example, if your app is documenting the wound image, wound width, and wound length, you need to make three separate API calls.

### Get Started

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## 2.5 JWT Authentication Troubleshooting

File: site/docs/02-playbooks/05-jwtauthtroubleshooting.html

Source: https://fhir.epic.com/Documentation?docId=jwtauthtroubleshooting

When developing Backend OAuth 2.0 applications, the issue developers most often run into is a HTTP 400 status code with an invalid_client error, which looks like this:

```
{
    "error": "invalid_client",
    "error_description": null
}
```

This error can be caused by a few different underlying issues. A few of the most common causes are listed below.

### Issues with the App Listing

* Ensure the backend systems consumer type and OAuth 2.0 boxes are checked on the app listing.

* In Epic on FHIR, the OAuth 2.0 box is always checked.

* Ensure you have uploaded a public key to your environment.

* If creating a user/patient facing app that uses JWTs, make sure that "Require Refresh Tokens" is checked, so that you can then upload your public key.

### Issues with the Request Body

* Ensure in the body of the POST, "client_assertion" is used and NOT "client-assertion".

* Ensure you aren't double-encoding the "client_assertion_type" parameter's value of "urn:ietf:params:oauth:client-assertion-type:jwt-bearer".

### Issues with the Token Request Formatting

* Ensure the request URL ends in /token and NOT /authorize.

* Ensure HTTP POST verb is used, NOT GET.

* Ensure the Content-Type of "application/x-www-form-urlencoded" is used.

* Ensure the content is in the body of the request, NOT the URL.

### Issues with the JWT

* Ensure the iat and nbf claims are NOT in the future.

* Ensure the exp claim is in the future.

* Ensure the exp is no more than 5 minutes in the future.

* Ensure the exp is not more than 5 minutes after the iat and nbf claims.

* Ensure the app's client ID is in both the iss and sub claims.

* Ensure the correct client ID is used. For the Sandbox, non-prod client ID should be used.

* Ensure the jti is no longer than 151 characters and contains a unique ID.

* Ensure the JWT was not expired at the time of use.

### Issues with the Signature

* Ensure the private key used to sign the JWT corresponds to the public key registered with the sandbox or Epic customer.

* If you re-upload your public key, you need to wait for the changes to sync. This might take up to 60 minutes to sync with the Sandbox, and 12 hours to sync at an Epic organization's Epic instance.

* Note the “Sandbox Public Key” on your app is for use against the Epic sandbox. When you are ready to integrate with an Epic customer, you will be asked to upload non-PRD and PRD public keys for each customer.

* If your app is cloud-hosted, it is OK to re-use public keys across customers. This assumes you can protect a single copy of the key and are using a different key for non-PRD and PRD.

* Ensure the signature algorithm indicated is the algorithm your code cryptographically signs the JWT with.

* Ensure the signature algorithm used is supported. RS256 and RS384 are RS384 is preferred.

If you are unable to identify the issue with your request and would like additional assistance, you can request optional technical support from Epic through Vendor Services at an hourly billable rate.

## 2.6 Payer Connections to Epic Provider Systems

File: site/docs/02-playbooks/06-payerintegration.html

Source: https://fhir.epic.com/Documentation?docId=payerintegration

Payers may want to connect to Epic provider systems to gain insights about their patients. A variety of use cases are supported and detailed below as well as general guidelines on how to limit the information received to just your patient population.

### Roster Management

Providers can import identifiers from your systems to patients in their Epic instance which can subsequently be used to accurately filter your data requests to just patients in your population. Flat files of patient demographics with one line per patient is the expected format.

Format

```
ID|Last|First|Middle|Suffix|Sex|Gender Identity|Sexual Orientation|Sex Assigned at
    Birth|DOB|SSN|Address1|Address2|City|State|Zip|County|Country Phone1|Phone2|Phone3|Last Update Date|Status|Date of
    Death|Aliases|Email|Maiden|Marital Status|Race|Religion|Language
```

Example

```
80347521|Roberts|Allison|Marie|Jr.|F|F|Straight|F|06/11/1956|111-11-1111|113 Park St.|#102|Madison|WI|53709|Dane|United
    States of America|608-271-9000|608-472-1900|608-888-9090|10/24/2011|Deceased|10/24/2011|Smith,Allison
    M^Jones,Allison M|example@epic.com|Stevens|Married|Unknown|Lutheran|English|
```

The demographics and identifier can be used in subsequent FHIR queries like Patient.$match to discover the FHIR Patient Resource ID and use that for additional FHIR API calls to gather data. Refer to our recommendations for strong patient matching when using FHIR APIs for additional details.

The identifier can also be used to filter information received over HL7v2 Interfaces such as the ones used to support Encounter Notification so that you only receive information for your patients.

### Clinical Data Exchange (CDE)

CDE is possible using FHIR. Payers can use the DocumentReference (Generated CCDA) FHIR Resource to obtain a CCDA of a patient or an encounter. Additionally Epic supports a broad library of FHIR APIs including and beyond USCDI v3 which can be used to gain more specific information you may need outside of the CCDA. Information for these can be found at fhir.epic.com.

Combined with Encounter Notification you can use this to get near real time clinical information about your patients.

### Encounter Notification

Epic supports the HL7v2 Outgoing ADT Interface and the 278N X12 Notice of Admission Interface which can be implemented by provider organizations to send you messages when patients have a visit.

For upcoming scheduled encounters Epic also supports the HL7v2 Outgoing Appointment Scheduling Interface which will send real time updates when appointments are scheduled or updated.

You can work with Epic provider customers to configure these interfaces to send you real time notifications for your patient roster.

### Member Insights and Care Gaps

Care Gaps can be ingested by Epic provider systems in bulk using the Care Gaps Exchange flat file interface. Patient, Event, and Medication Adherence Care Gaps are supported through this method.

The Community Specification for Care Gaps data is available here. Forming your data to this specification will allow provider customers to load this to their system and track care gaps for your patient population.

The CDS Hooks framework can call into your system at pre-defined triggers so that you can surface Care Gaps, Member Insights or other information to the provider while the patient is being seen.

You can work with Epic provider organizations to filter these CDS Hooks interactions to just your patient roster using the identifiers you would import when working through Roster Management.

Epic's support for CDS Hooks is built on a workflow engine and CDS rule engine. Organizations using Epic can configure native CDS alerts by defining the workflow point at which they are triggered, known as the hook, and the criteria to evaluate when they are triggered. These criteria determine whether a CDS alert appears to the clinician and, if it does, what information it contains. Epic's implementation of CDS Hooks enables native criteria to call a remote CDS Hooks API when evaluating whether to show a CDS alert.

CDS Hooks also supports feedback on the actions taken by the provider so that you can gain additional insights and refine the information presented at point of care.

We have an extensive CDS Hooks Tutorial that walks through the Epic implementation of CDS Hooks.

### Prior Authorization

Epic now supports the prior authorization FHIR APIs required by the CMS 0057 Final Rule, released in the February 2026 version of Epic. The Da Vinci 2.1 implementation guide provides general guidance, with API-specific details available on the API specifications pages at Vendor Services or Epic on FHIR. For guidance on connecting with other systems for CRD, DTR, or PAS, please refer to the CMS Prior Authorization Playbook. Groups can also join Vendor Services for general technical support and basic connectivity testing tools.

### Get Started

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## 2.7 Implantable Cardiac Devices Playbook

File: site/docs/02-playbooks/07-cardiac-devices.html

Source: https://fhir.epic.com/Documentation?docId=cardiac_devices

File discrete results for the device interrogation with In addition to any fields required by the interface specification, this message should include the following:

### Implantable Cardiac Devices

### Overview

Implantable cardiac device monitoring systems are used to streamline cardiac device data workflows, support informed clinical decisions, and deliver efficient, comprehensive care. The IHE Implantable Device – Cardiac Observations (IDCO) profile is used to communicate data from an implantable cardiac device (such as a pacemaker or ICD) with Epic and allows that data to be incorporated into the patient chart, helping ensure the legal medical record contains complete, accurate data. Device interrogations can happen in-clinic during a scheduled appointment or remotely where the patient initiates a device check from home and sends them in through the device vendor's system. From here, the general pieces to the device check workflow include the following:

* Ordering and scheduling
* Device interrogation and data uploads to the vendor-specific programmer
* Data analysis and any needed clinical action
* Study signed
* Data filed to Epic
* Billing

#### Workflow Diagrams

These diagrams illustrate two possible workflows. In both examples below, the order originates from your cardiac system and is sent to Epic with an Incoming ORM message. However, these orders could also be placed within Epic directly, removing the need for the Incoming ORM message. Your exact workflow may be a mixture of elements from both of these diagrams.

This workflow is designed to work well both for customers licensed for Cupid (Epic's cardiology application), and those not licensed for Cupid.

Device check study is finalized in Epic

Device check study is finalized in your cardiac system

### Initial Order

An imaging order must be placed in Epic for the device check. The order holds the entire workflow together, ensuring all the device data makes it to the right place and enabling other functionality such as automated scheduling of device checks. Orders can be placed via two equally valid methods.

* Your app can send an Incoming Orders from CPOE Systems message to create the order. This is typically done for remote device checks.

* Or Epic users can place the order in Epic. This is typically done for in-clinic device checks.

Your system can be notified of the successful device check order via an Outgoing Imaging Results and Orders message. This is either a rebound of the Incoming CPOE message you sent, or your notification of the order placed natively in Epic. You'll use the retrieved Epic order ID and patient identifiers when resulting and charging for the device check.

### Filing Device Data

Incoming Ancillary Results - Imaging interface.

* The Epic Order ID in the OBR-2 segment

* A PDF summary of the data

* This can be either an embedded Base-64-encoded PDF filed to Epic's Web BLOB, or a pointer to a document on a Document Management System (DMS).

* The Narrative & Impression (even if preliminary).

* As much discrete data from the IHE IDCO profile as is relevant for your app.

* The appropriate status of the result in OBR-25.

* If the result was finalized in your system, specify “F” for Final.

* If the result needs to be finalized in Epic, or your system will send final results later, send “P” for Preliminary.

### Finalizing Results

If the results are finalized in Epic, your app will be notified of the result with an Outgoing Imaging Results and Orders message.

Alternatively, if results are finalized in your system instead of in Epic, we recommend supporting SMART on FHIR to allow users to securely SSO into your system while also potentially giving you access to call additional APIs for necessary clinical data. Refer to the OAuth 2.0 tutorial. for more information.

Once the results are finalized, send charges for the device check with the Incoming Financial Transactions interface. Include the original order identifier in FT1-23.

### Get Started

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## 2.8 Computer Telephony Integrations (CTI)

File: site/docs/02-playbooks/08-cti-playbook.html

Source: https://fhir.epic.com/Documentation?docId=cti_playbook

Phone systems use the web service to route a call to a user working in Epic and automatically open the correct patient record and activity for the context of the call. When a payment collection screen appears to a user in Epic for collecting credit card information, Epic uses the web service to send a request to the phone system to pause a call recording. When the payment collection is complete, the same web service sends a request to resume the recording.

Computer Telephony Integration (CTI) apps allow Epic users to make and receive telephone calls in Hyperspace. Incoming calls from a phone system bring the user to the activity in Epic that matches the purpose of the call and with the caller identified. Outgoing calls allow the user to initiate a call in the phone system to a selected phone number with a single click.

### Incoming Calls to Epic

Incoming Computer Telephony Integration

### Outgoing Calls from Epic

Contact center agents and other Epic users can make telephone calls directly from Hyperspace with computer telephony integration (CTI), cutting down on misdials and spending less time dialing. When a user clicks a button or link in Hyperspace to call a patient, provider, or other phone number, Epic uses the Outgoing Computer Telephony Integration web service to pass the dialed number and information about the agent to the phone system and log a record of the call.

### Pause/Resume Call Recording

PhoneSystem

### Embedded Call Controls

Contact center agents interact with a phone system's controls for accepting, placing, and transferring calls directly in Hyperspace using the Active Guidelines (AGL) framework. Phone systems use the AGL Telephone Alert API to notify the agent of an incoming call and open embedded call controls.

### Call Analytics and Transcripts

Contact center agents and managers providing quality assurance can view analytics such as average call time and number of transfers, along with call transcripts in Hyperspace. Phone systems use the DocumentCallAnalytics and TranscriptIngestion APIs to file these items to Epic.

### Get Started

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## 2.9 Payment Device Integrations Playbook

File: site/docs/02-playbooks/09-paymentdeviceintegrations-playbook.html

Source: https://fhir.epic.com/Documentation?docId=paymentdeviceintegrations_playbook

The Payment Device Integration allows payments to be collected in real time. This is handled by outgoing requests sent to your endpoint during a payment workflow. The first request checks whether the device is initialized while the second request is needed for a payment transaction response. The payment is processed outside of Epic and through the payment processor.

### Payment Device Technology

The Credit Card and Check Device Interface resource contains specification details for both the InitializeDevice and ProcessTransaction calls. Additionally, the Generic Payment Gateway Interface is used for transactions such as voiding a payment or removing a stored token.

### Custom Transaction Fields

Epic supports sending custom key/value pairs in requests in addition to credit card transaction key/value pairs. Key/value pairs can either be static or dynamic. An example of a static value is the inclusion of a serial number for a device. Dynamic values can be used to send the logged in user (%USERID%), workstation id (%WORKSTATIONID%), or the logged in department id (%DEPTID%). Additional mnemonics for dynamic fields can be found in Appendix A of the Generic Payment Gateway Interface resource.

### Get Started

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## 2.10 Bayesian Medication Dosing Decision Support Playbook

File: site/docs/02-playbooks/10-bayesianmedicationdosing-playbook.html

Source: https://fhir.epic.com/Documentation?docId=bayesianmedicationdosing_playbook

A third-party decision support system used to provide personalized medication dosing recommendations. These products launch from Hyperspace using SMART on FHIR, pull patient and medication information, and provide the end user with optimized medication dosing recommendations. Bayesian analysis is commonly used for medications with narrow therapeutic

windows or medications requiring precise dosing based on patient factors.

### Technology Overview

A SMART on FHIR launch uses OAuth 2.0 to embed an iframe within Hyperspace for provider access. To support Bayesian analysis, begin by reviewing available public APIs instead of relying on manual provider entry. If using the MedicationRequest.Search API, apply data or status filters to retrieve only the relevant medication orders. The Launching Your App resource walks through details for SMART on FHIR launches, embedding an app in Hyperspace, and more.

### Get Started

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## 2.11 Medication Ordering Decision Support Playbook

File: site/docs/02-playbooks/11-medicationorderingdecisionsupport-playbook.html

Source: https://fhir.epic.com/Documentation?docId=medicationorderingdecisionsupport_playbook

This template is designed for developers creating a Medication Ordering Decision Support app. When you access this link, you will be directed to the app creation page with the audience, use case, and API fields pre-filled. You retain the ability to modify any field as needed to meet your application requirements.

A third-party decision support system used to suggest medication changes. These products launch from Hyperspace using CDS Hooks to pull medication order information and provide the end user with medication ordering recommendations.

### Technology Overview

CDS Hooks can be used at the point of provider ordering to provide medication recommendations. When a medication change may be needed, a suggested action can be used in the CDS Hooks response to show the recommendation. If additional data needs to be pulled, start by reviewing available public APIs rather than relying on manual provider entry. We have an extensive CDS Hooks Tutorial that walks through the Epic implementation of CDS Hooks.

### App Template

App Template - Medication Ordering Decision Support

### Get Started

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## 2.12 Fully Autonomous Coding Playbook

File: site/docs/02-playbooks/12-autonomouscoding-playbook.html

Source: https://fhir.epic.com/Documentation?docId=autonomouscoding_playbook

The incoming interface can also be used when there are cases that require a coder to review. If the feedback is variable, a free text comment can be returned. If there is a discrete list of reasons for the charge needing re-review, an informational modifier can be sent. Examples of APIs include, but are not limited to: This template is designed for developers creating an Autonomous Coding app. When you access this link, you will be directed to the app creation page with the audience and use case fields pre-filled. You retain the ability to modify any field as needed to meet your application requirements.

Fully Autonomous Coding applications utilize AI and machine learning to assign billing codes to patient charges or accounts. Applications receive patient information, clinical documentation, result and charge details from various outgoing interfaces and APIs. The autonomous coding app processes this information, sends updated charges back to the EHR, and they are filed to Professional Billing or Hospital Billing.

### Financial Transactions Interfaces

The autonomous coding flow uses the Outgoing Financial Transactions and Incoming Financial Transactions interfaces. The outgoing interface can be used in charge review mode to receive charges or placeholder charges present for an encounter. The incoming interface can be used in charge update mode in order to send automated coded charges back to Professional Billing and/or Hospital Billing. The incoming interface can be used to send the following:

* Update charge messages are used for charges needing updates to modifiers, procedure code, service date, billing provider, and service provider. Placeholder charges typically fall into this bucket if there is billable activity on that clinical document. Update charge messages can also be used with no updated data to acknowledge that a charge is good as is and should be billed without changes.

* New charge messages are used when adding net new charges that don't have an original Epic charge to match with. If a procedure note should generate two charges but only one placeholder charge exists, the first charge should be an update charge message to the placeholder charge and the second should be a new charge.

* Delete charge messages are used for charges that should not be billed.

### Additional Technology

The Outgoing Patient Administration interface is used to receive encounters for autonomous coding. Additional interfaces and APIs can be used to pull the data necessary for your product to complete autonomous coding. Examples of interfaces include, but are not limited to:

* Outgoing Documentation Interface

* Outgoing Results and Orders - LRI/ORM Interface

* Outgoing Imaging Results and Orders Interface

* (Hospital Coding) - Coding (Bidirectional) Interface

* DocumentReference.Search (Clinical Notes)

* Condition.Search (Encounter Diagnosis)

* Observation.Search (Labs)

### App Template

App Template - Autonomous Coding

### Get Started

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## 2.13 Automated Dispensing System Playbook

File: site/docs/02-playbooks/13-dispensing-playbook.html

Source: https://fhir.epic.com/Documentation?docId=dispensing_playbook

Automated Dispensing Systems (ADS) manage and include computerized devices that store, dispense, and track medications at the place of service. ADS commonly receive patient and order information to determine what medications are available for dispensing and send dispense information back to Epic when medications are removed from the ADS.

### Hardware

Vendors that support Automated Dispensing System functionality typically provide hardware to control and manage dispensing of medications.

### Patient Tracking

Epic supports the HL7v2 Outgoing Patient Administration (ADT) Interface. This interface enables ADS to track patient demographics and location. This information is used to create a patient profile in the ADS system such that providers are able to find the right patient when dispensing medications.

### Medication Tracking

Epic supports the HL7v2 Outgoing Verified Medication Orders Interface. This interface enables ADS to track the list of verified medication orders for a patient. Integration with this interface enables providers to select the right medication when dispensing medications.

### Dispensing Integration

Epic supports the HL7v2 Incoming Medication Dispenses and Load/Unload from ADS Interface. This interface enables ADS to communicate dispenses and waste to Epic. Additionally, the "load/unload" functionality of this interface is used to keep Epic in sync with what medication packages are currently stocked to the ADS hardware.

### Get Started

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# 2.14 Hemodynamics System Playbook

File: site/docs/02-playbooks/14-hemodynamics-system-playbook.html

Source: https://fhir.epic.com/Documentation?docId=hemodynamics_system_playbook

Hemodynamic Monitoring Systems are used in Cardiac Cath Labs, Cardiac EP Labs, and some Interventional Radiology Labs to monitor patients during invasive procedures. These systems monitor the patient’s vitals and pressures associated with specific cardiac structures. Epic’s procedure log is used in conjunction with the existing hemodynamic monitoring system, and the hemodynamic system writes relevant data into Epic’s procedure log to keep all the data integrated with the patient’s chart in Epic.

## Recommended Workflow

* Support the Outgoing Imaging Results and Orders Interface to receive the order from Epic.

* This will include the procedure log ID in ZPF-9 that you’ll use in later messages.

* Support the Outgoing Patient Administration – Registration and ADT Interface to receive the CSN (encounter identifier for the surgery).

* Support the Incoming Procedure Log Interface to file the hemodynamic data such as discrete vitals into flowsheet rows. You should file this data at regular intervals during the case (Example: every 5 minutes).

* Send the log ID to Epic in PID-18 or PV1-19.

* Include a timestamp in OBX-14. This is required for the flowsheet data to function in Epic.

* When the patient's condition changes, support filing that condition to a flowsheet row using the Incoming Procedure Log Interface.

* This can be filed in the same periodic batches as your other hemodynamic data, or it can be ad-hoc as the condition changes.

* Include the Log ID and Timestamp like you would with your other hemodynamic data.

* Support generating a waveform PDF summary. Send it either to Epic’s Web BLOB via the Incoming Scanned Document Link interface, or to a document management system (DMS).

* Include the CSN received from the Outgoing ADT interface.

## Get Started

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# 2.15 ECG Management System Playbook

File: site/docs/02-playbooks/15-ecgmanagementsystem-playbook.html

Source: https://fhir.epic.com/Documentation?docId=ecgmanagementsystem_playbook

Electrocardiogram (ECG) management systems store and interpret waveforms produced from an ECG machine, associate the waveform with an order for a patient, and file the data back to Epic. ECG management systems differ from point-of-care (PC-based) ECG and waveform visualization solutions in their ability to store data to a server and provide comprehensive workflow and interpretation support.

## Receiving Orders

ECG management systems use the Outgoing Ancillary Orders HL7v2 interface to receive ECG orders from Epic. Orders may be solicited (application receives an order from Epic and files a result to Epic) or unsolicited (application files an ECG result to Epic without receiving an order from Epic).

## Filing Results

To send ECG results to Epic, ECG management systems use the Incoming Ancillary Results HL7v2 interface. When filing results for unsolicited ECG orders, Epic rebounds an Outgoing Ancillary Orders HL7v2 message for this new order which is

accepted by the ECG management system and may be automatically associated with the corresponding study. If the ECG management system cannot automatically associate the order with the result, a user will need to manually associate the result in Epic.

The ECG management system will send two result HL7v2 messages to Epic: the preliminary result message (sent after acquiring the ECG) and the final result message.

Each result message includes the following:

* Patient demographics in PID segment
* Assigned Patient Location in PV1-3
* Visit Number in PV1-19
* Image availability in OBR-21 or OBX-5
* Time ECG was performed in OBR-7
* Accession number OBR-3
* Procedure code in OBR-4

The preliminary result message includes a preliminary result Status in OBR-25.

The final result message includes the following:

* Result text and discrete measurements in OBX segments:
* Impression
* Narrative
* Ventricular Rate
* Atrial Rate
* PR Interval
* QT Interval
* QRS Duration
* QTC Calc
* ST Interval
* TP Interval
* Cardiac axes (e.g. P-axis, R-axis, etc.)
* Actionable Findings, sent discretely in the result message as mapped OBX values
* Ex. Atrial fibrillation
* A discrete diagnosis in DG1
* Used for medical billing and coding

### SMART on FHIR App Launch

Allow users to access a view of the ECG by launching a web viewer from Epic using SMART on FHIR. You may choose to support study-level, patient-level, and/or work list mode integration.

### Responding to Release of Information Request

To support integration with Epic’s Release of Information module, send the PDF of the ECG result in the ORU^R01 message as a reference pointer. You can then use the EPS Document Retrieval web service to deliver the PDF on demand.

### Get Started

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## 2.16 Point-of-care Lab Middleware Integrations Playbook

File: site/docs/02-playbooks/16-pocmiddleware-playbook.html

Source: https://fhir.epic.com/Documentation?docId=pocmiddleware_playbook

Point-of-care lab middleware integrations connect lab hardware to Epic to facilitate receiving results from point-of-care devices into Epic. These integrations are necessary to translate data received by many different types of lab devices into a standard format to be consumed by Epic.

### Receiving Orders

Point-of-care tests (POCT) may be solicited (lab middleware receives an order from Epic and files a result to Epic) or unsolicited (lab middleware files a result to Epic without receiving an order from Epic). Most commonly, these tests will be unsolicited. In this scenario, the lab middleware will use the Outgoing Patient Administration HL7v2 interface to receive patient and encounter information related to a given test.

If a point-of-care test is solicited, the lab middleware will receive the order via the Outgoing Orders to Lab Instruments HL7v2 interface.

### Filing Results

Customers may choose to file POCT results to either EpicCare or Beaker. To file POCT results to EpicCare, use the Incoming Ancillary Results HL7v2 interface. To file POCT results to Beaker, use the Incoming Results from Lab Instruments HL7v2 interface.

In order to offer flexible translation capabilities across multiple instrument types, be sure your product is able to pass instrument error codes to Epic in the OBX-8 segment of the ORU result message.

### Get Started

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## 2.17 Standard Lab Middleware Integrations Playbook

File: site/docs/02-playbooks/17-standardlabmiddleware-playbook.html

Source: https://fhir.epic.com/Documentation?docId=standardlabmiddleware_playbook

Standard lab middleware integrations connect lab hardware to Epic to facilitate receiving results from lab instruments into Beaker. These integrations are necessary to translate data received by many different types of lab instruments into a standard format to be consumed by Epic.

### Receiving Orders

Use the Outgoing Orders to Lab Instruments HL7v2 interface to receive standard instrumentation lab orders from Epic.

### Filing Results

To file standard instrumentation lab results into Beaker, use the Incoming Results from Lab Instruments HL7v2 interface.

In order to offer flexible translation capabilities across multiple instrument types, be sure your product is able to pass instrument error codes to Epic in the OBX-8 segment of the ORU result message.

### High-volume Result Throughput

Some customers may process a large amount of standard lab orders and results each day, so it is important that your middleware is able to support a realistic volume of orders. Consider aiming to support processing:

* 15,000 specimen per day per organization

* At least 5,000 orders within one hour

### Get Started

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## 2.18 Waveform Visualization Playbook

File: site/docs/02-playbooks/18-waveformvisualization-playbook.html

Source: https://fhir.epic.com/Documentation?docId=waveformvisualization_playbook

Waveform visualization apps display waveform data in near real time via a web and/or mobile app that launches from Epic. They may support viewing historical data (ex. data from the past 24 hours).

### App Launch

Launch your waveform visualization app from Epic by passing patient context as part of an encrypted launch URL. Customers may use these products on with a variety of devices.

### Send Images to Epic

As part of this integration, you may choose to send images to Epic. You can do so via the Incoming Scanned Document Link HL7v2 interface.

### Get Started

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## 2.19 RTLS Playbook

File: site/docs/02-playbooks/19-real-time-location-systems-playbook.html

Source: https://fhir.epic.com/Documentation?docId=real_time_location_systems_playbook

The Application can support messaging to notify Epic of badge location changes through the . The Epic implementation of the HL7v3 specification also facilitates patient and provider co-location messages to allow the Application to indicate when a provider has spent a significant amount of time with the patient such that they should be recorded as having attended to the patient’s care.

Real-Time Location System (RTLS) is used within the healthcare industry to provide real-time visibility into where patients, staff, and assets are located within their facilities. RTLS uses a tag that gets assigned to the entity (patient/staff/asset) they want to locate and a sensory network that’s deployed within the facility to determine location of the tag. Epic integrates with RTLS leveraging an HL7v3 interface to communicate incoming patient, staff, and medical device locations, and tag associations. In Epic, the RTLS data is used to provide visibility and automate workflows.

### Location Updates

HL7v3 Incoming (to Epic) Real Time Location Services Interface

### Badge Associations and Disassociations

RTLS data exchange with Epic supports messaging to keep badge associations to patients in sync between Epic and the Application. For example, associations can be performed as part of the a user's workflow with in Epic so that the user does not need to navigate to a separate application. In addition, Epic supports messaging to keep badge disassociations to patients in sync between Epic and the Application. These associations and disassociations between systems can be accomplished through one of the following integration means:

* By HL7v2 Outgoing Patient Administration Interface supported z-segment, ZPV-14.

* By HL7v3 Incoming and Outgoing Real Time Location Services Interface.

### Get Started

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## 2.20 Surgical Materials Playbook

File: site/docs/02-playbooks/20-surgical-materials-playbook.html

Source: https://fhir.epic.com/Documentation?docId=surgical_materials_playbook

Epic supports integration with external Materials Management Systems to streamline surgical supply and instrument tracking. Epic recommends implementing the following functionalities to ensure accurate and timely information in both Epic and the

external system.

### Synchronizing Supply Definitions

You may use the Incoming Materials Management Interface to automate synchronization of supply definitions from an external materials management system to Epic. This message creates or updates a surgical inventory item record and corresponding balance records. It can contain multiple item segments to add/update multiple items in a single message.

### Surgical Resources

The Incoming Surgical Resource Data Interface can be used to synchronize the surgical instrument records maintained in an external instrument tracking system to Epic. The M16 event can create, update and inactivate surgical instrument records inside of Epic. A message can contain information about the status of the surgical instruments, and the locations where they are authorized.

The Outgoing Surgical Case Scheduling Interface should be used to plan surgical instrument usage from within Epic for specific encounters.

### Supply Depletion

Typically, planned usage will be sent from Epic to the materials management system the previous business day using the Outgoing Surgical Case Scheduling Interface. After the encounter occurs, actual usage and wastage will be reconciled against the planned usage using the Outgoing Inventory Depletion Interface. It indicates to the external system which supply was used, from where it was picked, and the number by which the external system should decrement the balance of the supply.

If using an external system where supply usage is being documented (such as a cabinet dispensing system), the Incoming Procedural Supply Usage Interface can be used to trigger usage messages through the external system to Epic. The interface works together with the previous two interfaces to reconcile planned supply usage for an encounter with the items pulled from the supply cabinet on the day of.

### Get Started

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## 2.21 Digital Pathology Playbook

File: site/docs/02-playbooks/21-digital-pathology-playbook.html

Source: https://fhir.epic.com/Documentation?docId=digital_pathology_playbook

An external organization may want to send their slides digitally to an organization on Beaker for consult or vice versa. To consider this workflow, you must be able to exchange images between two organizations.

### Overview

Digital Pathology is an integration between Beaker (Epic's laboratory application) and a 3rd party Image Management System (IMS) to allow pathologists to gain access via a hyperlink to high-resolution, digitized images of case slides for examination and interpretation. The hyperlink can be set up as one link for the overall case with all the associated slide images (Case-level) and/or as multiple links to the individual slide images (Slide-level).

Workflow: (A detailed diagram can be found in the Digital Pathology Specification)

* When a lab user orders a slide, Beaker will send an Outgoing Orders to Lab Instruments message to the image vendor for image availability.

* When each slide image is available, the vendor will send an message back into Beaker. Hyperlinks for the images on the case and/or individual slides are generated, which can be viewed by Pathologists.

Incoming Results to Lab Instruments

* This will open the vendor's IMS with the selected case/slide.

### Interface Messages

* When a lab user orders a slide, Beaker will send an message (OML^O21) to the image vendor for image availability.

Outgoing Orders to Lab Instruments

* Message elements are defined in the Digital Pathology Specification.

* When each slide image is available, the vendor will send an message back into Beaker - Specifically a Specimen Status Update (SSU^U03) message for each individual slide.

Incoming Results to Lab Instruments

* This is as opposed to sending multiple slides in the same message. It allows for faster image availability and avoids potential storage issues in Epic.

* You should allow linking to slide images using an LIS-generated barcode ID.

* We recommend sending additional slide status messages besides overall image availability in the SAC-8 field of your Incoming Results from Lab Instruments messages. Examples include:

* QC Complete

* Pathologist Reviewed

* To improve synchronization with your IMS, you can optionally receive an Outgoing Results and Orders Interface (ORU^R01) message when a case has been finished and use that to close out the case in the IMS.

### Your IMS System

* We recommend you display at least the following minimum amount of information that was received in the Outgoing Orders to Lab Instruments message in the IMS:

* Display patient demographics

* PID-3: Patient Identifier List (MRN)

* PID-5: Patient Name

* Display the current case/slide ID

* ORC-2/OBR-2: Case ID

* OBR-19: Container (Slide) ID

* Sync the selected case in the Beaker worklist to the images in your IMS. We recommend using a web-based application since it advantages over a local thick client method. It allows better synchronization between the IMS and Beaker. For example, switching cases in Beaker can switch cases and images in the IMS, or you can close case images in the IMS if the user is no longer viewing the case in Beaker. If you support a thick client application, it should follow the specification (note that this only works for case-level image links).

XML Context Synchronization

* Provide your customers a web URL format for your IMS for Beaker to launch with a hyperlink. This should be a deep link based off the case ID or slide ID depending on which formats you support.

* Ex. "https://viewer.vendor.com/pathology/cases/scanID=ID"

* In the incoming lab instruments message, send the string portion of the URL that can be used as an identifier for the case/container (not required if using Beaker container ID).

* ZSP-2.1: URL Identifier

* ZSP-2.3: IMS specific ID Type

### Consult Workflows

* This will require that the sending organization sends orders to Epic with case and slide external IDs by either:

* A paper requisition

* An order message across an Incoming Orders from CPOE Systems Interface

* The consult organization can create a case and send image availability messages using the external IDs to match the scanned images in the IMS.

* The consult organization will then send the results back to the sending organization by either:

* Fax

* A result message across an Outgoing Results and Orders Interface

### Get Started

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## 2.22 Wearable Cardiac Devices Playbook

File: site/docs/02-playbooks/22-wearable-cardiac-devices-playbook.html

Source: https://fhir.epic.com/Documentation?docId=wearable_cardiac_devices_playbook

### Wearable Cardiac Devices Overview

Wearable cardiac devices, like Holter monitors, event monitors, long-term continuous monitors and mobile cardiac telemetry, are used to monitor and record cardiac data while a patient is away from the office or hospital. This ECG data can be recorded over multiple days and analyzed by your system to produce preliminary reports for clinical review and follow-up. The general pieces to the wearable cardiac monitor workflow include the following:

* Ordering and scheduling
* Device placement and activation
* Data analysis from wearable device vendor
* Data filed to Epic
* Study signed
* Billing

### Initial Order

An imaging order must be placed in Epic for the cardiac monitor. The order holds the entire workflow together, ensuring all the device data makes it to the right place and enabling other functionality such as automated scheduling and billing.

Your system can be notified of the placement of a cardiac monitor via an Outgoing Imaging Results and Orders message. This is a notification of the order placed natively in Epic. You'll use the retrieved Epic order ID and patient identifiers when sending result data for the monitor.

### Filing Device Data

File discrete results for the device with the Incoming Ancillary Results - Imaging interface In addition to any fields required by the interface specification, this message must include the following:

* The Epic Order ID in the OBR-2 segment
* A PDF summary of the data
* This can be either an embedded Base-64-encoded PDF filed to Epic's Web BLOB, or a pointer to a document on a Document Management System (DMS).
* The Narrative & Impression (even if preliminary).
* As much discrete data from the IHE IDCO profile as is relevant to your app.
* The appropriate status of the result in OBR-25.
* If the result was finalized in your system, specify “F” for Final.
* If the result needs to be finalized in Epic, or your system will send final results later, send “P” for Preliminary.

### Finalizing Results

If the results are finalized in Epic, your app will be notified of the result with an Outgoing Imaging Results and Orders message.

Alternatively, if results are finalized in your system instead of in Epic, we recommend supporting SMART on FHIR to allow users to securely SSO into your system while also potentially giving you access to call additional APIs for necessary clinical data. Refer to the OAuth 2.0 tutorial for more information.

### Get Started

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## 2.23 Clinical Knowledge Set Apps Playbook

File: site/docs/02-playbooks/23-clinical-knowledge-set-playbook.html

Source: https://fhir.epic.com/Documentation?docId=clinical_knowledge_set_playbook

The Clinical Knowledge Set integration is based on the HL7 Infobutton standard, which allows clinicians to retrieve targeted information provided by third parties, specific to the context of the patient or clinical workflow. Rather than searching third party knowledge base, searches are performed automatically based off elements of the patient's chart like diagnoses or orders.

### Integrating Targeted Information into Clinical Workflows

During a variety of clinical workflows, clinicians can benefit from the addition of targeted information that is included automatically based on elements of the patient's chart. Using the Infobutton, third-party content groups can make their knowledge base readily available in Epic's system for relevant users.

Applications can follow the HL7 Infobutton specification for keys to send with the URL. Potentially sensitive data should be encrypted and limited to patient and provider demographic information.

If users would like to manually search for content, launching your application from Epic using the Smart on FHIR standard is possible.

### Get Started

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## 2.24 Total Parenteral Nutrition Compounding Playbook

File: site/docs/02-playbooks/24-tpn-compounding-playbook.html

Source: https://fhir.epic.com/Documentation?docId=tpn_compounding_playbook

Your application should help automatically compound sterile solution for a patient, adhering to the precise amounts provided in the order. The final output should be a complete or near-complete solution to be delivered to a patient.

Total Parenteral Nutrition (TPN) Compounding is the process of preparing sterile solutions for patients to receive intravenously. This process is conducted via TPN compounding machines (compounders), which are computerized medical devices that receive ingredient level information and compound precise amounts of each component. TPNs can be compounded in-house or offsite by a third-party compounding service.

### Accepting Orders from Epic

For Total Parenteral Nutrition compounding, you will need to be able to receive and process orders from Epic. Some examples of how this is done:

* Directly receive TPN orders via the Outgoing Verified Medication Orders interface.

* Have your customer export TPN orders from Epic to a flat file and import the flat file directly into the compounder.

* Print TPN orders from Epic and manually enter them into a program that processes and adjusts orders as necessary.

### Processing Orders

Other considerations as a third-party provider of TPN Compounding. Your application could:

* Handle barcode information with dispense messages

* Provide hardware that allows for effective clinician workflows

* Report on manually added ingredients

### Order Complete

### Get Started

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## 2.25 Encoder Playbook

File: site/docs/02-playbooks/25-encoderplaybook.html

Source: https://fhir.epic.com/Documentation?docId=encoderplaybook

Encoder integrations involve using a third-party system to determine and file diagnosis, procedure, and DRG codes based on clinical documentation. Coders or Clinical Documentation Specialists (CDS) launch from Epic into the encoder, which interfaces with Epic to ensure accurate coding and data storage. Encoders can integrate for both facility coding workflows and Clinical Documentation Improvement (CDI) workflows.

### DRG Support

Encoders must support filing multiple DRG types in a single launch. If supporting CDI workflows, the encoder can provide an auto-suggested DRG via a backend interface for automated CDI reviews, while still supporting multiple DRG types for filing.

Integration occurs through a front-end workflow by using a SMART on FHIR EHR launch with either of the two methods below. In both cases, the payload is defined in the Coding – Bidirectional Interface Technical Specification, which applies to both CDI and coding workflows:

* Front-End Coding via Web Services

* Front-End Coding Integration via Subspace

### Clinical Data Access

To support auto-suggested DRGs for CDI workflows, encoders can retrieve clinical documentation from the patient's chart using FHIR APIs or outgoing interfaces such as:

* DocumentReference.Search (Generated CCDA) (R4)

* DocumentReference.Search (Clinical Notes) (R4)

* Outgoing Documentation Interface

* Outgoing Results and Orders – LRI/ORM Interface

* Outgoing Imaging Results and Orders Interface

### Backend Integration

Encoders use the backend portion of the Coding – Bidirectional Interface Technical Specification, to file autonomous suggested DRGs for CDI workflows. A Backend Systems app is required to make asynchronous FHIR API calls and to send interface messages (e.g., automated CDI reviews).

### Get Started

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## 2.26 Medication Carousel Playbook

File: site/docs/02-playbooks/26-medicationcarouselplaybook.html

Source: https://fhir.epic.com/Documentation?docId=medicationcarouselplaybook

Medication Carousels/Robots are used in inpatient pharmacies to store, retrieve, and automate dispenses to pharmacists. Carousels unlike ADS act more as an inventory hub in a central pharmacy which allows for space optimization and improve picking efficiency for high-volume storage.

### Hardware

Vendors that support Medication Carousels and Robots provide the hardware to control and manage dispensing of medications.

### Medication Tracking

Epic supports the HL7v2 Outgoing Verified Medication Orders Interface. This interface enables medication carousels and robots to track verified orders associated with pharmacy workflows. Integration with this interface ensures that the correct medications are retrieved and dispensed from the carousel inventory.

### Inventory Integration

Epic supports the HL7v2 Load/Unload (ZPM) Interface. This interface enables carousels to communicate physical stock movements such as load, unload, restock, destock, expiration, and return bin actions. Using this interface keeps Epic and Willow Inventory in sync with the current stock levels within the carousel hardware.

### Optional Integration

Medication carousels may also use the Interdepartmental Transfer (IDTR) Interface to automate non-patient stock movements, such as replenishment or ADS restocks. This supports coordinated transfers between the central pharmacy carousel and other care locations, while Load/Unload continues to maintain accurate inventory counts.

### Get Started

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your product’s strengths. Explore the list of eligible categories and share your interest with us through the Technology Guidance form.

# 2.27 End-User Authentication Playbook

File: site/docs/02-playbooks/27-enduserauthentication-playbook.html

Source: https://fhir.epic.com/Documentation?docId=enduserauthentication_playbook

All users are required to authenticate when logging in to Hyperspace, but there are several other workflows that might demand the extra security that additional authentication provides. End-User Authentication includes two subcategories, Login & Session Management and Reauthentication. Login & Session Management applications provide secure and efficient mechanisms for users to access Epic’s applications such as Hyperspace, Haiku, Canto, or Rover. Reauthentication applications verify the identity of end-users before performing certain actions within Epic’s applications such as Hyperspace, Haiku, Canto, or Rover.

## User Login

By default, Hyperdrive and Epic mobile applications require users to authenticate by entering their User ID and password. However, additional authentication devices can be configured to use either Generic Desktop Authentication with SAML tokens or OpenID Connect to authenticate the user. If subspace APIs will be used to manage the state of the users Hyperdrive session, Generic Desktop Authentication can be used to gather the needed information to connect to the subspace hub.

## Session Management

Epic supports the Subspace Communication Framework for asynchronous, semi-event-driven integrations between Epic and external applications. These should be used to manage the state of the user's Hyperdrive session such as securing the workstation or logging the user out of Hyperdrive.

## Reauthentication and Witnessing

There are common workflows in Epic which require a user to reauthenticate or have a witness to complete the action. Epic uses OpenID Connect to compete reauthentication workflows in most contexts. Passive authentication, such as ED Narrator, will require using Desktop Generic Authentication.

In “witnessing” workflows or “dual sign” workflows, a second user is witnessing the user doing an event (example: disposing of narcotics or prescribing controlled substances) and both users need to sign off on the event having happened. In this case, the logged-in user will re-authenticate in their session and a second, separate user will also authenticate within the user’s session to sign off on the event. In other workflows, there are expectations to “default in” the currently logged-in user, but other users are acceptable as sources of signoff in these workflows.

## Get Started

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# 2.28 Bedside Dining Playbook

File: site/docs/02-playbooks/28-bedsidedining-playbook.html

Source: https://fhir.epic.com/Documentation?docId=bedsidedining_playbook

Bedside Dining applications present patients dining menus and options to place dining orders within MyChart Bedside.

## Launching from MyChart Using SMART on FHIR

Patients can access your dining menu directly from MyChart Bedside and Bedside TV. When launching third-party applications, Epic uses SMART on FHIR to pass patient context along with other information to facilitate your ordering workflow.

## Dietary Restrictions

Epic can share dietary orders in real-time using the Outgoing Diet Orders interface to send you important nutritional information to assist the patient with meal selections.

## Get Started

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including the ability to request individualized assistance with your app's data exchange. In addition, some product categories qualify for Toolbox support and designation on Showroom, which can accelerate implementation and enable you to highlight your product’s strengths. Explore the list of eligible categories and share your interest with us through the Technology Guidance form.

# 2.29 Printing Output Management Playbook

File: site/docs/02-playbooks/29-oms-playbook.html

Source: https://fhir.epic.com/Documentation?docId=oms_playbook

## Output Management

All Epic community members use the Epic Print Service (EPS) to print documents generated by various applications from their instance of Epic. While EPS is not the only method of printing from Epic, it processes the majority of print jobs exiting the system. Third-party products can integrate with EPS to collect the jobs exiting from the service that are bound for physical printer devices, which would normally be done through Windows print queues.

Print jobs contain two components, the metadata and the data to be printed. The metadata will include important information needed to complete the print job (e.g., the printer name) as well as information that will be useful for reporting (e.g., scenario name).

### Batch Printing

EPS batch printing allows multiple print jobs in any combination of formats to be printed together on the printer without other batched jobs coming between them. Epic will send a Batch Open request followed by all relevant print jobs in that batch and then a Batch Close request.

### Large Documents

Requests are sent using a chunked transfer encoding, which allows requests to be streamed directly to and from the disk. EPS is capable of sending any size of document over the output API so it’s important to account for memory limitations on these documents, often imposed by web servers.

### Status Updates

Output Management systems can file back a status to Epic on specific print jobs using the StatusUpdateURL provided in the metadata of the print job. Filing back these statuses will help users keep track of which printouts were successful and which have failed.

## Get Started

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# 2.30 Wayfinding Playbook

File: site/docs/02-playbooks/30-wayfinding-playbook.html

Source: https://fhir.epic.com/Documentation?docId=wayfinding_playbook

Wayfinding applications can use a patient’s appointment information, such as the encounter department, to help users navigate healthcare organizations.

## Launching from MyChart

Patients may launch your application from MyChart and use your navigational tools to reach their destination. When launching third-party applications, patient context, such as the Department ID of an upcoming appointment, can be passed to your wayfinding application. If you require PHI, a SMART on FHIR launch must be used. Using this context, wayfinding applications can appropriately navigate the patient from their current location to where they need to go.

## Blue Dot Wayfinding Using BLE APIs

Wayfinding applications embedded in MyChart Mobile can support blue dot navigation by integrating with Epic's MyChart BLE APIs, which provide access to nearby BLE beacon and scan data. These APIs handle permission checks and enable apps to start monitoring for specific devices, delivering a continuous stream of detected BLE data. Your wayfinding application's positioning engine then translates this incoming signal data into a computed user location.

Copy Template App ->

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# 2.31 Infusion Pump Playbook

File: site/docs/02-playbooks/31-infusion-pumps-playbook.html

Source: https://fhir.epic.com/Documentation?docId=infusion_pumps_playbook

Use the interface to send the infusion order from Epic to the pump. Use the interface to send an acknowledgement from the pump to Epic indicating the pump successfully accepted the order or send a rejection from the pump indicating why the order was rejected.

Bi-directional infusion pump integration enables Epic to exchange information directly with your pumps. Clinicians can send order details from Epic to auto-program the pump, minimizing manual entry and potential errors. Your pump then transmits documentation data such as rate changes and infused volumes back to Epic, where it is filed seamlessly into the patient’s chart. Follow the IHE-PCD standard, using the interface specifications below.

## Outgoing Interfaces (Epic to Pump)

Outgoing Infusion Orders

## Incoming Interfaces (Pump to Epic)

Incoming Infusion Verification

Use the Incoming Infusion Documentation interface to send discrete measurements from a running infusion pump into Epic.

* Send event-based PCD-10 and periodic PCD-01 messages via the incoming infusion documentation interface.

## Get Started

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# 2.32 Patient-Facing / Consumer Health Apps

File: site/docs/02-playbooks/32-patient-facing-apps.html

Source: https://fhir.epic.com/Documentation?docId=patient_facing_apps

Developers can create without coordination with a health system, either through TEFCA IAS or Epic on FHIR. Some patient-facing applications are designed as part of a larger product that involves the implementation of a clinician or backend systems app where coordination with an Epic customer is needed. Your app may also need to use APIs that are outside the scope to qualify for automatic download. In these situations, you can create your app using any of the available resources from and customers can request it through the . You can work directly with a healthcare organization who is interested in your application, or make your app more visible by .

## Patient-Facing Consumer Health Apps

Patient-authorized health apps allow people to access, track, and share their medical records and personal health information, connecting their MyChart credentials to healthcare provider data. Third-party developers play a large role in this process, creating patient-facing apps that patients can choose to retrieve the desired data from instances of Epic where they receive care.

### Connect Your Health Apps and Devices

Patient-facing apps use OAuth2.0 so that patients can authenticate using their MyChart credentials. When patients use your app to access their health record, they are redirected to their healthcare provider's MyChart page and log in with their credentials. Patients authorize the app to pull specific data types that the app requests via FHIR APIs (authorization can be revoked at any time). Screenshots of the patient experience are detailed in our step-by-step Patient Authentication tutorial.

### Encouraging Patient Data Sharing

We have found that complex or confusing workflows can discourage patients from sharing their health data. To help address this, we've compiled a few strategies to encourage successful data exchange with patient-facing apps in our Design Strategies to Increase Patient Data Sharing tutorial.

### Client Record Registration and Distribution

To complete this workflow, each app must first pre-register with Epic and receive a client record (client ID). This client ID includes the app's requested FHIR APIs. An app can only request data through the FHIR APIs that are both registered to its client ID and authorized by the patient.

As part of registration, the app developer is expected to complete and attest to a Data Use Questionnaire for the client ID. The questionnaire determines what patients see when they are asked to authorize data sharing.

The client ID must then be distributed to each customer environment the app connects to. Developers of patient-facing health apps have several options for distributing their client ID to customer environments, as described below:

## Automatic Client ID Distribution

Patient-Facing Apps Using FHIR

### TEFCA for Individual Access Services (IAS)

What is TEFCA?

The Trusted Exchange Framework and Common Agreement™ (TEFCA) is a U.S. government-sponsored exchange framework for healthcare data interoperability that aims to provide a single network on-ramp for a wide variety of health information exchange use cases. Read more about TEFCA on open.epic.

What role do third-party app developers have with TEFCA for Individual Access Services?

One of the interoperability pathways TEFCA supports is the ability for patients to use apps of their choice to retrieve copies of their medical records from participants in the framework. In TEFCA, this is referred to as Individual Access Services (IAS). The idea is that individuals can use TEFCA-connected IAS apps to augment their access to patient portals with their health information.

Benefits of TEFCA IAS:

* There are over 200 Epic provider organizations participating in TEFCA and responding to IAS requests as of September 2025.

* As Epic on FHIR connections do not have record location capabilities, patient-facing apps must query all network endpoints for data retrieval. In comparison, TEFCA IAS can identify a list of organizations where a patient likely has data stored, allowing for efficient querying.

* Future: TEFCA IAS will allow for broader connection to health systems beyond Epic.

* Future: Epic is committed to growing TEFCA membership with a goal of onboarding 100% of the Epic provider community.

* Future: Patients using TEFCA IAS apps will be able to authorize sharing their health data from one central location using MyChart Central. This allows providers to continue managing federal, state, and organizational privacy requirements and gives patients a unified workflow for managing how their data is shared.

### Patient-Facing Apps through Epic on FHIR

Apps created through Epic on FHIR that adhere to specific technology use requirements, allow the Client IDs to be automatically distributed to Epic organizations with auto-downloads enabled. The types of auto-download apps through Epic on FHIR fall into three categories:

* USCDI Apps using only USCDIv1 or USCDIv3 APIs, depending on the Epic version.

* CCDS (MU3) Apps using only a subset of DSTU2 FHIR APIs.

* CMS Payer Apps using only ExplanationOfBenefit and/or a subset of FHIR APIs

Details on which APIs are available for the automatic client distribution app types listed above can be found here.

Benefits of Epic on FHIR apps:

* All US provider organizations currently receive client IDs that meet the automatic distribution criteria that are activated through Epic on FHIR.

* Client ID registration steps are self-service to app developers through the Epic on FHIR website.

## Client Record Distribution by Customer Request

Epic on FHIR

App Request Process

requesting a listing on Showroom

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### Client Record Registration

How do I register my app with TEFCA IAS?

After creating a qualifying product that meets the conditions described here, developers follow registration steps in order to exchange data with TEFCA participants. Epic Nexus Participants then respond to FHIR-based queries from IAS apps through

this framework.

We offer a testing plan to ensure that apps are well prepared for a successful production data exchange. When your application is ready for production exchange, it will be automatically distributed to Epic community members who are live on TEFCA and IAS via FHIR.

How do I register my app with Epic on FHIR?

Check out our step-by-step Developer Guide, where you can find details on Creating a Client ID and more.

How do I connect my app to customer environments?

Once your app is active (Ready for Production) and synced to customer environments, either through the automatic or customer-requested distribution methods, you can connect to customer-specific endpoints included in our User-access Brands Bundle, which includes details that can help patients find the right endpoint.

### Developer Support

Want a little extra help? Vendor Services offers an optional suite of benefits including the ability to request individualized assistance with your app's data exchange.

In addition, some product categories qualify for Toolbox support and designation on Showroom, which can accelerate implementation and enable you to highlight your product’s strengths. Explore the list of eligible categories and share your interest with us through the Technology Guidance form.

# 2.33 Endoscope Data Capture Playbook

File: site/docs/02-playbooks/33-endoscope-data-capture-playbook.html

Source: https://fhir.epic.com/Documentation?docId=endoscope_data_capture_playbook

Endoscope data capture integrations allow vendors to gather data from a scope, format it, and then send it downstream. Currently, Epic is focused on image data, though that will expand to video, event, and other data in the future. This data is used throughout Lumens workflows, e.g., documenting findings or writing a procedure note.

## Processing an Order

Customers send an ORM Message from the Outgoing Imaging Orders Interface to provide a list of orders per room/device being utilized so an end user can select an order to acquire data against.

Sample message:

```
MSH|^~\&|IMG_ARRIVE_APPT||||20250110141123|1|ORM^O01|334314.48|T
PID|1||205042^^^EPI^MR||ZZTEST^SAG^^^^^L||19650110|M|||^^^^^US^PERM|||||||2282|374-75-8473
PV1|1|OUTPATIENT|^^^^^^^^EMH GASTROENTEROLOGY^^||||1004^GASTROENTEROLOGY^PHYSICIAN^^^^^^PROVID^^^^PROVID|1004^GASTROENTE
    ROLOGY^PHYSICIAN^^^^^^PROVID^^^^PROVID||||||||||2282|||||||||||||||||||||||||20250110133709|||||||V
ORC|NW|1065621^EPC|AccNum8||Arrived||^^^20250110133758^^R||20250110141123|1^EPIC^USER^^||1004^GASTROENTEROLOGY^PHYSICIAN
    ^^^^^^PROVID^^^^PROVID|^^^^^^^^EMH GASTROENTEROLOGY|(555)555-5555^^^^^555^5555555|||||||||||||||I
OBR|1|1065621^EPC|AccNum8|GI2^EGD^IMGEAP|R|20250110141122|||||O|||||1004^GASTROENTEROLOGY^PHYSICIAN^^^^^^PROVID^^^^PROVI
    D|(555)555-5555^^^^^555^5555555|||||||ES|Arrived||^^^20250110133758^^R|||||||||20250110134000
ZPF|1|ENDO^ENDOSCOPY PROCEDURE ORDERABLES|||||^^^^^^^^EMH GASTROENTEROLOGY||1846
```

We recommend that the device and this interface should not be replaced with a DICOM MWL capable product. DICOM MWL products pull from a database, rather than read information from a message sent in real-time. As such, there is always a chance of data being desynced in the data capture system from the EMR based on when a change is made in the EMR versus when a DICOM MWL checks for a change. This can have interface filing implications if you use an interface method to file data that checks against procedure, provider, or diagnosis.

## Sending Data Back to Epic

Customers accept data from your data capture system in one of three ways:

* Send data to Epic’s Gallery Product through a Windows upload directory:

* Send files to a directory to then be absorbed by Gallery via a file sweep.

* Send one file per image sent with a title in the format &ltACCNO&gt_&ltENTERPRISE MRN&gt_&ltSCAN DATE&gt_&ltSCAN TIME&gt_&ltgeneric file name&gt where ACCNO is the accession number, Enterprise MRN is the unique Epic MRN for a patient, Scan Date is the day the image was taken, and Scan Time is the time the image was taken. Images are sent as files in tiff, gif, or jpeg format.

* Send data to the Epic Web Blob Server through Base64 encoded HL7

* Send data in MDM T02 messages through the Incoming Scanned Document Link Interface with base64 encoded image data. Images are encoded from a file in the tiff, gif, or jpeg format. We recommend one message per image with a unique identifier for each image.

* Sample Message:

```
MSH|^~\&|PACS|PACS|PACS|PACS|20200916175205||MDM^T02|16002787236963|P|2.3|3
PID|1||205042^^^EPI^MR||ZZTEST^SAG^^^^^L||19650110|M|||^^^^^US^PERM|||||||374-75-8473
PV1|1|OUTPATIENT|^^^^^^^^EMH GASTROENTEROLOGY^^|||||1004^GASTROENTEROLOGY^PHYSICIAN^^^^^^PROVID^^^^PROVID|||||||||||||
    ||||||||||||||||||||20250110133709|||||||V
ORC|RE||AccNum8||SC||^^^20200916110000^20200325133000^R|||||59889^ DOCTOR ^DOC^^^^^^PROVID|||||101094130^EPIC
    ENDOSCOPY^SYSTEM|||||||IMAGES_AVAILABLE
TXA||Endoscopy Image||20200916175205|1004^GASTROENTEROLOGY^PHYSICIAN^^^^^^PROVID|||||||1.2.826.0.1.3680043.9.6362.667445
```

```
    708373935139286584129118690289 1|||^^AccNum8||AU||AV|
OBX|1|ED|Endoscopy Image^Endoscopy Image|1|^JPG^^^&ltencoded image&gt
```

* Send data to a 3rd party image storage vendor that integrates with Lumens:

* The data capture vendor partners with an image storage vendor to send media to.

### Timing

You should aim to hit minimum benchmark of sending 30 image files or messages to Epic in under 50 seconds to ensure the media files appear in an Endoscopist's report. The "clock" starts when an end user indicates to data capture to send images and ends when the last image hits the relevant Windows upload directory or Incoming Interface.

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## 2.34 Alert Manager Playbook

File: site/docs/02-playbooks/34-alert-managers-playbook.html

Source: https://fhir.epic.com/Documentation?docId=alert_managers_playbook

Alert Communicator is included in EpicCare Inpatient and requires a third-party Alert Manager. Alert Managers ingest alerts from a variety of devices, such as inpatient vitals monitors, and determine who should get notified for each alert. They then send Epic's Alert Communicator a standardized PCD-06 message including the alert recipient and the relevant alert details. We then display it to the appropriate users as a toast notification in Hyperspace or a push notification in Haiku, Canto, and Rover. They also handle escalation in scenarios where the original recipient does not respond in an appropriate amount of time.

### IHE ACM PCD Profile

Epic supports the IHE ACM PCD profile for receiving device-initiated alerts. This standard helps facilitate communication of alarm information to caregivers. To connect with Epic's Alert Communicator, the alert manager should be approved by the FDA via the 510(k) process for reviewing medical devices and we recommend implementing the following fields as specified in version 1.3 (or newer) of the IHE ACM profile:

Field | Used for...
IHEPCDACMParticipants | discrete patient mapping
IHEPCDACMLocations | discrete patient mapping
IHEPCDACMObservation alertTimestamp | discrete patient mapping
IHEPCDACMObservation eventIdentification | alert subtyping/display
Wctp-ChoicePair | sending discrete follow up actions for the user to take on an alert
Wctp-Recipient | specifying the recipient of the alert
Wctp-MessageControl deliveryPriority | alert notification priority mapping
Embedded IHEPCD04 HL7 Messages | displaying inline waveform information

### Outgoing HL7v2 ADT

It is recommended to have an Outgoing ADT (Patient Administration) interface in scope so that the alert manager system can receive pertinent patient information such as demographics, location, and treatment team to inform alarm routing.

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Copy Template App ->

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## 2.35 Telehealth Providers Playbook

File: site/docs/02-playbooks/35-telehealth-providers-playbook.html

Source: https://fhir.epic.com/Documentation?docId=telehealth_providers_playbook

Alternatively or additionally, you can when you'd like to use OAuth 2.0 for SSO, but you do not need your app to be launched from Epic. Your app initiates a login on its own, and Epic presents the patient with a MyChart login screen where they can enter their credentials. are typically used between two clinical systems looking to exchange patient-specific information. They include both the transport standards for clinical information exchange and the format and content of the information exchanged. Check out our to learn more.

Healthcare organizations may look to supplement their own telehealth operations for the purposes of 24/7 access or increased capacity of on-demand video visits.

### MyChart Extensibility & SSO

After patients or proxies have logged in to MyChart, the Epic customer's IT team can give them access to additional features from the menu within the existing MyChart mobile app or website. This menu launch could open a separate app or website for the patient to interact with. MyChart can also pass along information about the logged in user so that the other app or website can provide user-specific content, potentially eliminating the need for the patient or proxy to log in again. For example, you can identify a patient or proxy based on tokens that pass the patient ID, MyChart account ID, name, or date of birth. An OAuth 2.0 access token can also be used to call FHIR APIs on behalf of the patient to read, search, or write data that the application is authorized to interact with. This approach is an implementation of an EHR Launch.

implement the standalone launch workflow

### Clinical Data Exchange

EHR-to-EHR clinical information interfaces

supported standards

The below list represents some of the interfaces which may be useful to implement:

* Incoming and Outgoing Patient Discovery (Link)

* The Direct Standard® (Link)

* Clinical Document Architecture (CDA) (Link)

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## 2.36 Vitals Gateway and Middleware Device Integration Playbook

File: site/docs/02-playbooks/36-vitalsgateway-playbook.html

Source: https://fhir.epic.com/Documentation?docId=vitalsgateway_playbook

Vendors in this category receive vitals data from devices like bedside monitors or anesthesia carts. Vendors support either a Spot Check Vitals integration, a Continuous Vitals integration, or both. Spot Check Vitals devices receive discrete data that is validated on the device and then sent to EpicCare Inpatient Clinical Documentation. Continuous Vitals devices receive a constant stream of data which is then sent to EpicCare Inpatient Clinical Documentation or Epic’s Anesthesia application.

### Spot Check Vitals

Use the Outgoing Patient Administration (ADT) Interface paired with the Incoming Clinical Documentation Flowsheet Data Interface:

* The Outgoing ADT Interface sends patient data from Epic to the gateway or middleware device.

* The Incoming Flowsheet Interface files the pre-validated data from the device’s system directly into flowsheet rows in the patient’s chart.

### Continuous Vitals

Use the Incoming Device Data Interface to send a constant stream of unvalidated data from the device to Epic.

Send a PCD-04 message from the device’s system to a customer’s Alert Management System when an abnormal vital is recorded.

### Get Started

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# 2.37 Remote Patient Monitoring with Devices Playbook

File: site/docs/02-playbooks/37-rpmtutorial.html

Source: https://fhir.epic.com/Documentation?docId=rpmtutorial

Remote Patient Monitoring (RPM) empowers clinicians to monitor their patient's chronic conditions from home. Patients submit data to Epic flowsheets such as vitals, via direct device connection with the Bluetooth® Generic Health Sensor Profile or software connections with APIs and interfaces

## Bluetooth® Connections

If you manufacture devices used in RPM you can send device data directly into MyChart following the Bluetooth® Generic Health Sensor Profile

## RPM Software

If you are not a device manufacturer, or you have a software component to your RPM offering, follow our Remote Patient Monitoring Tutorial to file device data with HL7v2 Interfaces or FHIR APIs.

## Home Dialysis

If you provide Home Dialysis software, follow our Remote Patient Monitoring Tutorial to file device and/or patient-entered data with HL7v2 Interfaces or FHIR APIs.

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# 2.38 External Health Records for Patient Treatment

File: site/docs/02-playbooks/38-external-health-records-for-patient-treatment.html

Source: https://fhir.epic.com/Documentation?docId=external_health_records_for_patient_treatment

To provide the right care for a patient, it’s critical that any provider can access the right information for the right patient at the right time. We support many interoperability standards widely used around the world and have created additional tools to make sure there is always a path to accessing the information you need. Electronic health record vendors can connect providers to national data exchange frameworks using industry-standard technologies.

## Carequality

Like the framework and agreements that enable roaming across cellular networks and providers, Carequality enables nationwide coordination of care throughout the US. Carequality’s common framework includes trust policies, implementation-level functionality, standards-based technical and testing requirements, and operational practices for faster connectivity with other network members. All US-based Epic customers are members of the Carequality network, providing access to coordinated care for millions of patients across hundreds of healthcare organizations. To get started, contact Carequality to become a member.

## The Trusted Exchange Framework and Common Agreement™

The Trusted Exchange Framework and Common Agreement™ (TEFCA™) is a U.S. government-sponsored exchange framework for healthcare data interoperability that aims to provide a single network on-ramp for a wide variety of health information exchange use cases. Epic customers connect to the TEFCA network through the Epic community’s Qualified Health Information Network® (QHIN™): Epic Nexus. The Epic Nexus QHIN connects to other QHINs, allowing for advanced interoperability with minimal setup. The Epic community is in the process of getting connected to TEFCA. You can see the organizations that are live (or planning to go live) here. If you’re an Epic customer, contact your BFF to get started. If you’re not an Epic customer, you can learn more about connecting to Epic organizations for TEFCA’s supported use cases here.

## Direct Messaging

Direct Messaging provides the ability to send patient charts to another provider or organization and works similar to email. The organization that is currently seeing the patient can send a standardized C-CDA summary to another organization for

referrals and other transitions of care. It can also be used for provider-to-provider communications between organizations, say for consults. Epic organizations have also used Direct Messaging to improve social care outcomes with groups like United Way, and to share information with Health Information Exchanges (HIEs), government repositories, and public health reporting databases. Check out more details about how to set this up in our open.epic specifications.

### Health Information Exchanges

In addition, many organizations that use Epic are also connected to other networks or Health Information Exchanges. Reach out to a specific organization to understand the different additional paths that may be available to you.

## 2.39 Building a Patient Roster

File: site/docs/02-playbooks/39-patient-roster-playbook.html

Source: https://fhir.epic.com/Documentation?docId=patient_roster_playbook

Patient Rostering describes a broad range of use cases that typically start with trying to establish a roster of patients from Epic that are applicable to your product. This article describes a few common ways you might be able to accomplish this. There may be other niche APIs or technologies that were designed for specific use cases. We recommend being both flexible and specific with your roster definition. This will allow healthcare organizations the best chance of meeting your requirements and open the possibility of using those more niche options.

### Event Based Interfaces

HL7v2 and other standards-based interfaces are a tried-and-true method to push information based on an event in Epic in real time. The simplest example is an Outgoing Patient Administration (ADT) Interface which can push a message for admissions, discharges, and other encounter or patient events. Organizations using Epic commonly have middleware which can filter messages for you. For example – you might want to roster all patients that are being seen in a particular department. Make sure you are clear with customers in your filtering requirements to reduce the amount of noise you receive. In addition to the Outgoing Patient Administration (ADT) Interface, other events can also be leveraged to fill your roster. Examples include Outgoing Appointment Scheduling, Outgoing Imaging Orders, Outgoing Ancillary Orders, Outgoing Case Scheduling, and many more.

### Bulk FHIR

Before Bulk FHIR can be used, a patient roster must be defined within Epic. The roster within Epic is defined by a set of logical rules that determines if the patient meets those requirements. A common example is all patients with a certain diagnosis. If all you need is basic patient demographic information, you can simply leverage Patient.Search (R4) to construct the Bulk FHIR payload return with only the Patient resource for every individual in the roster. You can then follow up with RESTful FHIR calls – or use Bulk FHIR itself to retrieve the corresponding FHIR resources for each patient. See our Bulk FHIR Tutorial for more information on using Bulk FHIR and best practice recommendations for being a good steward of system performance. For example, Bulk FHIR would not be a good option for large patient rosters, or if you require up to the minute real time data. Bulk FHIR could be a good option if you infrequently need to update your roster for a small number of patients. We also strongly recommend leveraging _typeFilter to improve system performance.

### Extracts

Epic has always provided a myriad of reporting tools that can often double as a way to extract information for third party integrations. This can be a good option if you can define your patient roster in a way that can be translated into an Epic report. Depending on the extraction tool used, these reports can be run anywhere from daily to as frequently as a couple times an hour. Simplifying your requirements could lead to performance gains and the ability to increase extraction frequency. Customer staff business intelligence developers to create ad hoc queries to meet your needs and deliver them to you. Project priority and timelines for these requests vary greatly across customers. Alternatively, you can choose to take on the report complexity and license Kit or Clarity through Vendor Services.

### FHIR List.Search API

The List.Search (Patient List) (R4) FHIR API can retrieve predefined patient lists from Epic. The two types of Patient Lists in Epic are System Lists and My Lists, both of which are created and maintained by customer staff. System Lists are commonly used to populate a list of patients who are admitted to or recently discharged from a specific department. My Lists are commonly used by end users to quickly see a list of patients that they are caring for. If a pre-existing list doesn’t fit your needs, you can work with the Epic organization to determine if a list could be built for you. Make sure that your requirements are clear and consider being flexible in order to work within the Patient List framework in Epic.

### Get Started

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Guidance form.

# 2.40 Reference Labs

File: site/docs/02-playbooks/40-reference-labs-playbook.html

Source: https://fhir.epic.com/Documentation?docId=reference_labs_playbook

Reference labs and other specialty diagnostic labs can connect to Epic provider systems via point-to-point interfaces for the tests they perform. Common lab, pathology, specialty diagnostic, and genetic testing specimens may be collected at the external lab, by the originating healthcare organization, or directly by the patient.

## Lab Test Compendium

External labs work directly with the ordering healthcare provider organization to configure and map lab procedure identifiers in their system that match the orders your lab results. For example, they will configure procedure names and identifiers, result components, order questions, and other discrete data , and. Epic customers can find additional Epic configuration details and project management recommendations in Galaxy, our customer documentation portal, in the Third-Party Reference Labs Strategy Handbook.

## Receiving Outgoing Orders from Epic

Healthcare providers using Epic can transmit lab orders to the external lab LIS with the Outgoing Ancillary Orders interface to be notified when an applicable lab test is placed. The interface message will include details such as procedure, source, type, and collection date/time.

## Sending Incoming Results to Epic

When the external lab processes the test, the results then flow back into Epic, giving providers timely access to discrete data within the patient’s chart. Though critical results should be communicated synchronously such as with a phone call, when results are received via interface, Epic provides workflows and routing mechanisms to the appropriate customer-defined staff responsible for acting on results.

### Common labs, pathology, and specialty diagnostics

Labs use the Incoming Ancillary Results interface to send these results back to the ordering organization. Results are sent as discrete result components (e.g., WBC count, sodium, glucose, as opposed to sending textual documents) and defined normal ranges, units, and flags. Narrative-only results are generally appropriate for pathology results, though narratives can also add interpretation of other labs with discrete result components.

### Genomic Results

Discrete genomic results also file to Epic via the Incoming Ancillary Results interface. To include discrete genomic data as a part of the test's result, specifically formatted OBX segments must be included on the result message. The supported data elements are outlined in our genomics-specific interface specifications.

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# 2.41 Perioperative Workflows Playbook

File: site/docs/02-playbooks/41-periop-playbook.html

Source: https://fhir.epic.com/Documentation?docId=periop_playbook

## Perioperative Workflow Data Sharing Playbook

Epic supports multiple data sharing technologies to support the exchange of perioperative data for surgeries, anesthesia, and related visits. This playbook outlines common data types and workflows. Your product’s connection should account for how users interact with your app at each stage of the patient’s perioperative journey, because timing directly influences how your app exchanges data with organizations using Epic. Additionally, work with your customers to understand their perioperative workflows to create a robust connection.

## Before The Day Of Surgery

### Surgical Consult visits and Pre-Anesthesia Testing (PAT) appointments

Patients could discuss the surgical plan with their surgeon in an inpatient or outpatient consult appointment prior to the case request order being placed or surgery being scheduled. Similarly, they may also see their anesthesiologist for a

pre-anesthesia testing visit in an inpatient or outpatient appointment up to 30 days prior to the scheduled surgery.

Your app can pull appointment data via the Encounter FHIR resource. Outgoing Appointment Scheduling (SIU) HL7v2 interfaces can push appointment information to your app. Work with your customers to determine how to use Surgical Consult or PAT provider information, visit type information, or department information to identify these types of visits.

### Case Request Orders

Surgeons use case request orders to electronically request OR schedulers to add a surgical case to the schedule. Includes information like requested date of surgery and surgical procedure. Work with your customers to determine how their scheduling department processes case requests and determines how their surgical schedule is made.

Your app can pull Case Request order information via the ServiceRequest (Orders) FHIR resource. You can use the category field to identify these orders. Your customers may need to do additional setup to include Case Request orders in ServiceRequest. Additionally, the Outgoing Surgical Case Scheduling (SIU) HL7v2 interfaces can push case request order information to your app. Your customers may need to do additional interface setup to trigger outgoing messages when Case Request orders are signed.

### Scheduled Surgical Cases

When a surgery has been added to the surgical schedule. Surgeries will be scheduled with information like projected start and end times, procedure, and staff.

Your app can pull scheduled surgery information via the Appointment (Scheduled Surgeries) FHIR resource. The scheduled case will also show up in Encounter albeit with less information. Additionally, the Outgoing Surgical Case Scheduling (SIU) HL7v2 interfaces can push surgery scheduling information to your app.

### Preference Cards

Preference cards help OR staff efficiently prepare for surgeries by pulling in information that usually stays the same for a procedure, such as the pick lists and patient positioning used.

There is not currently an industry standard way of exchanging data for preference card information.

## Day Of Surgery

On the day of surgery, there are many clinicians, such as nurses, surgeons, and anesthesiologists, that will interact with the patient and document information about the surgery. Work with your customer to understand their day of surgery workflows in order to create a robust connection.

### Surgery Progress and Staff Times

Your app can pull surgery progress and surgeon timing information via the Encounter FHIR resource. Additionally, the Outgoing Surgical Case Scheduling (SIU) HL7v2 interfaces can push real-time surgery progress, surgeon, surgical staff, and anesthesia staff information to your app. The outgoing messages can be configured to send at various key surgical events like Procedure Start or Out of Room.

If your app will be the source of truth for surgery progress, your app can use the Incoming Surgical Case Tracking HL7v2 Interface to write key surgical events to the patient chart. Note this will need to be used in conjunction with the Outgoing Surgical Case Scheduling HL7v2 interface.

### Surgical History

Your app can pull information about a patient’s historical surgical procedures using the Procedure (Surgical History) FHIR Resource.

### Surgical Consent

Your app can pull information about a patient’s surgical consents using the Consent FHIR Resource.

### Orders

Your app can pull pre-operative, intra-operative, or post-operative order information via the ServiceRequest (Orders) FHIR resource. You can use the category and encounter fields to identify these orders. Your customers may need to do additional setup to include all types of orders that your app may care about in ServiceRequest.

### Labs

Your app can pull pre-operative, intra-operative, or post-operative lab information via the DiagnosticReport (Results) FHIR resource. Those resources may contain component-level lab results, which your app can pull via the Observation (Labs) FHIR resource.

### Imaging

Your app can pull pre-op, intra-operative, or post-operative diagnostic imaging report information via the DiagnosticReport (Results) FHIR resource which will contain references to Observation or Binary FHIR resources for impression and narrative results.

### Notes

Your app can pull note information via the DocumentReference (Clinical Notes) FHIR resource and text information via the Binary (Clinical Notes) FHIR resource. Work with your customers to understand the note types their users create for documents like H&Ps, Operative Notes, and Anesthesia notes.

Work with your customers to determine if they may store discrete information in SmartData Elements on their perioperative notes. If you need to pull particular SmartData Elements on a particular type of note, work with your customers to identify these scenarios and use the Observation (SmartData) FHIR resource to pull this information into your app.

Your app can write PDF or RTF note information via the Incoming Transcriptions HL7v2 interface or plain text note information via the DocumentReference FHIR API. Work with your customers to determine how their users will incorporate your note into their existing workflows.

### Vitals

Your app can pull vitals information via Observation (Vitals) FHIR resource. Preop and postop vitals are often taken by nursing staff. Intraoperative vitals often come from vitals monitor devices.

Device data flows to the patient chart via the Incoming Device Data HL7v2 interface in an intraoperative setting. If your app must read real time intraoperative vitals, many customers will have a way to split an incoming feed in their interface engine. Develop your app to consume the incoming HL7v2 feed.

If your app will write real time intraoperative vitals to the patient chart, use the incoming feed for large amounts of data or the Observation (Vitals) FHIR Resource for small amounts of data.

### Lines, Drains, Airways, and Wounds

Your app can pull pre-operative, intra-operative, or post-operative Lines, Drains, Airway, and Wound information via the Observation (LDA-W) FHIR resource.

Additionally, your app can create or update Lines, Drains, and Airway information via the Observation (LDA) FHIR resource.

### Medications

Your app can pull pre-op, intraoperative, and post-op medications that were given via the MedicationRequest (Orders) FHIR resource. This will return information like infusion start and stop times but not totals.

### Implants

Your app can pull information about implants via the Device (Implants) FHIR resource.

### Specimens

Your app can pull information about specimens that were taken during a surgery via the ServiceRequest (Orders) resource which will contain references to Specimen FHIR resources.

### Blood Products

Your app can pull blood products that were given via the DiagnosticReport (Results) FHIR resource. This will return information like start and stop times but not totals.

### Supply Usage

The Outgoing Inventory Depletion HL7v2 interface can send information about supplies documented as used in Epic to your app. The Incoming Materials Management HL7v2 interface can bring supply data into Epic from your app.

## After The Surgery

After the day of surgery, there is a window for clinicians to finalize their documentation and findings. Once the documentation is finalized, charges are posted and sent for billing. Work with your customer to understand how long it typically takes for them to finalize and post their surgeries.

### Outcomes

After the surgery documentation is finalized, your app can pull surgery information including outcomes information like complications from Procedure (Surgeries).

### Patient Location

Your app can pull information about a patient’s inpatient location after surgery via the Encounter FHIR resource. Additionally, the Outgoing Patient Administration HL7v2 interface can send your app information about patient movement after surgery such as admissions to inpatient, transfers, or discharges.

### Charges

We do not currently support industry standard ways of reading which charges were posted to a surgery or on a claim related to a surgery.

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# 2.42 Deploying Predictive Models Playbook

File: site/docs/02-playbooks/42-predictive-models-playbook.html

Source: https://fhir.epic.com/Documentation?docId=predictive_models_playbook

Epic offers several methods for with connecting predictive analytics models with user workflows. Customers licensed for Epic's can deploy your model such that it appears natively in Epic workflows. You can provide your model directly to Epic customers who license Epic's . These customers will map inputs from their Epic system to your models, and either upload the models to their cloud instance or import them to their operational database. Either way, the models look just like the Epic-provided models they're used to seeing. Customers should contact their Epic representative to determine if they have the Cognitive Computing Developer Platform licensed, and to confirm their readiness for custom cloud-based models. Nebula is Epic's managed cloud platform, one of two options for deploying models with the Epic Cognitive Computing Developer Platform. When models are mapped, uploaded, and turned on, customers' Epic systems will securely transport inputs to our managed cloud platform, where the complex calculations take place. Cloud calculated scores are delivered back to the customer's local instance of Epic and are displayed to end users. This option works well with models created in tools like Pytorch or XGBoost, and has the most processing power suitable for complex models. Any Python package that can be used for modelling can be installed and accessed for real-time calculation. Nebula and the Cognitive Computing Platform is the most full-featured option for providing predictive models to customers. Your data scientist can implement a variety of algorithms using standard python packages, and customers' users benefit from the clear score displays in Epic. Our native visualization of contributing factors helps users identify which factors are most significantly impacting a score. Nebula also provides the most processing power for complicated algorithms. For simpler models and for customers who don't have Nebula, you can export your model as a PMML file for customers to import. They'll import the model and map inputs from their Epic system to allow your model to run. The PMML option also requires Epic's Cognitive Computing Developer Platform. We recommend using Nebula instead of PMML. The PMML option is limited to file sizes of less than 4MB, and supports Naive Bayes, decision tree, decision forest, and certain regression algorithms. For Epic customers without the Cognitive Computing Platform, model vendors will need to handle calculations and filing scores into Epic. This option is the least helpful to clinical users, as it's difficult to see the factors that went into the final score, making it difficult for users to validate and trust scores. Using this option, you will gather inputs needed for your model to calculate the scores using FHIR APIs, interfaces, or other methods. You can then calculate the model score, and file the model result using an API. Often, an Observation.Create API is used as the mechanism to file data to flowsheets. Starting in the November 2026 version of Epic, you can use the ; prior to that version you can use . From there, customers can configure their system to display your model’s result where they would like to see it in their workflow, such as a column or configuring alerts to trigger.

## Deploying Predictive Models to Epic Provider Systems

Cognitive Computing Platform

### Epic's Cognitive Computing Platform

Cognitive Computing Developer Platform

### Nebula

### PMML

### Limitations

### Filing scores using FHIR APIs

Flowsheets sub-resource

Vital Signs sub-resource

## Get Started

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# 2.43 Clinical Decision Support Playbook

File: site/docs/02-playbooks/43-cds-playbook.html

Source: https://fhir.epic.com/Documentation?docId=cds_playbook

Epic uses CDS Hooks, an HL7 standard, for in-workflow, real-time, dynamic, provider-facing decision support provided by external decision support services.

### Applicable Use Cases

CDS Hooks should be used only to trigger real-time, end user-facing decision support. Epic recommends against using this feature for other purposes. If you need only a notification, we instead recommend using an event-based interface.

### Developer Considerations

CDS Hooks is a powerful integration technology because it directly interacts with clinician workflows. As a CDS developer, you share responsibility for a positive user experience. Your CDS service must be fast, avoid alert fatigue, and improve over time. Review our Designing an Efficient Application tutorial for additional performance recommendations. Make sure to incorporate CDS Hooks Feedback into your processes in order to monitor acceptance of your decision support suggestions and to continuously improve over time.

### Workflow Overview

Epic's support for CDS Hooks is built on a workflow engine and CDS rule engine. Organizations using Epic can configure native CDS alerts by defining the workflow point at which they are triggered, known as the hook, and the criteria to evaluate when they are triggered. Epic's implementation of CDS Hooks enables native criteria to call a remote CDS Hooks API when evaluating whether to show a CDS alert. When a hook is triggered and criteria are met, a CDS Hooks call is made to your endpoint with FHIR Prefetch data, both of which are configured by customer analysts when configuring your connection. Your CDS service evaluates context, makes additional FHIR API calls when necessary, and returns a CDS Hooks card response, which can include textual information, suggest discrete actions to be taken, or a link to your SMART app, all of which are embedded directly in the provider's workflow.

### Technology Details

For step-by-step details of a CDS Hooks connection, start with our CDS Hooks Developer Tutorial. For API-specific details of Epic's implementation of the CDS Hooks standard and supported APIs, see our Epic's CDS Hooks Support specification and search for CDS Hooks on the Epic on FHIR Specifications page.

### Get Started

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## 2.44 ED Pre-Arrival EMS Playbook

File: site/docs/02-playbooks/44-ed-pre-arrival-ems-playbook.html

Source: https://fhir.epic.com/Documentation?docId=ed_pre_arrival_ems_playbook

Emergency Medical Services (EMS) integrations help emergency departments prepare for incoming patients by exchanging identity, status, and clinical information before arrival. A common starting point is receiving a PDF "run sheet" for clinical review; more advanced connections add real-time ED "expect" notifications, discrete prehospital vitals, and continuous ETA updates so teams can allocate rooms, resources, and clinicians ahead of time.

This playbook outlines recommended practices using public interoperability standards (HL7® v2, FHIR®, SMART on FHIR) so EMS platforms can connect with healthcare provider organizations that use Epic.

### Identity Matching

Use the FHIR Patient.$match (demographics) (R4) API to search for an existing high confidence patient match and retrieve the hospital MRN. If no match is returned, maintain the patient identity in your EMS system and plan to reconcile the patient information between systems after arrival—once the patient and Contact Serial Number (CSN) for the encounter has been created in Epic. CSN (is the unique identifier Epic assigns to each patient encounter and once created it does not change. See fhir.epic more information on patient matching and ID Types in FHIR APIs.

### Pre-Arrival Notification

During transport, send HL7 v2 ADT A05 Incoming Patient Administration messages to create an expected ED encounter. When sending these messages, include PID demographics, ZAM-16 populated with your EMS Run ID, and a brief NTE that captures the chief complaint or mechanism of injury if available. Upon receiving a successful message, Epic will automatically generate the Epic CSN. To receive the Epic CSN value, process the associated outgoing (from Epic) ADT message for the patient encounter creation.

### Encounter Linking

If a pre-arrival notification is not triggered from the EMS system, allow the health system to have already created the ED encounter through its user-based registration/ADT workflows and link to it in your system using the Epic CSN. Creating the ED encounter will trigger an outgoing ADT message which should be consumed from the Outgoing Patient Administration interface. After the CSN exists, include both the EMS Run ID and the CSN in all subsequent messages, and consume ADT A08 messages (demographic updates) and ADT A40 (merges) messages to keep your copy of the patient data synchronized with their system. ADT A31 messages will represent patient demographic updates and can be used to keep demographic information up-to-date in your (EMS) system.

### Filing Vitals and Assessments

Transmit transport vitals and assessments via ORU^R01 messages using the Incoming Clinical Flowsheet Data interface as they are documented in your EMS system, ensuring OBR-7 reflects the collection time and OBX-14 is present for each measurement with units; as an alternative, it is acceptable to use FHIR Observation.Create (Vital Signs) (R4) resources to deliver the same content. When using FHIR to file vitals you will need to work with the corresponding health system to obtain the FHIR IDs for the relevant flowsheets during implementation.

### Documents

File run sheets, ECGs, and images, to the health system or the health system's document management system (DMS) using an MDM^T02 Incoming Scanned Document Link interface. You can send these assets after the health system assigns a Contact Serial Number (CSN) encounter identifier; if CSN is not yet available, queue the document messages and transmit them one once the CSN is available. Use descriptive titles such as 'EMS Run Sheet – Unit 42' to aid in clinical review of the assets.

### Post Arrival Feedback

Allow the health system to return outcome data to the EMS system by consuming messages such as A08, A31, and A03, from Outgoing Patient Administration ADT interface for diagnoses, disposition, and the deceased indicator.

### Identifier Patterns

Dual ID: Carry both the EMS Run ID and the CSN in all inbound and outbound messages to maintain reliable matching across systems and to simplify downstream troubleshooting.

Single ID: For workflows that rely solely on the CSN, hold any messages that require the encounter identifier—such as document links—until the patient is banded, and then send those messages using the CSN only.

### Operational Notes

Security: Use Backend OAuth for server-to-server FHIR access and network configuration, such as VPNs and adding IPs to an allowlist for connecting to healthcare organizations via an interface. For more information about interfaces and connecting to healthcare organizations via an interface, see the Interfaces (HL7v2) article.

Testing: Engage the hospital's interfaces team and include the EMS feeds in interface readiness testing as well as clinical mapped record testing to confirm correct filing and display.

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## 2.45 Specialty Registries Playbook

File: site/docs/02-playbooks/45-registries-playbook.html

Source: https://fhir.epic.com/Documentation?docId=registries_playbook

Specialty registries are used to track and analyze defined patient populations with specific conditions or procedures to improve care quality and patient outcomes or for research. This Playbook outlines the most common workflows used to extract data from Epic so that you can connect to a specialty registry or even build and maintain your own registry.

### Getting Started

There are a few things to consider when building a specialty registry:

* Some data may reside in another system of record, outside of Epic (e.g. diagnostic images). To access data that does not reside in Epic, you may need to connect to other products.

* There may be data that exists in Epic but is not discretely documented. You may need to pull non-discrete data from Epic, such as notes, and then abstract the desired data from the note into discrete values that conform to the value set defined by the registry.

* Customer workflows and configuration can vary so your data extraction methods may need to be flexible. For example, customers could capture data in a custom flowsheet row that needs to be mapped in your solution.

## Cohort and Trigger Definition

The first step when creating a specialty registry is identifying the target cohort. Start by writing a clear definition, such as “all patients with an active diabetes diagnosis” or “patients between the ages of 18 and 40 scheduled for orthopedic surgery.” Then, go a level deeper, for example, patients with specific ICD-10 codes, defining relevant time ranges. Depending on the data extraction method, you may need to deliver these details to the customer. Once you have a clear population definition, evaluate whether your criteria are triggered by real-time workflow events such as supported HL7v2 interfaces (e.g. new orders placed), or if the registry is looking for past data (e.g., all patients with a specific condition that may have been entered years ago). Additionally, determine what will trigger you to collect new data. This could be a workflow event like “annual diabetic check-up”, or you may decide to pull data on the patient population one time each quarter.

## Data Extraction Methods

After defining the registry cohort, the next step is pulling the data that represents the relevant clinical event or timeframe. This could be done using various technologies, including SQL , event-based triggers, and bulk FHIR.

### SQL Reporting

Epic provides a myriad of reporting tools that can often double as a way to extract information for third party integrations. This can be a good option if you can define the patient population in a way that can be translated into an Epic report. Depending on the extraction tool used, these reports can be run anywhere from daily to as frequently as a couple times an hour. Reporting tools are a good choice for a broad array of use cases; however, it’s important to understand that it can be a lot of work for customers. You will need to provide customers with your inclusion criteria and the details of the patient data needed for the registry, and they will need to write a report to extract the corresponding data. Some customers have many business intelligence developers that can write queries to meet your needs, whereas other customers may not be able to prioritize these needs for some time. Optionally, you can choose to take on the report complexity, licensing Kit or Clarity through Vendor Services, and writing the SQL query to connect to customer environments and minimize the work each customer needs to complete.

### Event-Based Interfaces and Additional APIs

HL7 and other standards-based interfaces can be used to trigger data extraction when certain events occur in Epic. These interfaces can send alerts for events such as admissions, discharges, or orders, as well as future events like scheduled appointments or surgeries. Using these messages as triggers allows you to know when to pull relevant data for specific patients. For example, if the defined cohort includes patients recovering from surgery, you can use the surgery scheduling event as a trigger. Once that event occurs, you can use FHIR APIs to retrieve the relevant clinical information such as lab results, observations, and encounters. Event-based interfaces are most effective when your use case depends on real-time or near-real-time data. They are less suitable for condition-based registries that do not match to a triggered workflow event, such as those defined by diagnoses alone. If your use case requires timely updates tied to clinical activity, interfaces can be a good trigger method, complemented by FHIR queries to read supplemental data.

### Bulk FHIR

Bulk FHIR is another option for extracting registry data, particularly when you need to pull information for a defined cohort on a set schedule (e.g. every two weeks for pediatric patients with type 1 diabetes). Using Bulk FHIR requires configuration from the health system, as they will need to create an analytics registry around specific inclusion criteria, within Epic. Once the registry is in place, you can use Bulk FHIR to query data for that cohort at regular intervals, in this case retrieving the most recent blood-sugar results. Bulk FHIR can be a powerful tool, but it’s not intended for particularly large-scale, nor high-frequency extractions. We recommend keeping the cohort size around 1,000 patients or fewer to ensure good performance. For detailed setup steps and guidance on how to optimize your queries, refer to the Bulk FHIR Tutorial and accompanying best practice recommendations.

## Get Started

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# 2.46 Validating AI Features Playbook

File: site/docs/02-playbooks/46-validating-ai-features-playbook.html

Source: https://fhir.epic.com/Documentation?docId=validating_ai_features_playbook

Validating AI features is essential to ensuring that models are used appropriately, provide measurable value, and operate fairly across patient populations. While many health systems have governance processes in place, meaningful validation requires testing AI performance on local patient populations and workflows, as these can vary widely across care settings. Validation should also be ongoing rather than a one-time assessment, with continuous monitoring for accuracy and equity over time.

This playbook highlights how app developers can use Epic's open-source, Python-based AI Trust and Assurance Suite to support standardized, repeatable evaluation of your AI models.

## Connecting to the Data

To validate your AI feature, you'll need access to the underlying data that reflects model inputs, real-world outcomes, and performance between relevant subgroups. Depending on your use case and the customer's preference, this data can be provided through SQL-based reports or retrieved directly from Epic using FHIR APIs.

### SQL

SQL can be used to extract the relevant data for validating the AI feature. You will need to provide customers with a list of data elements required to validate the AI feature; from there, they will need to write a report to extract the corresponding data. Once the report is complete, they will share the resulting dataset with you for use in your validation workflow.

Customers have varying levels of resources that they will be able to devote to writing these reports. Some customers may have several business intelligence developers to write the requested queries, while other customers may not be able to prioritize these needs for some time. Optionally, you can choose to take on the report complexity, licensing Kit or Clarity through Vendor Services, and writing the SQL query to connect to customer environments and minimize the work each customer needs to complete.

### FHIR

For specialized models that require smaller amounts of data, you may choose to retrieve the necessary validation data directly from Epic using FHIR APIs. You can query specific resources such as Observations, Conditions, or Encounters, depending on what your AI feature requires.

## Setup and Installation

Once you have extracted the necessary data from Epic and are storing it in your system, you can get set up with the Trust and Assurance Suite by following our installation instructions. there are many ways to customize the Seismometer notebook, including creating configuration files, creating a data dictionary, creating usage configuration, and more. For more information, refer to our Customizing the Notebook topic.

## Using the Tools

Most health systems will expect you to conduct in-depth review to assess model performance, fairness, drift, and associated impact or outcomes. To conduct this review, you can utilize Seismometer for aggregating line-level metrics into healthcare-specific analysis, along with evaluation instruments that include healthcare-specific LLMs as a judge. For more information on the available tools and recommended workflows, check out the Evaluation Instruments page.

## Workflow Considerations

Local organizational practices and workflows can heavily influence real-world performance of AI products. You should work directly with the customer on their local intake and validation processes to optimize workflow integration.

## Get Started

If you're using FHIR, check out our step-by-step Developer Guide. Are you interested in using kit or Clarity, or looking for a little extra help? Vendor Services offers an optional suite of benefits including the ability to request individualized assistance with your app's data exchange. Follow the steps above to install and use the AI Trust and Assurance Suite. If you create guidance on how to extract data from your system for use with the Trust and Assurance Suite, and would like to publish it with our materials, feel free to contact us. In addition, some product categories qualify for Toolbox support and designation on Showroom, which can accelerate implementation and enable you to highlight your product's strengths. Explore the list of eligible categories and share your interest with us through the Technology Guidance form.

# 2.47 CMS-0057 Prior Authorization Playbook

File: site/docs/02-playbooks/47-cms-prior-auth-playbook.html

Source: https://fhir.epic.com/Documentation?docId=cms_prior_auth_playbook

## CMS-0057 Prior Authorization APIs

We have released a comprehensive suite of APIs in support of the CMS-0057 regulation and its goal of modernizing prior authorization processes. These APIs enable real-time, electronic exchange of prior authorization requests and decisions, reducing administrative burden and accelerating access to care for patients.

Prior authorization is a process used by payers to determine whether a requested healthcare service, procedure, or medication meets coverage requirements before the service is delivered. Traditionally, this process has relied on manual workflows including phone calls, faxes, and portal submissions, often resulting in delays, incomplete documentation, and extended turnaround times.

Improving prior authorization efficiency benefits both healthcare systems and payers by reducing administrative costs, minimizing rework, and streamlining utilization management workflows. For patients, faster authorization decisions mean reduced delays in accessing necessary care, fewer appointment cancellations, and improved health outcomes. The CMS-0057 regulation mandates that payers respond to expedited authorization requests within 72 hours and standard requests within 7 calendar days, making automated workflows essential for compliance.

## The Three Components of Electronic Prior Authorization

Prior authorization using FHIR APIs is built on three integrated Da Vinci implementation guides that work together to create an end-to-end electronic workflow:

### Coverage Requirements Discovery (CRD)

CRD enables providers to receive real-time coverage information directly within their EHR workflow. When an order is placed or an appointment is booked, CRD automatically checks with the payer to determine whether prior authorization is required, what documentation is needed, and whether there are any coverage restrictions. This proactive approach helps providers understand authorization requirements before submitting a request, reducing denials and rework.

Our CRD Response Formatting Guide can be found here.

### Documentation Templates and Rules (DTR)

DTR streamlines the collection of required documentation by presenting dynamic, payer-specific questionnaires directly within the EHR. DTR uses embedded clinical logic to ask context-specific questions, ensuring that providers supply complete and accurate documentation the first time. Once documentation is complete, the EHR compiles this clinical information into an authorization request.

### Prior Authorization Support (PAS)

PAS automates the electronic submission and tracking of prior authorization requests. The payer system confirms receipt and returns either an approval with an authorization number, a denial with specific reasons, or a pended status requesting additional information. Providers can track authorization status in real-time without phone calls or portal logins.

## Accessing Epic Prior Authorization APIs

Epic's prior authorization APIs are available through both Vendor Services and Epic on FHIR via the self-service app creation process.

Epic community members can register their applications on either site to access these APIs and begin integration with Epic customers. Non-community members can join Vendor Services to create applications, receive general technical support, and access limited connectivity testing. Epic on FHIR offers free app creation services and API access to all developers.

Prior Authorization API functionality is released in the February 2026 version of Epic. We recommend working closely with health systems/health plans to ensure alignment on configuration, testing timelines, and go-live readiness.

The APIs were constructed around the Da Vinci 2.1 FHIR IG, though it does not follow it exactly. For more information, please view individual APIs on the "API Specifications" tab of Epic on FHIR or Vendor Services. Our documentation will continue to evolve and upgrade to provide clarifying information as we adapt to this space.

Note: Some APIs have recently changed names - this will not affect functionality. This list should reflect the most updated API names

### I am a...

### Health System Connecting with an Epic Payer

Example: I'm a health system sending CRD, DTR, and PAS requests to a payer that runs on Epic.

Copy Template App ->

One click creates a copy of this app with the context, use case, and all scopes below already selected.

Create an application with a "Backend Systems" context, select the Use Case of "CMS Prior Auth", and add the following Incoming and Outgoing scopes:

CRD

Incoming:

* Coverage Requirements Discovery (R4)

* Coverage Requirements Discovery Endpoint Discovery (R4)

Outgoing:

* CRD Context Resource

* CRD Prefetch Resource

* CRD System Action

DTR

Incoming:

* Questionnaire.$questionnaire-package (R4)

* Questionnaire.$next-question (R4)

* ValueSet.$expand (DTR Payer Guidelines) (R4)

* Questionnaire.$log-questionnaire-errors (R4)

PAS

Incoming:

* Claim.$submit (Prior Auth) (R4)

* Claim.$inquire (Prior Auth) (R4)

* $submit-attachment (Prior Auth) (R4)

Outgoing:

* PAS Outgoing Binary Fetch

* PAS Outgoing Claim Response Notification

* PAS Claim Response Submit

-> Final step: Share this Client ID with the Epic payer so they can download the app in Vendor Services.

Note: If you're making an Epic-to-Epic connection without Payer Platform, you'll also need to download a Client ID from the Epic payer.

### Payer System Connecting with an Epic Provider

Example: I'm a payer responding to CRD, DTR, and PAS requests that come from a health system running on Epic.

Application #1: Clinicians, Staff, or Administrative Users app

Copy Template App #1 ->

One click creates a copy of this app with the context, use case, and all scopes below already selected. Both apps are required.

Create an application with a "Clinicians, Staff, or Administrative Users" context, select the Use Case of "CMS Prior Auth", and add the following Incoming and Outgoing scopes:

CRD

Required:

Incoming:

* ServiceRequest.Update (Prior Auth) (R4)

* MedicationRequest.Update (Prior Auth) (R4)

* ServiceRequest.Read (Prior Auth) (R4)

* ServiceRequest.Search (Prior Auth) (R4)

* Appointment.Read (Prior Auth) (R4)

* Appointment.Search (Prior Auth)

* MedicationRequest.Read (Prior Auth) (R4)

* MedicationRequest.Search (Prior Auth) (R4)

* Coverage.Read (Patient Insurance Information) (R4)

* Coverage.Search (Patient Insurance Information) (R4)

* Patient.Read (Demographics) (R4)

* Patient.Search (Demographics) (R4)

Outgoing:

* CRD Request

* CDS Hooks Framework

Additional scopes to consider (will vary based on your implementation and the resources your CRD service needs to access):

Incoming:

* Encounter.Read (Patient Chart) (R4)

* Encounter.Search (Patient Chart) (R4)

* Location.Read (Organizational Directory) (R4)

* Location.Search (Organizational Directory) (R4)

* Practitioner.Read (Organizational Directory) (R4)

* Practitioner.Search (Organizational Directory) (R4)

* PractitionerRole.Read (Organizational Directory) (R4)

* PractitionerRole.Search (Organizational Directory) (R4)

* Organization.Read (Organizational Directory) (R4)

* Organization.Search (Organizational Directory) (R4)

Review the complete list of available scopes to determine which additional resources your implementation requires.

Application #2: Backend Systems App

Copy Template App #2 ->

One click creates a copy of this app with the context, use case, and all scopes below already selected. Both apps are required.

Create an application with a "Backend Systems" context, select the Use Case of "CMS Prior Auth", and add the following Incoming and Outgoing scopes:

DTR

Outgoing:

* DTR Outgoing Questionnaire.$questionnaire-package

* DTR Outgoing ValueSet.$expand

* DTR Outgoing Questionnaire.$next-question

* DTR Outgoing Questionnaire.$log-questionnaire-errors

PAS

Incoming:

* PAS Incoming Claim Response Notification

Outgoing:

* PAS Outgoing Claim.$submit

* PAS Outgoing Claim.$inquire

* PAS Outgoing Parameters.$submit-attachment

-> Final step: Share both Client IDs with the Epic health system so they can download the apps in Vendor Services.

Note: If you're making an Epic-to-Epic connection without Payer Platform, you'll also need to download Client IDs from the Epic health system.

### Delegate for CRD/DTR/PAS Connecting with an Epic Payer

Example: I run DTR on behalf of a payer that uses Epic, so I answer the DTR requests that come to me under that delegation arrangement.

If your organization handles CRD, DTR, or PAS on behalf of an Epic payer under a delegation arrangement, you'll connect using the same applications and scopes as a payer system connecting with an Epic provider, since you're acting as the payer-side responder for the delegated component(s).

Application #1: Clinicians, Staff, or Administrative Users app

Copy Template App #1 ->

One click creates a copy of this app with the context, use case, and all scopes below already selected. Both apps are required.

Create an application with a "Clinicians, Staff, or Administrative Users" context, select the Use Case of "CMS Prior Auth", and add the following Incoming and Outgoing scopes:

CRD

Required:

Incoming:

* ServiceRequest.Update (Prior Auth) (R4)

* MedicationRequest.Update (Prior Auth) (R4)

* ServiceRequest.Read (Prior Auth) (R4)

* ServiceRequest.Search (Prior Auth) (R4)

* Appointment.Read (Prior Auth) (R4)

* Appointment.Search (Prior Auth)

* MedicationRequest.Read (Prior Auth) (R4)

* MedicationRequest.Search (Prior Auth) (R4)

* Coverage.Read (Patient Insurance Information) (R4)

* Coverage.Search (Patient Insurance Information) (R4)

* Patient.Read (Demographics) (R4)

* Patient.Search (Demographics) (R4)

Outgoing:

* CRD Request

* CDS Hooks Framework

Additional scopes to consider (will vary based on your implementation and the resources your CRD service needs to access):

Incoming:

* Encounter.Read (Patient Chart) (R4)

* Encounter.Search (Patient Chart) (R4)

* Location.Read (Organizational Directory) (R4)

* Location.Search (Organizational Directory) (R4)

* Practitioner.Read (Organizational Directory) (R4)

* Practitioner.Search (Organizational Directory) (R4)

* PractitionerRole.Read (Organizational Directory) (R4)

* PractitionerRole.Search (Organizational Directory) (R4)

* Organization.Read (Organizational Directory) (R4)

* Organization.Search (Organizational Directory) (R4)

Review the complete list of available scopes to determine which additional resources your implementation requires.

Application #2: Backend Systems App

Copy Template App #2 ->

One click creates a copy of this app with the context, use case, and all scopes below already selected. Both apps are required.

Create an application with a "Backend Systems" context, select the Use Case of "CMS Prior Auth", and add the following Incoming and Outgoing scopes:

DTR

Outgoing:

* DTR Outgoing Questionnaire.$questionnaire-package

* DTR Outgoing ValueSet.$expand

* DTR Outgoing Questionnaire.$next-question

* DTR Outgoing Questionnaire.$log-questionnaire-errors

PAS

Incoming:

* PAS Incoming Claim Response Notification

Outgoing:

* PAS Outgoing Claim.$submit

* PAS Outgoing Claim.$inquire

* PAS Outgoing Parameters.$submit-attachment

-> Final step: Share both Client IDs with the Epic payer so they can download the apps in Vendor Services.

Note: If you're making an Epic-to-Epic connection without Payer Platform, you'll also need to download Client IDs from the Epic payer. You should also confirm with the delegating payer which of CRD, DTR, and PAS you are responsible for under the delegation arrangement.

### Payer Connecting with an Epic Delegate

Example: I'm a payer that has delegated PAS to an organization running Epic, so I send prior authorization requests along to them.

Copy Template App ->

One click creates a copy of this app with the context, use case, and all scopes below already selected.

Create an application with a "Backend Systems" context, select the Use Case of "CMS Prior Auth", and add the following Incoming and Outgoing scopes:

CRD

Incoming:

* Coverage Requirements Discovery (R4)

* Coverage Requirements Discovery Endpoint Discovery (R4)

Outgoing:

* CRD Context Resource

* CRD Prefetch Resource

* CRD System Action

DTR

Incoming:

* Questionnaire.$questionnaire-package (R4)

* Questionnaire.$next-question (R4)

* ValueSet.$expand (DTR Payer Guidelines) (R4)

* Questionnaire.$log-questionnaire-errors (R4)

PAS

Incoming:

* Claim.$submit (Prior Auth) (R4)

* Claim.$inquire (Prior Auth) (R4)

* $submit-attachment (Prior Auth) (R4)

Outgoing:

* PAS Outgoing Binary Fetch

* PAS Outgoing Claim Response Notification

* PAS Claim Response Submit

-> Final step: Share this Client ID with the Epic delegate so they can download the app in Vendor Services.

Note: If you're making an Epic-to-Epic connection without Payer Platform, you'll also need to download a Client ID from the Epic delegate.

### Additional Settings

These are our suggested selections for the additional settings.

Does your app use OAuth 2.0?

Select "Uses OAuth 2.0"

SMART on FHIR Version

Select "R4"

SMART Scope Version

SMART scopes are returned in the scope parameter of the OAuth 2.0 token endpoint response and determine which resources an application has permission to access and what actions the application is permitted to take. The scopes provided are based on the scopes that your app requested in your authorize request as well as the incoming APIs you have selected on your app page.

The actions defined by SMART v1 scopes are .read and .write. SMART v2 scopes use CRUDS syntax and actions include: .c – create, .r – read, .u – update, .d – delete, and .s – search.

For example, the Observation.Create resource corresponds to either:

* SMART v1: {user-type}/Observation.write

* SMART v2: {user-type}/Observation.c

Which version should you select?

If your app uses SMART v1 scopes or if your app does not use SMART scopes at all, you should select SMART v1.

If your app uses SMART v2 scopes, you should select SMART v2.

Endpoint URI

For apps with a user-facing context ("Clinicians, Staff, or Administrative Users"), you can enter your FHIR base URL. For apps with a backend context, this field can be left blank.

### Get Started

These connection practices have enabled many successful connections across our community. As every product has unique goals, we encourage you to review this approach with your customers to confirm the best fit. If you're just getting started, check out our step-by-step Developer Guide. Want a little extra help? Vendor Services offers an optional suite of benefits including the ability to request individualized assistance with your app's data exchange. In addition, some product categories qualify for Toolbox support and designation on Showroom, which can accelerate implementation and enable you to highlight your product's strengths. Explore the list of eligible categories and share your interest with us through the Technology Guidance form.

## 2.48 CMS-0057 Access Playbook

File: site/docs/02-playbooks/48-cms-access-playbook.html

Source: https://fhir.epic.com/Documentation?docId=cms_access_playbook

Note. The “Outside Record” subresources return data that payers receive by data exchange. For a given data type, it is recommended to select “Outside Record” because it is most likely to return data. Note. The “Outside Record” subresources

return data that payers receive by data exchange. For a given data type, it is recommended to select “Outside Record” because it is most likely to return data. Note. The “Outside Record” subresources return data that payers receive by data exchange. For a given data type, it is recommended to select “Outside Record” because it is most likely to return data.

We have released 70+ APIs to support the CMS-0057 regulation and its goal of improving interoperability between payers, providers, and patients. These APIs enable secure, standardized access to payer-held data that is required for patient access, provider access, and payer-to-payer data exchange.

These APIs are designed to simplify how payer information is exchanged, making coverage and prior authorization processes more transparent and predictable while helping support timely access to care. Standardized APIs allow payer data to be exchanged electronically and in real time, reducing reliance on manual processes.

These capabilities also support patient access by enabling individuals to electronically access their own health and coverage information through their application of choice. Tapestry customers with Autodownloads turned on will automatically connect with patient access applications. Patients just need to authenticate with MyChart and then they will be able to access their own data.

For Epic customers, these APIs provide a scalable, standards-based way to exchange payer data without requiring payer-specific custom integrations. Customers benefit from fewer manual workflows, improved patient transparency, and an interoperability model that aligns with current and future federal requirements.

## Patient Access

Patient Access enables individuals to use patient-facing applications to electronically retrieve their payer-held data through a defined set of patient-facing FHIR APIs. These applications connect to Epic payer systems to access data that is available through the patient access APIs.

Patient-facing apps allow patients to review the data being requested, grant access to the app, and view their own data. Patients can revoke that access at any time, giving them ongoing control over how their information is shared.

### Automatic Client Distribution

Epic supports an auto-download capability for eligible patient-facing applications. When auto-download is enabled, qualifying apps are automatically made available to Epic payer environments without requiring manual customer configuration.

### If you are a patient-facing application connecting to an Epic payer to retrieve a patient's data on their behalf:

* Review the Patient-Facing Apps Using FHIR tutorial.
* Create an app on Vendor Services or Epic on FHIR.
* Select "Patients" as the primary user.
* Select "Enable Autodownload".
* Select the "CMS Patient Access API" use case.
* Scope the APIs that represent the data you would like to receive.
* Toggle on OAuth 2.0.

## Provider Access

Provider access supports system-to-system exchange of payer data. This allows providers to access payer-held data required for coverage determination, prior authorization workflows, and continuity of care.

### If you represent a Health System connecting with an Epic Payer.

* Create an app on Vendor Services or Epic on FHIR.
* Select "Backend Systems" as the primary user.
* Select the "CMS Provider Access API" use case.
* Check "Incoming APIs" and scope the following APIs:
* Bulk Data Delete Request
* Bulk Data File Request
* Bulk Data Kick-off
* Bulk Data Status Request
* Patient.Read (Demographics) (R4)
* Patient.Search (Demographics) (R4)
* Any other APIs that represent the data you want to receive.
* Toggle on OAuth 2.0.

### Retrieving Patient Data as a Provider

To retrieve patient data, contact the Epic payer you wish to connect to so that they can download and configure your application created above. Once configured, the Epic Payer will provide a FHIR Group ID (or IDs) representing the attributed patient population for your organization. Refer to the Bulk Data Kick-off specification for guidance on using this ID for a Bulk

FHIR $export call. Epic does not currently support $provider-member-match or $davinci-data-export operations for usage with the Provider Access API.

## Payer-to-Payer Access

Payer-to-payer access supports system-to-system exchange of payer data. This use case enables payers to share information with other payers.

### If you represent a non-Epic payer connecting with an Epic payer.

* Create an app on Vendor Services or Epic on FHIR.
* Select "Backend Systems" as the primary user.
* Select the "CMS Payer-to-Payer Access API" use case.
* Check "Incoming APIs".
* Scope all available APIs.
* Toggle on OAuth 2.0.

### Retrieving Patient Data as a Payer

To send or retrieve member data, contact the Epic payer you wish to connect to so that they can download and configure your application created above.

To retrieve member data, use the $bulk-member-match operation to determine a FHIR Group ID to use for the relevant member(s). Refer to the Bulk Data Kick-off specification for guidance on using this ID for a Bulk FHIR $export call. Epic does not currently support $davinci-data-export operations for usage with the Payer-to-Payer API.

To send member data, the Epic payer will perform a $bulk-member-match operation request to retrieve a FHIR Group ID to use in a $export operation request. Epic will not make $davinci-data-export operation requests for the Payer-to-Payer API.

## Testing

You can test the Patient, Provider, and Payer-to-Payer Access workflows in the Hyperdrive Client Testing Harness.

If you need a little extra help, Vendor Services offers an optional suite of benefits including the ability to request individualized assistance with your app's data exchange.

### Configuration

If you are enrolled in Vendor Services, open a Sherlock ticket with the following information to get connected to the Vendor Services sandbox:

* CMS Use Case
* Non-Production Client ID

### Bulk Member Match

Payer-to-Payer apps can test $bulk-member-match with the patient demographics below.

Name | Member ID | DOB | Legal Sex | Address | SSN | Phone Number
CMSFHIRONE, TEST | E11111 | 03/01/2000 | M | 123 Anywhere St, Verona WI 53593 | 444444444 | 9999999999
CMSFHIRTWO, TEST | E22222 | 03/01/1960 | F | 123 Anywhere St, Verona WI 53593 | 666666666 | 8888888888
CMSFHIRTHREE, TEST | E33333 | 04/01/1950 | F | 123 Anywhere St, Verona WI 53593 | 999999999 | 5555555555
CMSFHIRFOUR, TEST | E44444 | 06/03/1996 | U | 123 Anywhere St, Verona WI 53593 | 123459876 | 2222222222

### Clinical Data APIs

When testing the ExplanationOfBenefit and OutsideRecord subresource APIs, you can use the following test patients to retrieve the corresponding data:

Patient Name | Data Type
CMSFHIRONE, TEST | Authorizations, Claims
CMSFHIRTWO, TEST | Authorizations, Claims
CMSFHIRTHREE, TEST | Authorizations, Claims
CMSFHIRFOUR, TEST | Diagnoses, Encounters, Immunizations, Problems, Procedures, Providers

## Get Started

These connection practices have enabled many successful connections across our community. As every product has unique goals, we encourage you to review this approach with your customers to confirm the best fit. If you're just getting started, check out our step-by-step Developer Guide. In addition, some product categories qualify for Toolbox support and designation on Showroom, which can accelerate implementation and enable you to highlight your product’s strengths. Explore the list of eligible categories and share your interest with us through the Technology Guidance form.

# 2.49 Radiation Therapy Treatment Management System Playbook

File: site/docs/02-playbooks/49-radiation-therapy-tms.html
Source: https://fhir.epic.com/Documentation?docId=radiation_therapy_tms

## Radiation Therapy Treatment Management System Playbook

A radiation therapy Treatment Management System (TMS) is a central hub for patients' radiotherapy treatment plans and manages sending those plans to treatment modalities (e.g., linear accelerators and brachytherapy systems) and receiving and storing records of those treatments. A TMS can focus on plan management, safety, and tracking, leaving other tasks like scheduling, note-taking, and charging to the EHR. This Blueprint describes the recommended integration between a TMS and Epic for smooth workflows between the two systems.

## Getting information from Epic

* Use the Outgoing Patient Administration Interface (ADT) to receive patient information (MRNs, encounter IDs, demographics) from Epic.

* Use Patient.$match (Patient Match) (R4) to retrieve additional patient demographic information.

## Sending information to Epic

* Use Exchange of Radiotherapy Treatment Summaries (XRTS) FHIR APIs to report treatment planning and delivery for each patient from the TMS to Epic.

* Refer to our XRTS FHIR API Tutorial for additional details.

## Additional features

* If your product captures charges, use the Incoming Financial Transactions Interface to send dropped charges to Epic.

* If your product creates notes or other documents, use the Incoming Scanned Documents Interface to send these documents to Epic.

* If your product has user interface for clinicians, use SMART on FHIR EHR launch to enable single-sign-on behavior from Epic.

* If your product retrieves cancer staging information, use the Condition.Search (Problems) (R4) to retrieve cancer staging information from patient problem lists in Epic.

## Get Started

These connection practices have enabled many successful connections across our community. As every product has unique goals, we encourage you to review this approach with your customers to confirm the best fit. If you're just getting started, check out our step-by-step Developer Guide. Want a little extra help? Vendor Services offers an optional suite of benefits including the ability to request individualized assistance with your app's data exchange. In addition, some product categories qualify for Toolbox support and designation on Showroom, which can accelerate implementation and enable you to highlight your product's strengths. Explore the list of eligible categories and share your interest with us through the Technology Guidance form.

# 2.50 XRTS FHIR API Tutorial

File: site/docs/02-playbooks/50-xrts-fhir-tutorial.html

Source: https://fhir.epic.com/Documentation?docId=xrts_fhir_tutorial

## Overview

While some Treatment Management Systems(TMS) support limited data exchange using custom HL7 v2 interfaces for radiation treatments, the industry standard for this type of communication is now XRTS (Exchange of Radiotherapy Summaries), a FHIR implementation guide designed jointly by members of CodeX and IHE.

## FHIR Resource Structure

The key elements of XRTS are variations on three FHIR resources: BodyStructure, ServiceRequest, and Procedure.

The XRTS profile of BodyStructure, Radiotherapy Volume, is used to describe a volume of patient anatomy which needs to be specifically identified as part of radiotherapy planning or delivery. The other resources point to Radiotherapy Volume to identify patient-specific targets of radiation.

The core of the XRTS model is a representation of radiotherapy treatment with three levels of specificity.

* A plan is the most specific: a single series of treatments with fixed details (total dose, fractionation, technique, and volumes to be treated).

* A phase is a container for a plan, so that if a plan adaptation must occur, when the plan is replaced, continuity is maintained by making the original plan and adapted plan part of the same phase.

* A course is a container for one or more phases which the oncologist considers related, such as multiple lines of treatment for a single disease or several treatments for related targets, like metastatic sites.

Typically, a patient's radiotherapy regimen is captured in a single course and organized into one or more phases. In the simplest case, each phase is instantiated in just one plan (no change to dosing or fractionation, all fractions identical), but if plan adaptation is required, then the sequence of plans is collected within the phase that represents that one series of fractions over time.

These three levels of detail can be represented in three different stages of development:

* The prescribed stage is when an oncologist has declared an intention to treat a given patient with radiotherapy. At this stage, the course and phases may contain information about the disease and anatomy to be targeted, the expected treatment technique and fractionation, and so on.

* When a treatment plan is generated for the prescription, then the resources are in the planned stage. At this point, more definite detail is expected in the resources (dosing and fractionation, exact technique, etc.). A resource in the planned stage should contain a reference to the prescribed resource that it corresponds to (course to course, phase to phase; it is not expected that plan resources exist before the planned stage, though it is not prohibited).

* Once the first treatment starts, the resources are in the treated stage. Each treated resource carries information about what radiotherapy doses have actually been delivered, linking back to the planned resources corresponding to each.

In total, these six aspects work out into nine different representations:

| Prescribed | Planned | Treated
Course | Course Prescription | <- Planned Course | <- Course Summary
Phase | Phase Prescription | <- Planned Phase | <- Treated Phase
Plan | Plan Prescription | <- Treatment Plan | <- Treated Plan

The diagrams the FHIR implementation guide help to further clarify these relationships.

In FHIR, the prescribed and planned resources are instances of ServiceRequest, and the treated resources are instances of Procedure. The XRTS implementation guide describes in detail how variations on ServiceRequest and Procedure correctly represent each specific scenario and how Procedures and ServiceRequests may be reference one another because of their relationships. For instance, a Procedure may be based on a ServiceRequest (treatment as a fulfillment of a plan), and a ServiceRequest may be part of another ServiceRequest (a phase within a course). To indicate on each resource exactly what is meant, use the prescribed profile URLs and values for the category, code, and intent elements. These are the values that distinguish course, phase, and plan.

All XRTS APIs:

BodyStructure | Create | Read | Search | Update
Procedure | Create | Read | Search | Update
ServiceRequest | Create | Read | Search | Update

## Workflow for Data Exchange

### Basic Interactions

As part of XRTS, the resources listed above can be sent to Epic (FHIR create and update) and can then be retrieved from Epic (FHIR read and search). These resources include references to other resource types, like Patient, Practitioner (i.e., provider), and Location (e.g., department). These other resources will already exist within Epic, and you will receive them through the Outgoing Patient Administration Interface (ADT).

Use Patient.$match with shared demographic information (received through ADT) to get the patient's FHIR ID and additional patient demographics, if desired. Use Practitioner.Search with shared practitioner information to get the practitioner's FHIR ID and practitioner information. Use Location.Search with shared location information to get the location's FHIR ID and location information. We recommend that you cache these existing FHIR resource IDs instead of looking them up for every interaction, because they are almost always static and only change in rare circumstances (like merging duplicate patient records).

### Order of Operations

When a TMS first provides a set of resources to Epic, it's necessary to send the resources in the right order, since some are dependent on others because of reference chains. For example, a TMS must know the resource ID of the Patient before executing a BodyStructure.Create, since the BodyStructure resource must contain a FHIR reference to the patient whose anatomy is described. Likewise, a TMS must create the BodyStructure before a ServiceRequest or Procedure since each must contain a reference to the BodyStructure that is being treated.

With your prerequisite identifiers established, call BodyStructure.Create first. When the create interaction is successfully executed, Epic will return its resource ID to your app, which you can cache with the volume in your database and use for all FHIR references to that object.

When creating a course, whether prescribed, planned, or treated, the course resource must be sent first, followed by its phases, followed by its plans. Each plan contains a reference to the phase that it belongs to, and each phase contains a reference to the course that it belongs to, so the resource must exist in Epic and have a resource ID assigned before you construct and send a resource that references it.

Usually, a treated-type resource also points to the planned-type resource which it fulfills, which also works only if the planned resource exists in Epic and has been assigned a resource ID before the treated resource is sent. (For emergency treatment where the standard treatment planning process is not used, it is acceptable to send the treated resources with no planned resources. However, anytime planned resources exist, the treated resources must include references to the planned resources that they correspond to.)

For update interactions, the order of operations only matters if the updated resource contains references to new resources. For instance, you can update multiple phases and plans in any order if you are only changing dose values and fractionation. But you cannot update a course to include a new target volume before creating the BodyStructure resource in Epic.

### Expected Timing of Communication

Resources are generally expected to be sent to Epic approximately in real time after the relevant event occurs in the TMS:

* When an oncologist approves a prescription, create the target volumes, course, phases, and plans in Epic through XRTS so that the patient’s expected radiation is visible right away.

* When a treatment session is completed, create or update the treated course, phases, and plans in Epic through XRTS right away so that the record of the patient’s treatment progress is up to date and can be correctly included in, for example, notes that must be written for accreditation purposes.

* When a phase or course is completed, update the treated and planned resources with the appropriate status change so that clinicians can tell in Epic that treatment is ended.

# 2.51 Histocompatibility Laboratory Information System Playbook

File: site/docs/02-playbooks/51-histocompatibility-laboratory-information-system-playbook.html

Source: https://fhir.epic.com/Documentation?docId=histocompatibility_laboratory_information_system_playbook

Want a little extra help? offers an optional suite of benefits including the ability to request individualized assistance with your app's data exchange. In addition, some product categories qualify for support and designation on Showroom, which can accelerate implementation and enable you to highlight your product’s strengths. Explore the list of and share your interest with us through the .

These integrations support transplant programs by performing critical tests that ensure compatibility between organ or tissue donors and recipients and sharing test results with health systems. A health system will place an order for specimen, accension, or laboratory test which will then be performed by the lab. The lab then sends these results via HL7 interface to the health system. Any health system performing Solid Organ or Blood Marrow transplants will use Human Leukocyte Antigen (HLA) crossmatch labs tests.

## Design Elements

* Utilize the Outgoing Ancillary Orders interface to be notified when an HLA test is ordered at the health system

* Utilize the Incoming Ancillary Results interface to send back the result to the health system

### Vendor-Provided Values

You can fill out this table with the specific values you’ll need to provide to health systems. These will be referenced in the longer implementation guide below.

Reference ID | Description | Value
V1 | App non-production launch URL
V2 | App production launch URL
V3 | Vendor HTTPS Endpoint for receiving Outgoing Ancillary Orders messages or IP address (if using TCP/IP)

### Customer-Provided Values

Ask the customer to fill out this table with the values you will need for the install.

Reference ID | Description | Value
C1 | Customer endpoint for receiving Incoming Ancillary Orders interface or IP address (if using TCP/IP)

## Get Started

These connection practices have enabled many successful connections across our community. As every product has unique goals, we encourage you to review this approach with your customers to confirm the best fit. If you're just getting started, check out our step-by-step Developer Guide.

### Creating a Histocompatibility Laboratory Information System App

Copy Template App ->

One click creates a copy of this app with the context, use case, and all scopes below already selected.

Create an application with a Backend Systems context, select the Use Case of General and add the following Incoming scopes:

[Incoming/Outgoing/Provide Content/Kit/Subspace]

* HL7v2

Vendor Services

Toolbox

eligible categories

Technology Guidance form

# 2.52 Staff Duress Playbook

File: site/docs/02-playbooks/52-staff-duress-playbook.html

Source: https://fhir.epic.com/Documentation?docId=staff_duress_playbook

Both workflows begin with registering the device with Epic through the same Real-Time Location System (RTLS) badge mechanism. A badge worn by an individual is registered (associated) with a user. A fixed device is registered (associated) with a device record. Staff Duress supports messaging to keep badge (device representation) associations and disassociations for staff in sync between Epic and your application, so staff do not manage the association in more than one system. Sync associations and disassociations through the , either directly, or indirectly by sending the associations and disassociations to a customer's RTLS, which then relays them into Epic over the same interface. Obtain the user's current location from a source such as the customer's RTLS location provider. Epic is not the source of truth for a user's current location, so query the location system directly rather than reading it back from Epic.

Duress alerts let staff reach out for help during workplace violence or other safety emergencies, whether triggered from a badge worn by an individual or a fixed button mounted in a space. Staff Duress enables third-party panic button devices to send these location-aware alerts into Epic through FHIR-based APIs. When an alert is triggered, Epic identifies the user (if one is associated with the device) and the location, then notifies designated security response and informational groups through Epic's Push Notification framework. Designated responders can acknowledge or decline the alert from within Epic, while other staff who need situational awareness, such as a charge nurse, receive an informational notification.

### Badge Associations and Disassociations

Staff Duress supports two trigger workflows:

* From a badge worn by an individual, such as a nurse

* From a fixed device, such as a wall button, mounted in a patient room

HL7v3 Incoming and Outgoing Real-Time Location Services Interface

### Triggering a Duress Event

When a user triggers a duress event on their device, send the event to Epic using the FHIR Device.$auxiliary-function operation, including:

* The date and time the event occurred, as a timestamp

* An event type of staff-duress

* A device identifier for the user signaling the event

* A discrete, up-to-date location identifier

### Monitoring Alert Response

After sending a duress event, poll FHIR Task.Read every five seconds, using the FHIR ID returned from Device.$auxiliary-function, to check the push notification status for up to 15 minutes.

If every available responder declines, or no responder confirms within that window, escalate the event through another channel and stop polling Task.Read. Optionally, use the Task.Read status to notify the user under duress on your own device or application, for example when a responder accepts the event or when no staff are available to respond. This notification happens entirely within your application; no additional information is sent to Epic.

### Authentication

Authenticate FHIR API calls using Backend OAuth 2.0.

### Get Started

To ensure that your application is configured correctly, start with a copy of the Staff Duress App Template. These connection practices have enabled many successful connections across our community. As every product has unique goals, we encourage you to review this approach with your customers to confirm the best fit. If you're just getting started, check out our step-by-step Developer Guide. Want a little extra help? Vendor Services offers an optional suite of benefits including the ability to request individualized assistance with your app's data exchange. In addition, some product categories qualify for Toolbox support and designation on Showroom, which can accelerate implementation and enable you to highlight your product's strengths. Explore the list of eligible categories and share your interest with us through the Technology Guidance form.

Copy Template App ->

One click creates a copy of this app with the context, use case, and all scopes below already selected.

## 2.53 Diagnostic Image Exchange Playbook

File: site/docs/02-playbooks/53-diagnostic-image-exchange-playbook.html

Source: https://fhir.epic.com/Documentation?docId=diagnostic_image_exchange_playbook

### Diagnostic Image Exchange Playbook

When a patient arrives at one healthcare organization with imaging performed at another, clinicians often need the original images, not just the report. With Diagnostic Image Exchange, electronic health record (EHR) and image exchange vendors can retrieve full diagnostic-quality DICOM® images on demand from healthcare provider organizations that use Epic, and share their own images in return.

Each exchange has a requester and a responder. As the requester, your system finds the patient's imaging studies at the organization, requests the images it needs, and stores or displays them. As the responder, the organization's Epic system handles each request and retrieves the images from the organization's DICOM archive using standard DICOM, so your system never connects to that archive directly. When Epic is the requester, the roles reverse: Epic requests images from your system and stores them in the organization's DICOM archive.

This playbook outlines recommended practices using the following public interoperability standards: IHE XCPD and XCA, HL7® C-CDA, FHIR® R4, and DICOMweb™ WADO-RS.

## Before You Begin

Diagnostic Image Exchange builds on the existing treatment-based exchange of clinical records between your organization and a healthcare organization that uses Epic. Before you start, make sure that:

* You already exchange clinical documents (C-CDA) with the organization using IHE Cross-Community Access (XCA), for example through Carequality, TEFCA, or a point-to-point connection. Imaging studies are associated with the imaging orders and results shared in those documents. The FHIR and WADO-RS requests currently require a point-to-point connection with the organization, even if you also exchange documents through Carequality or TEFCA.

* You can link patients between your system and Epic using Cross-Community Patient Discovery (XCPD).

* Your requests are for treatment. Diagnostic Image Exchange isn't available for other purposes of use, such as payment or operations.

* Your system supports HL7 FHIR R4 and OAuth 2.0 backend (system-to-system) authorization.

* Your organization and the organization that uses Epic are in the same country. Diagnostic Image Exchange across country borders is not supported.

## Connecting with OAuth 2.0

Epic uses OAuth 2.0 to authorize both the FHIR ImagingStudy search and the WADO-RS image retrieval. Your system obtains an access token using the client credentials grant with a signed JSON Web Token (JWT) assertion, as described in Epic's Backend OAuth 2.0 guide. The organization that uses Epic registers your system as a client. To do that, they need the following from you:

* Public key used to verify your JWT signature, provided as either a JSON Web Key Set (JWKS) URL or a PEM-encoded RSA public key. A JWKS URL is strongly recommended and makes key rotation easier.

* FHIR R4 base URL, so Epic can request images from you. Supporting this is optional, but we encourage it so that images can follow patients in both directions. See "When Epic Requests Images from You" below.

* If you support requests from Epic, your OAuth 2.0 token URL and a client ID for the organization. Epic requests access tokens from you the same way you request them from Epic: the client credentials grant with a signed JWT.

* Technical contact for connection testing and troubleshooting.

In return, Epic gives you:

* A client ID. Use it as the iss and sub claims of your JWT.

* Their Interconnect base URL. Your token endpoint is <base>/oauth2/token and the FHIR R4 base URL is <base>/api/FHIR/R4.

Your JWT must also include the purpose of use for the request in a careeverywhere_context extension. Epic rejects requests without it. For Diagnostic Image Exchange, the purpose of use is always TREATMENT. An example JWT payload:

```
{
  "iss": "<client ID>",
  "sub": "<client ID>",
  "aud": "<base>/oauth2/token",
  "jti": "<unique value for each token request>",
  "exp": <no more than 5 minutes in the future>,
  "extensions": {
    "careeverywhere_context": {
      "PurposeOfUse": "TREATMENT"
    }
  }
}
```

## Recommended Workflow When Retrieving Images from an Organization Using Epic

Retrieving a patient's images from Epic takes four steps. Step 2 is optional, but recommended.

* Create a patient link. Send a Cross Gateway Patient Discovery (ITI-55) request as described in the Cross-Community Patient Discovery (XCPD) specification. When the patient is matched, keep the patient identifier the organization returns (its OID root and extension). You use this identifier in step 3.

* Retrieve the patient's clinical documents (optional, but recommended). Use Cross-Community Access (XCA) to query for and retrieve the patient's C-CDA documents, as you do today. These documents contain the imaging orders, results, and reports. Each imaging result includes a result reference ID that matches a secondary identifier on the corresponding ImagingStudy, so you can show each study with its report.

* Search the resource using the identifier from step 1, and include the WADO-RS Endpoint:

Get the list of available imaging studies.

ImagingStudy (R4)

```
GET <base>/api/FHIR/R4/ImagingStudy?patient:Patient.identifier=urn:oid:<root>|<extension>
    &_include=ImagingStudy:endpoint
```

Follow the Bundle's next links until all pages have been retrieved. Each ImagingStudy contains the Study Instance UID, its series and instances, and a reference to an Endpoint whose address is the WADO-RS base URL to use in step 4.

* For each instance you want, send a WADO-RS Retrieve Instance request as described in the specification, using the Endpoint address and the Study, Series, and SOP Instance UIDs from step 3:

Retrieve the DICOM instances.

WADO-RS Retrieve Instance

```
GET <Endpoint.address>/studies/<Study UID>/series/<Series UID>/instances/<SOP Instance UID>
Accept: multipart/related; type="application/dicom"
```

Epic retrieves the instance from the organization's DICOM archive and returns it as a DICOM Part 10 file.

## When Epic Requests Images from You

Organizations that use Epic can also retrieve images from your system using the same steps in reverse: an XCPD request, an XCA document query and retrieve, an ImagingStudy search against your FHIR R4 server, then WADO-RS Retrieve Instance requests to the Endpoint you return. To support this direction, your system should:

* Respond to the XCPD request and accept the patient identifier you returned as a chained patient:Patient.identifier search parameter on ImagingStudy.

* Respond to the XCA document query and retrieve requests.

* Accept the OAuth 2.0 access tokens Epic gets from your token URL, sent in the Authorization: Bearer header of each FHIR and WADO-RS request.

* Return ImagingStudy and Endpoint resources that meet the requirements in the "When Epic is the Requester" section of the ImagingStudy (R4) specification. Resources that don't meet those requirements are ignored.

* Include a secondary identifier on each ImagingStudy that matches the result reference ID of the imaging result in the C-CDA documents you send, so Epic can link the study to the right order.

* Host a WADO-RS Retrieve Instance service at an https address, as described in the WADO-RS Retrieve Instance specification.

## Connecting to DICOM Archives

Diagnostic Image Exchange relies on the organization's DICOM archive. The organization's Epic system connects to the archive over the organization's local network using standard DIMSE services. The organization and its archive vendor configure this connection together. To take part, the archive should support:

* Query (C-FIND) at the study, series, and instance levels, so Epic can find a patient's studies and the series and instances in each. Include the modality and SOP Class UID in responses and report an instance's availability accurately.

* Retrieve (C-MOVE) of individual instances.

* Storage (C-STORE) of instances that Epic receives from other organizations, when the organization requests images through Diagnostic Image Exchange. Received instances keep their original UIDs, so the archive should handle instances it already has, such as from an earlier CD import.

* HL7v2 Image availability notifications. An interface that notifies Epic when an order's images are available in the archive. This notification allows Epic to accurately represent a study's availability for exchange.

* Unique Study Instance UIDs. Epic identifies each study by its Study Instance UID, so each UID should identify exactly one study in the archive. Epic does not modify the Study UID during exchange.

## Operational Notes

Security: Use OAuth 2.0 backend authorization for every FHIR and WADO-RS request. Epic only returns studies that the organization's sharing rules allow for your organization, and it audits each search and retrieval.

Implementation and Testing: Work with the organization's interoperability and imaging teams to test with patients that have imaging results in both systems. If you also support requests from Epic, ask the organization to enable its outgoing Diagnostic Image Exchange requests to your organization. Confirm that the studies you receive match the results in the C-CDA documents and that the images file correctly in your system.

## Get Started

If you're just getting started, see External Health Records for Patient Treatment for how organizations that use Epic connect to exchange networks.

Want a little extra help? Vendor Services offers an optional suite of benefits including the ability to request individualized assistance with your app's data exchange.

In addition, some product categories qualify for Toolbox support and designation on Showroom, which can accelerate implementation and enable you to highlight your product's strengths. Explore the list of eligible categories and share your

interest with us through the Technology Guidance form.

# 2.54 Payer Medical Necessity Guidelines Playbook

File: site/docs/02-playbooks/54-med-necessity-guidelines-playbook.html

Source: https://fhir.epic.com/Documentation?docId=med_necessity_guidelines_playbook

Payers apply care guidelines to decide whether a requested service is medically necessary. Utilization reviewers often need to work through those guidelines while looking at the patient's chart. A common starting point is launching a guideline app from Epic so reviewers can select criteria, document their answers, and send the completed review back to the authorization.

This playbook outlines recommended practices using public interoperability standards (SMART on FHIR, FHIR®) so guideline vendors can connect with payers that use Epic.

## Launch from the Reviewer's Workflow

Use a SMART on FHIR EHR launch so the reviewer can open your app directly from Epic with the patient already in context. This also signs the reviewer in without a separate login. See the OAuth 2.0 tutorial for how to set up the launch and token exchange.

## Present Criteria for Review

Once the app launches, show the reviewer the guideline criteria that apply to the case. Let them select the criteria they want, evaluate them against the patient's information, and document their findings. A review is the result of applying one set of criteria to one patient.

## Send Reviews to Epic

When the reviewer is finished, send each completed review into Epic with the CriteriaReview.Create web service so it is saved with the patient's authorization.

## Group Reviews for the Same Authorization

If a reviewer applies more than one set of criteria to the same authorization request, group those reviews together with the ReviewCollection.Create web service. Epic then presents them as a single set.

## Confirm Completion

After the reviews have been sent, show a page letting the reviewer know they can close the browser window.

## Update a Review

If a reviewer changes a review after it was sent, update it with CriteriaReview.Update. Also call ReviewCollection.Update so the group of reviews stays in sync.

## Operational Notes

Security: Use SMART on FHIR (OAuth 2.0) for user-launched access and Backend OAuth for server-to-server calls.

Testing: Work with the payer's team to test launches and review filing end to end, and confirm that reviews appear correctly in the reviewer's workflow.

## Get Started

These connection practices have enabled many successful connections across our community. As every product has unique goals, we encourage you to review this approach with your customers to confirm the best fit.

If you're just getting started, check out our step-by-step Developer Guide.

Want a little extra help? Vendor Services offers an optional suite of benefits including the ability to request individualized assistance with your app's data exchange.

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# 2.55 Hospital Utilization Review Creation Playbook

File: site/docs/02-playbooks/55-utilization-review-playbook.html

Source: https://fhir.epic.com/Documentation?docId=utilization_review_playbook

Hospital Utilization Review Creation apps provide guidelines for the treatment of various medical conditions and the associated reimbursement. These apps integrate with the Utilization Review activity in Clinical Case Management. Utilization managers launch the app from Epic using a SMART on FHIR EHR launch with patient and encounter context, then document a patient's symptoms, events, and treatments to determine whether the patient meets sufficient criteria for admission or discharge and what level of care is needed in the hospital. The app then sends the criteria and notes added by the utilization manager into the Utilization Review activity in Epic.

## Filing Reviews

A criteria review is the result of applying a single set of criteria to a patient. Reviews are filed into Epic using the CriteriaReview.Create web service and include the criteria that were evaluated, whether each was met, the overall determination, and notes from the utilization manager.

To support editing a previously created review, the app should accept the review identifier from Epic on subsequent launches and use it to open the existing review. Previously created reviews can be updated by first calling CriteriaReview.Read to get the most up-to-date version of the review, then calling CriteriaReview.Update. Reviews created in error can be removed by updating their status to entered-in-error.

If you have created multiple reviews for the same case, such as an admission review and later continued stay reviews, these can be managed as a group using the ReviewCollection.Create, ReviewCollection.Read, and ReviewCollection.Update web services. The reviews will need to be created using CriteriaReview.Create before they can be grouped together.

After a review is sent to Epic, the app can close the browser window by calling AGL CloseActivity.

## Automated Review Creation

Reviews can also be created without a user launching the app. Review creation can be triggered by events in Epic or by patient census information fetched from Epic, and the resulting unsolicited reviews are filed using the CriteriaReview web service. Automatically created reviews that are still in progress should remain available for users to complete or delete.

## Clinical Data

FHIR APIs can be used to read discrete data from the patient's chart, such as labs, medications, vitals, and procedures. For user-launched reviews, this data can be used to pre-fill the review so the utilization manager doesn't need to enter findings that are already documented in Epic. For automated reviews, clinical data should be used to evaluate criteria before the review is filed.

## Get Started

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